THE GEOPHYSICAL INSTITUTE OF ISRAEL המכון הגיאופיסי לישראל
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סקר אלקטרומגנטי

(TDEM)

לאיתור וניטור הפן-הביני באקויפר החוף

פברואר, 2000 דו"ח מס' 904/62/99

מאת

מרק גולדמן, דרור גלעד, מיכאל יזרסקי, גנאדי שקליאר

הוכן עבור:

השרות ההידרולוגי - נציבות המים

1. הקדמה

המכון הגיאופיסי לישראל ביצע ב-1999 / 11-12 מדידות אלקטרומגנטיות בשיטת TDEM עבור השרות ההידרולוגי. במסגרת זו נערכו 61 מדידות בתוך הים התיכון, מזיקים בדרום ועד ראש הנקרה בצפון. חלק מהמדידות בוצעו בתחנות נטור קבועות בשנים 1989, 1993, 1995 ו-1997 (ראה טבלה מס' 1, שבה מוצגת המדידה האחרונה שבוצעה באתר). בנוסף, הטבלה כוללת גם מידע כללי על המדידות כגון מיקום, רום קרקע, גודל הלולאה ומרחק מהים.

מטרת הסקר במדידות חוזרות הייתה לעקוב אחר התנודות בעומק הפן-הביני לאורך זמן. התנודות הללו באות לידי ביטוי בשינוי בעומק השכבה הגיאוחשמלית המוליכה ביותר בחתך (התנגדות חשמלית נמוכה מ-3 אוהם-מטר).

מטרת הסקר במדידות באתרים חדשים הייתה לקבוע העומק עד שכבה (או שכבות) מוליכה/ות, לצורך מיפוי חדירת מי הים לאקויפר החוף באזורים בהם אין קידוחי תצפית. בנוסף לכך, מספר מדידות בוצעו בקרבת קידוחי תצפית פן-ביני לצורכי כיול וביקורת מדידות ה-TDEM.

מרבית תחנות המדידה אוזנו ב-GPS דיפרנציאלי. הסטייה המרבית במיקום וברום הקרקע נמוכה מ-1 מ'.

תחנות המדידה שלא אוזנו במהלך סקר זה שייכות לקבוצות הבאות:

א). מדידות חוזרות שבוצעו בדיוק באתרי מדידות קודמות כאשר גודל וכיוון לולאת המשדר נשארו ללא כל שינוי.תחנות אלא כונו "Exact monitoring".

אם רק המיקום לא השתנה, התחנות כונו "Exact same location" (ראה טבלה מס' 1).

ב). מדידות בהן היעדרות פו-ביני הובחנה ישר בשדה עקב ערכי התנגדות מדומה גבוהים למדי ("No interface"), או עקב עוות מוחלט של המדידה ("Completely distorted").

ג). תחנה HS502 לא אוזנה.הרום נקבע על פי איזון קידוח חדש באתר ("New borehole").

כל התחנות שלא אוזנו ע"י ה-GPS הדיפרנציאלי אוזנו ב-GPS רגיל, כאשר הסטייה המרבית במיקום אינה עולה על 30 מ'. נתוני רום הקרקע בנקודות אלו התקבלו ממפה טופוגרפית בקנה-מידה של 1:50000. במקרה זה, נקודות הציון מופיעות בטבלה מס' 1 כבנות חמש ספרות בלבד.

2. השיטה

שיטות העבודה בנטור ובאיתור הפן-הביני מתוארות בפרסומים ודו"חות מקצועיים קודמים (Goldman et al., 1988, רבינוביץ ואחרים, 1993), כך שאין צורך לפרטן שוב.

3. תוצאות

תוצאות המדידות ופענוח נתוני השדה שבהם הובחנה יחידת התנגדות נמוכה מ3- אוהם-מטר המתאימה להתנגדות מי-ים (Goldman et al., 1988), מוצגות בטבלה מס' 2 ובנספח לדו"ח זה. חשוב לציין שבמספר מקרים התנגדות היחידה הייתה גבוהה במקצת מ-3 אוהם-מטר, אך עקב שיקולים הידרוגיאולוגיים היא בכל זאת שויכה ל שכבת מי-ים.

כמו כן בטבלה מוצגות תחנות TDEM בהן לא נמצאו מי ים ואלו שלא ניתנו לפענוח עקב רעש אלקטרומגנטי ו/או עוות תלת-מימדי. הנתונים המוצגים בטבלה זו כוללים רק חלק מתוצאות הפענוח (התוצאות המלאות מוצגות בנספח לדו"ח).

בטבלה הוגדרו שתי יחידות התנגדות נמוכה בחלק מהמדידות. היחידה העליונה בעלת ערכי התנגדות בין 1 ל-3 אוהם-מטר, המתאימה בדרך כלל למי הים באקויפר. היחידה התחתונה בעלת ערכי התנגדות דומים לאלו שהתקבלו בשכבה העליונה המתאימה בד"כ לחרסית הסקיה שבעומק. בחלק מהנקודות הובחנה יחידת התנגדות נמוכה אחת בלבד. כאשר נמצאה היחידה בחלק העליון של החתך, היא שויכה ליחידת התנגדות נמוכה עליונה, בעוד שכאשר היא נמצאה בחלק התחתון, היא שויכה ליחידה התחתונה.
במספר נקודות מדידה נמצאה יחידת התנגדות גבוהה יחסית, המונחת על-גבי יחידת התנגדות נמוכה ממנה. תופעה זו מצביעה על תחום ערבוב של מים מליחים הנמצאים מעל למי-ים. במקרים אלה ערכי התנגדויות ועומקים על שתי השכבות הללו מוצגים בטבלה 2 באותה עמודה כאשר הם מחולקים על-ידי סימן "/".

מדידות HS518, HS540, HS545 מעוותות ע"י הפרעות תלת-מימדיות חזקות בלתי ניתנות לפענוח באמצעות מודלים חד-מימדיים. אמנם ניתן לפענח את החלק ההתחלתי של המדידות שלכאורה אינו מעוות או מעוות הרבה פחות מהחלק האמצעי או החלק הסופי שלהן, אך רק במדידה אחת בלבד (HS540) הצלחנו לפענח את החלק ההתחלתי של המדידה ולאתר מים מלוחים בחלק העליון של החתך (HS540A). חשוב לציין שאמינותו של הפענוח נמוכה למדי, כך שהתוצאות הכמותיות של הפענוח יכולות לשמש לצורך הערכה בלבד.

רשימת פרסומים

רבינוביץ, ב., גולדמן, מ., רבינוביץ, מ., רונן, ע., 1993. סקר אלקטרומגנטי לאיתור הפן הביני באקויפר החוף, דו"ח מס' 735/296/92, המכון הגיאופיסי.

Goldman, M., Arad, A., Kafri, U., Gilad, D. and Melloul, A., 1988. Detection of freshwater/ seawater interface by the Time Domain Electromagnetic (TDEM) method in Israel, 10th SWIM, Ghent.

TABLE 1 General data on the TDEM stations

TDEM Station HS...	Location	Loop size (m)	Easting	Northing	Elevation (m)	Distance from sea (m)	Previous Station HS...	Comments
501	Ashqelon	100	107822	119369	34.6	750		GPS
502	Ziqim	100	10314	11232	4	800	279	New borehole
503	Barnea	100	109842	122800	28.2	600		GPS
504	Ashqelon	100	109552	123046	19.6	250	320	GPS
505	Nitzanim	100	110828	124710	16.6	550	324	GPS
506	Nitzanim	100	111000	124642	15.5	700	82	GPS
507	Nitzanim	100	112871	128064	6.8	650	327	Exact monitoring
508	Nitzanim	100	112445	128120	4.6	250	328	Exact monitoring
509	Ashdod	100	114330	130891	17.5	650	301	GPS
510	Hof Rishon	100	124977	156167	4.2	250		GPS
511	Hertzliya	100	131690	174374	18.3	700		GPS
512	Shefaim	100	133906	181147	28.2	900		GPS
513	Yaqum	100	133954	184130	41.2	150	336	Exact monitoring
514	Yaqum	100	134582	184231	16.5	750		GPS
515	Poleg	100	135212	186986	14.2	700		GPS
516	Hofit	100	138625	199356	10.2	1250		GPS
517	Michmoret	100	139479	202392	21.0	1300		GPS
518	Giv'at Olga	100	139756	203934	12.2	1350		GPS
519	Giv'at Olga	100	139295	205860	10.6	530	264	GPS
520	Giv'at Olga	100	139929	205986	23.8	1150		GPS
521	Sdot Yam	50	139750	209851	2.1	50	252	GPS
522	Caesarea	100	140271	210199	5.5	500	251	GPS
523	Caesarea	100	140933	213671	5.7	330	263	GPS
524	Caesarea	50	140632	213363	4.4	100	253	GPS
525	Nahsholim	50	142888	225304	3.7	100	285	Exact same location
526	Neve Yam	100	143539	229343	4.8	75	286	GPS
527	Neve Yam	100	143991	229259	5.9	550	287	GPS
528	Neve Yam	50	143707	231164	4.3	75		GPS
529	Atlit	50	145185	236104	4.4	50		GPS
530	Kfar Galim	100	146051	240800	15.1	300		GPS
531	Kfar Shamir	100	146525	244091	15.6	450		GPS
532	Hahotrim	100	145966	239222	9.9	350		GPS
533	Habonim	50	143098	227199	3.9	100		GPS
534	Ga'ash	100	134329	182796	23.0	850		GPS
535	Caesarea	200	140302	211513	7.0	400	459	Exact same location
536	Jisr-Al-Zarka	100	14139	21594	5.0	450		No interface
537	Jisr-Al-Zarka	200	14255	21601	10.0	1500		No interface
538	Rosh Ha-Nikra	100	160794	276209	6.5	550	357	GPS
539	Betzet	50	160707	275063	7.1	550	354	GPS
540	Ahziv	50	15960	27120	5.0	200		Completely distorted
541	Nahariya	100	158134	265841	4.9	150		GPS
542	Nahariya	100	158520	265782	5.8	550		GPS
543	Hatzrot Yosef	50	157491	262631	1.9	100	351	GPS
544	Geva Carmel	100	146959	230378	35.1	3400		GPS
545	Carmel Park	100	148323	235546	56.9	3250		GPS
546	Shvei Zion	50	157813	264233	2.6	200	352	GPS
547	Shvei Zion	100	158073	264255	3.0	450		GPS
548	Betzet	100	161084	274871	17.5	900		GPS
549	Hatzrot Yosef	100	157750	262561	2.2	400		GPS

TDEM Station HS...	Location	Loop size (m)	Easting	Northing	Elevation (m)	Distance from sea (m)	Previous Station HS...	Comments
550	Hatzrot Yosef	100	158125	262487	3.6	750		GPS
551	Bustan Hagalil	100	157835	261625	3.4	550	350	GPS
552	Bustan Hagalil	100	158350	261006	4.1	1150	349	GPS
553	Kfar Galim	100	146436	240662	20.0	700		GPS
554	Neve Yam	100	144244	231191	6.1	600		GPS
555	Einot Shumi	200	145827	216075	18.3	4700		GPS
556	Shefaim	100	133787	180958	33.8	850	333	GPS
557	Palmahim	50	119737	143936	3.2	150	1	GPS
558	Palmahim	50	119874	143763	13.9	350		GPS
559	Palmahim	50	120988	146965	3.5	100	2	GPS
560	Palmahim	100	119416	142330	9.6	500		GPS
561	Palmahim	100	121194	146902	13.4	800	4	GPS

Table 2
Resistivities and Depths to the Low Resistivity Unit(s) Associated with Seawater Intrusion and/or the Saqiya Clays

TDEM Station No.	Hydrological Strip	Upper Layer(s)		Lower Layer(s)		Comments
		Resistivity (Ohm-m)	Elevation BMSL (m)	Resistivity (Ohm-m)	Elevation BMSL (m)	
502	101	2.9/0.8	90/132			
501	106	1.5	63	1.3	170	
503	108	1.4	80	1.1	158	
504	108	2.3/0.9	39/54			
505	109	2.5	62	1.2	166	
506	109	2.6	81	1.0	178	
507	111	1.0	61	0.7	181	
508	111	1.5	48	1.0	163	
509	113	1.9	52	1.5	156	
557	119	1.4	24	1.0	205	
558	119	2.1	48	3.3	156	
560	119	2.4/0.5	73/125			
559	121	1.3	28	2.5	173	
561	121	3.1*	68	2.1	137	
510	126	3.9*	40	3.6	119	
511	136	0.3	28			
512	139	1.6	64	1.6	155	
556	139	1.3	55	1.2	151	
534	140					Extremely distorted, NI
513	141	1.9/1.1	32/54	2.0	138	
514	141	1.5	50	2.1	153	
515	142	2.5	55	1.9	145	
516	148	3.2/1.0	46/146			
517	150	2.9	60	1.8	122	
518	151					3D distortion, NI
519	152	4.6/1.7	43/80	1.1	120	
520	152	1.2	53	0.7	142	
521	154	1.9/1.4	29/52			
522	154	3.1*	51			Noisy, LR
535	155	1.3	43	2.2	97	

TDEM Station No.	Hydrological Strip	Upper Layer(s)		Lower Layer(s)		Comments
		Resistivity (Ohm-m)	Elevation BMSL (m)	Resistivity (Ohm-m)	Elevation BMSL (m)	
523	156			2.3	153	
524	156	2.9*	45			
536	157					No interface
537	157					No interface
525	The Carmel Coast	2.3	15			LR
526		2.8/1.5	20/45			
527		3.5*	18			
528		3.3*	18			
529						No interface
530		2.3	36			
531		1.4	10 (AMSL)	2.5	148	LR
532		2.2	3.3 (AMSL)			LR
533		3.8*	25			
544				1.0	293	
545						3D distortion, NI
553						No interface
554				4.3/2.0	152/311	
555				3.0*	550	
538	The Western Galilee Coast	3.9*	27			
539		1.8/0.7	34/133			
540						3D distortion, NI
540A		2.6	17			3D distortion, LR
541		3.3*	33			
542		1.7	63			
543		1.2	14	1.8	32	
546		1.3	18	3.0	150	
547						No interface
548		2.5	34	2.4	133	
549		1.0	32	2.3/0.5/0.2	49/83/133	
550		1.3	2	2.1	32	
551		2.7/1.0	41/63	1.8/1.0	83/122	
552		1.3/0.9	50/90			

* = Abnormally high resistivity for seawater intrusion
LR = Low Reliability; NI = Not Interpreted

DATA SET: HS501

CLIENT: 100x100m DATE: 22-Nov-99
LOCATION: Ashqelon SOUNDING: z
COUNTY: 107822/119369 (+34.6m) ELEVATION: 34.60 m
PROJECT: FITTING ERROR: 2.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 85.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.181 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			34.60	
1	152.4	55.69	-21.09	0.365
2	25.58	41.91	-63.00	1.63
3	1.51	44.96	-107.9	29.72
4	5.91	62.07	-170.0	10.49
5	1.31			

ALL PARAMETERS ARE FREE

נספח

DATA SET: HS502

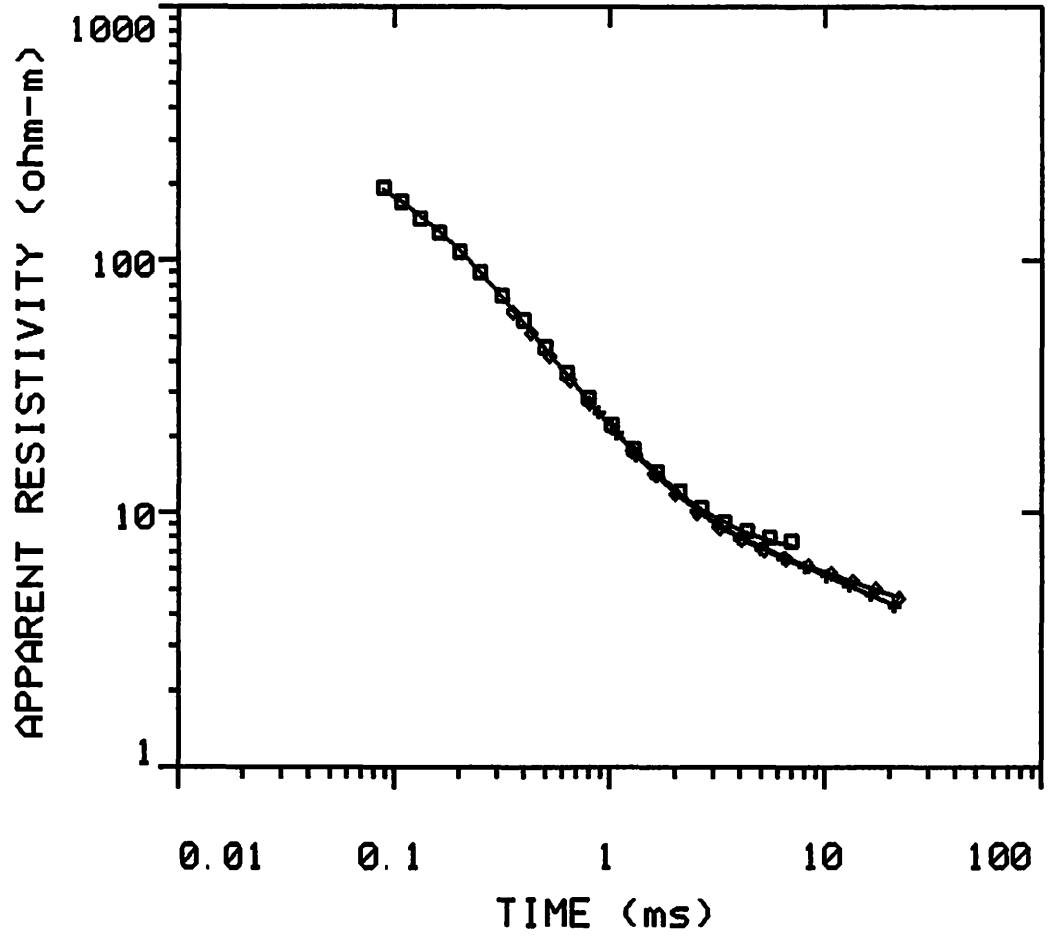
CLIENT: 100x100m DATE: 22-Nov-99
LOCATION: Ziqim SOUNDING: z
COUNTY: 103140/112320 (+5m) ELEVATION: 5.00 m
PROJECT: FITTING ERROR: 3.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 86.0000 SLOPE: NONE

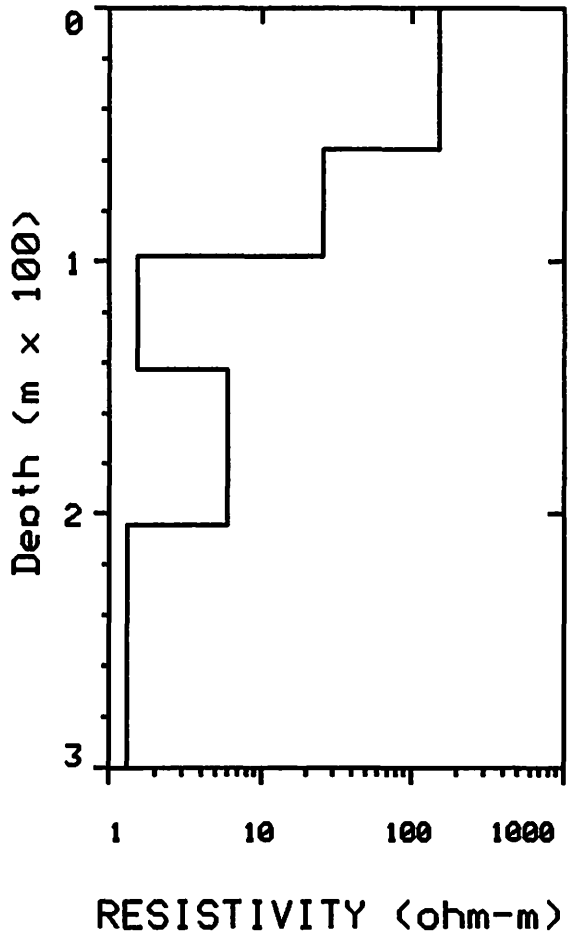
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.289 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			5.00	
1	20.90	3.51	1.48	0.167
2	592.5	14.84	-13.35	0.0250
3	17.32	76.44	-89.79	4.41
4	2.87	42.71	-132.5	14.86
5	0.801	54.28	-186.7	67.71
6	27.54			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.2%
I.P.R.G.	Ashqelon 107822/119369 (+34.6m)
Data Set: HS501 / Date: 22-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS503

CLIENT: 100x100m
LOCATION: Barnea
COUNTY: 109842/122800 (+28.2m)
PROJECT: FITTING ERROR: 3.0%
LOOP SIZE: 100.000 m by 100.000 m
COIL LOC: 0.000 m (X), 0.000 m (Y)
SOUNDING COORDINATES: E: 1.0000 N:

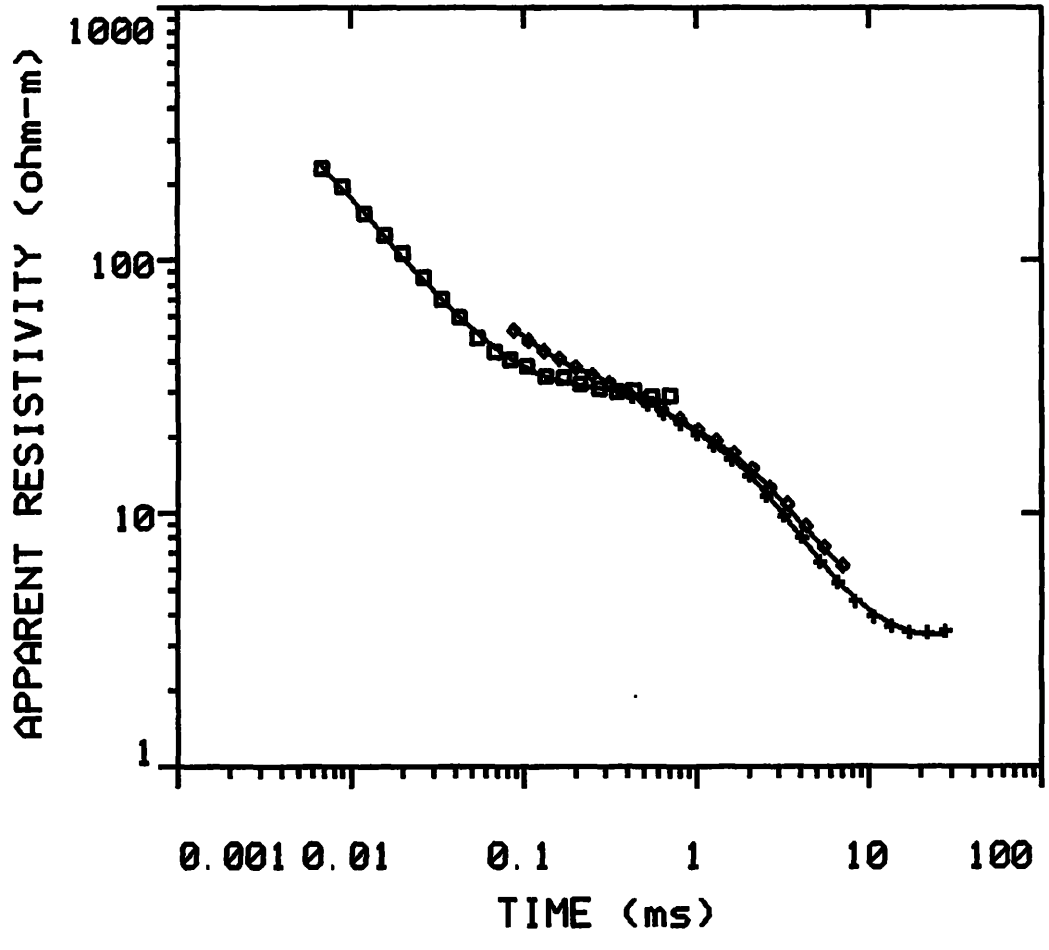
DATE: 22-Nov-99
SOUNDING: z
ELEVATION: 28.20 m
EQUIPMENT: Geonics PROTEM
AZIMUTH:
TIME CONSTANT: NONE
87.0000 SLOPE: NONE

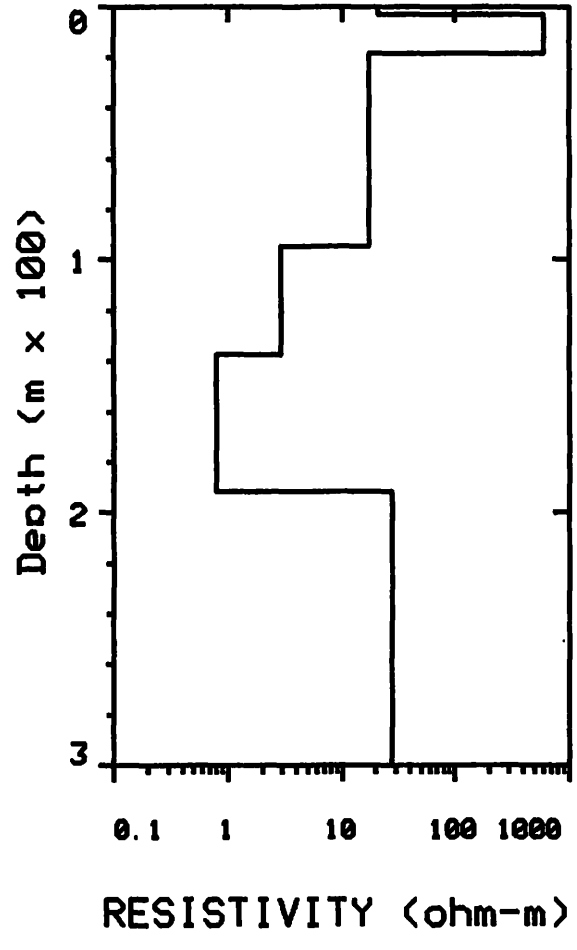
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.993 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			28.20	
1	34.30	15.81	12.38	0.461
2	85.54	43.83	-31.45	0.512
3	15.73	23.62	-55.08	1.50
4	100.3	25.17	-80.25	0.250
5	1.44	51.75	-132.0	35.75
6	8.90	25.80	-157.8	2.89
7	1.08			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.3%
I.P.R.G.	Ziqim 103140/112320 (+5m)
Data Set: HS502 / Date: 22-Nov-99	
Equipment: Geonics PROTEM / Sounding: 2	Azimuth:

DATA SET: HS504

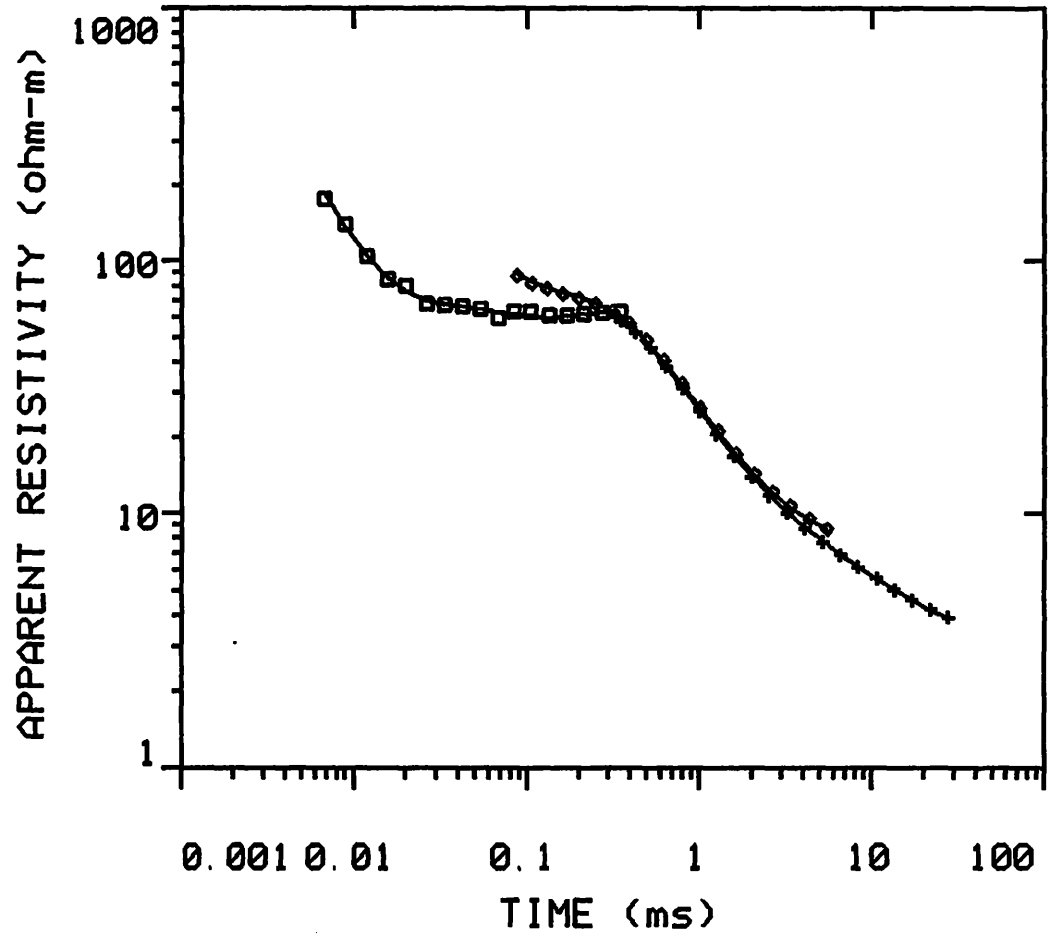
CLIENT: 100x100m DATE: 23-Nov-99
LOCATION: Ashqelon SOUNDING: z
COUNTY: 109552/123046 (+19.6m) ELEVATION: 19.60 m
PROJECT: FITTING ERROR: 3.4% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 88.0000 SLOPE: NONE

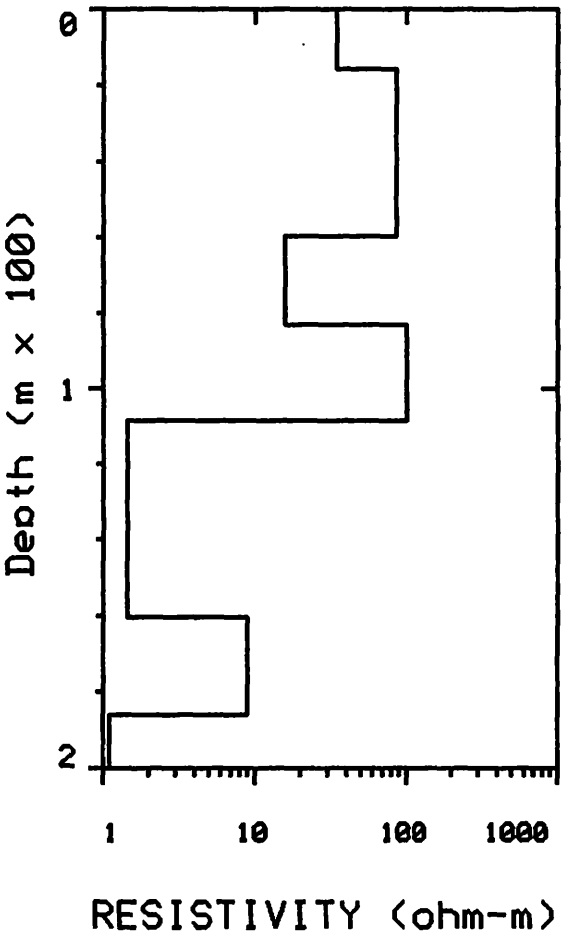
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.453 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			19.60	
1	86.18	14.71	4.88	0.170
2	40.20	44.23	-39.34	1.10
3	2.34	14.80	-54.15	6.32
4	0.931			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.0%
I.P.R.G.	Barnea 109842/122800 (+28.2m)
Data Set: HS503 / Date: 22-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS505

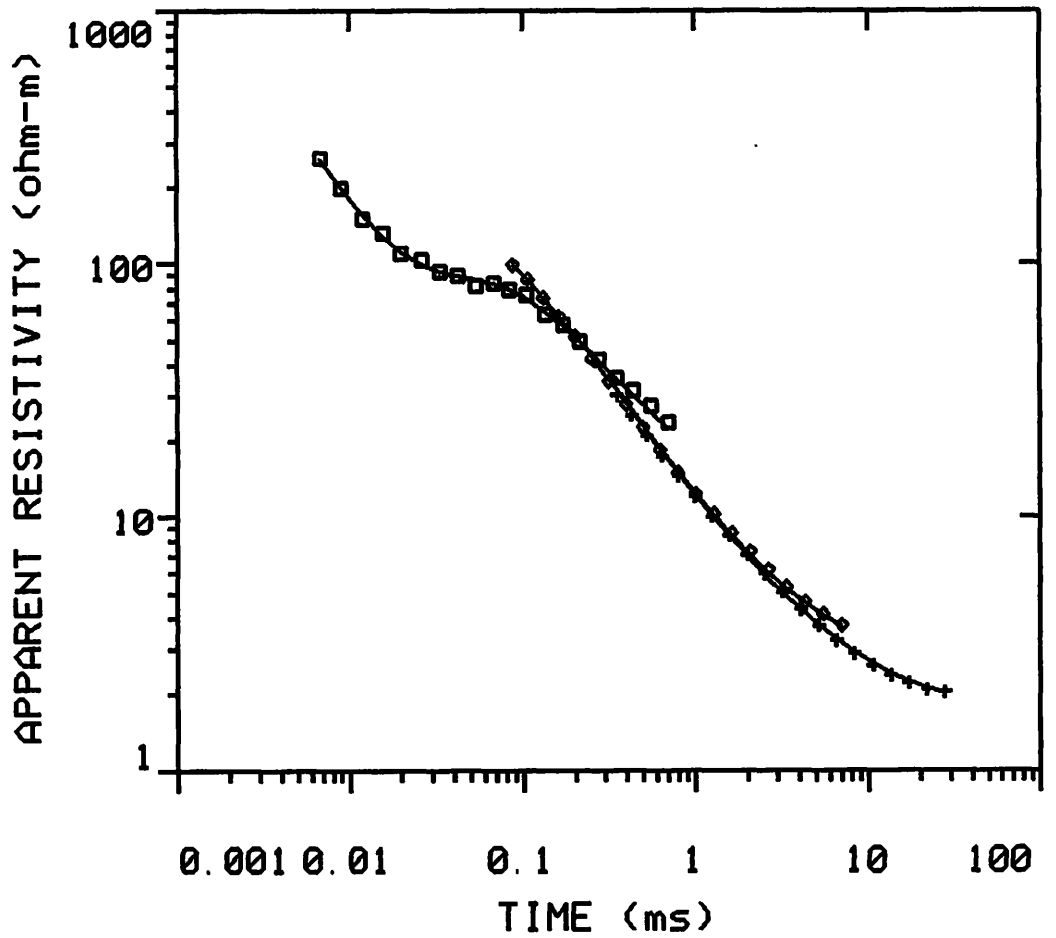
CLIENT: 100x100m DATE: 23-Nov-99
LOCATION: Nitzanim dunes SOUNDING: z
COUNTY: 110828/124710 (+16.6m) ELEVATION: 16.60 m
PROJECT: FITTING ERROR: 5.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 89.0000 SLOPE: NONE

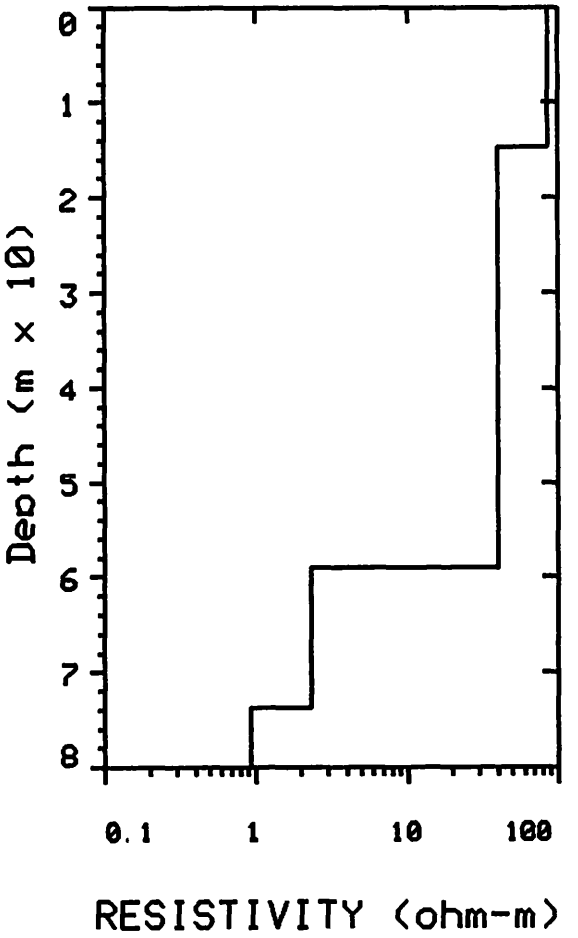
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 5.233 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			16.60	
1	87.76	78.72	-62.12	0.897
2	2.46	74.21	-136.3	30.13
3	8.86	29.65	-166.0	3.34
4	1.18			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.4%
I.P.R.G.	Ashqelon 109552/123046 (+19.6m)
Data Set: HS504 / Date: 23-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS506

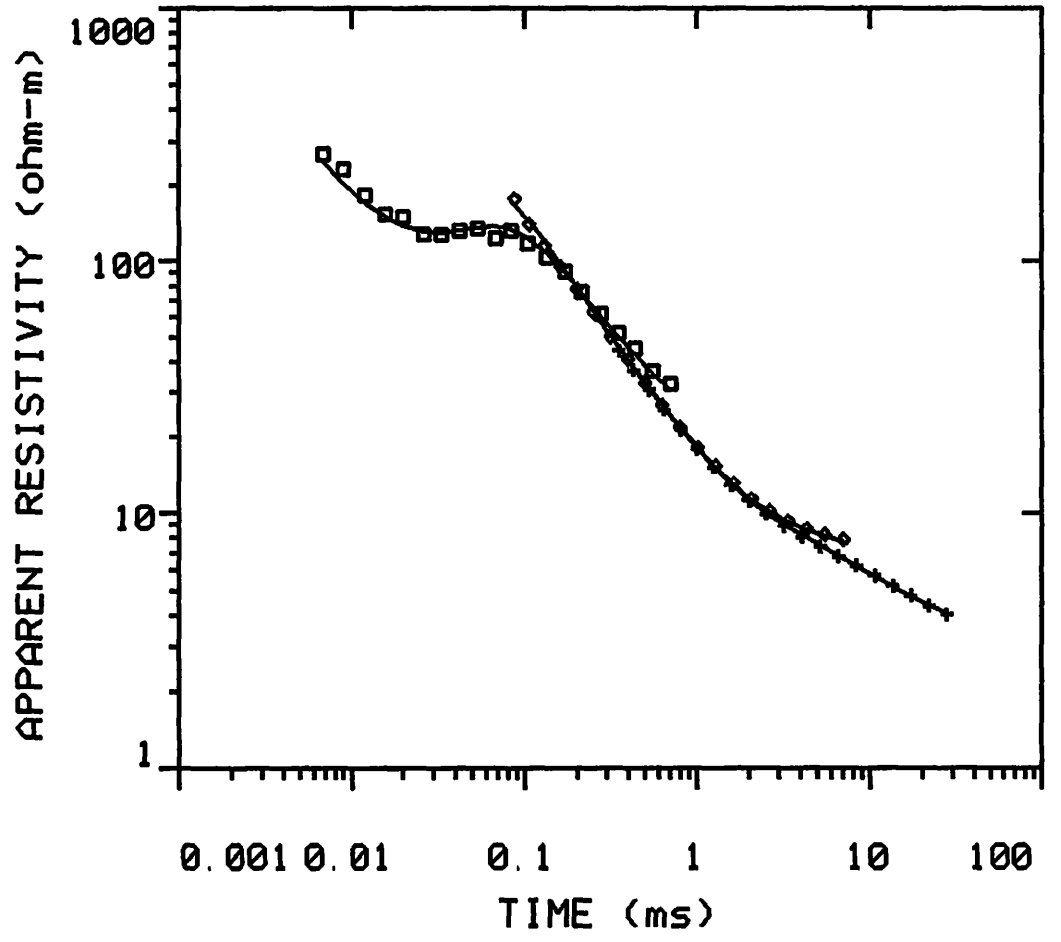
CLIENT: 100x100m DATE: 23-Nov-99
LOCATION: Nitzanim dunes SOUNDING: z
COUNTY: 111000/124642 (+15.5m) ELEVATION: 15.50 m
PROJECT: FITTING ERROR: 2.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 90.0000 SLOPE: NONE

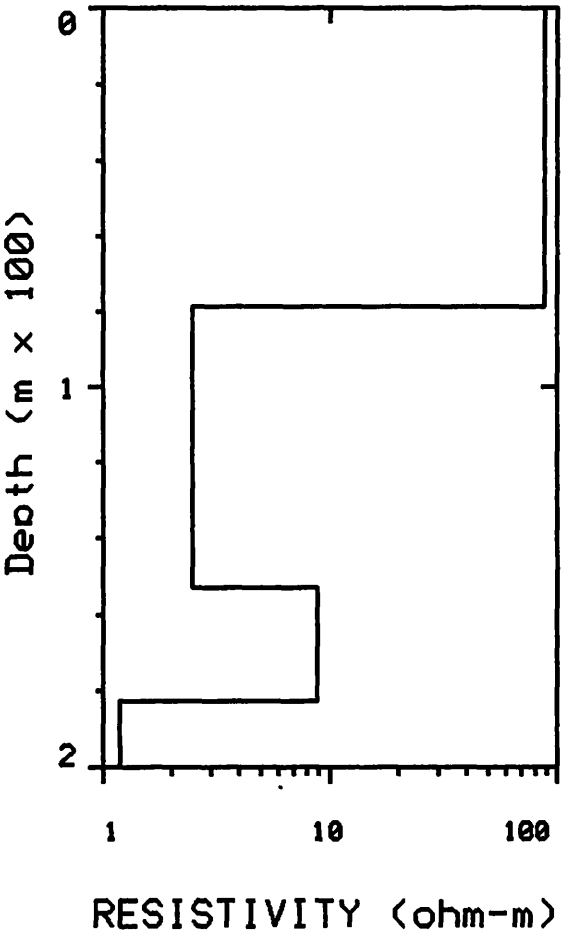
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.283 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			15.50	
1	22.41	12.35	3.14	0.551
2	54.82	59.18	-56.04	1.07
3	10.63	24.74	-80.79	2.32
4	2.65	61.75	-142.5	23.29
5	9.41	35.98	-178.5	3.82
6	1.02			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 5.2%
I.P.R.G.	Nitzanim dunes 110828/124710 (+16.6m)
Data Set: HS505 / Date: 23-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS507

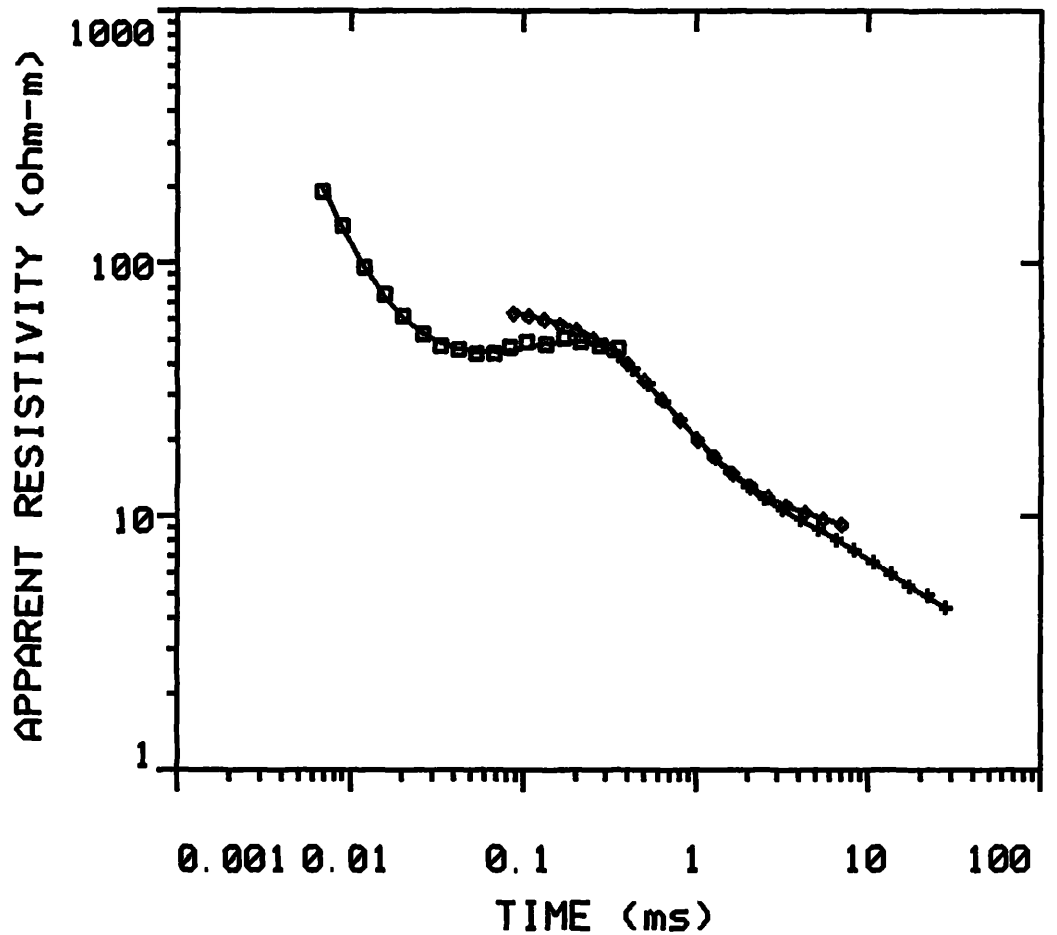
CLIENT: 100x100m DATE: 23-Nov-99
LOCATION: Hof Nitzanim SOUNDING: z
COUNTY: 112871/128064 (+6.8m) ELEVATION: 6.80 m
PROJECT: FITTING ERROR: 3.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 91.0000 SLOPE: NONE

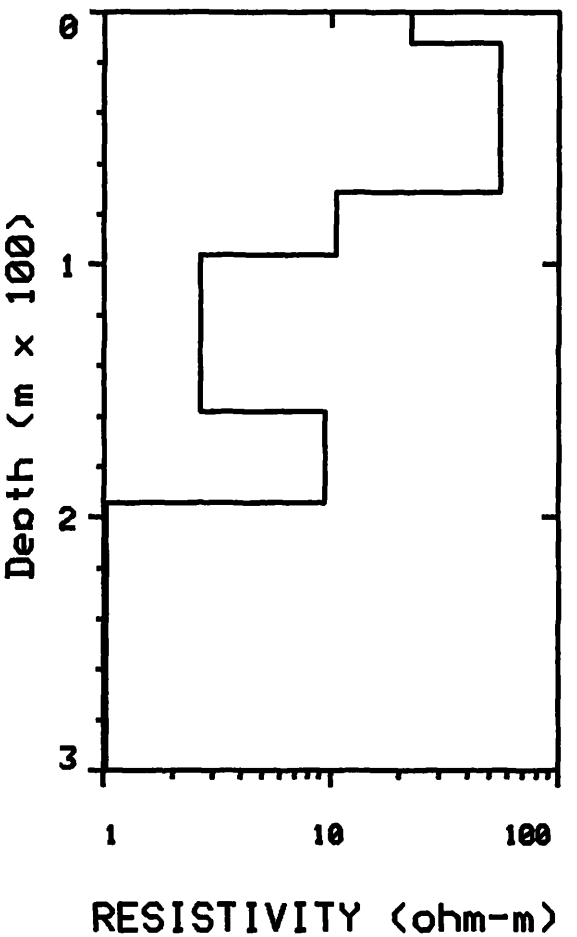
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.253 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			6.80	
1	62.03	12.36	-5.56	0.199
2	32.81	55.66	-61.23	1.69
3	1.03	38.09	-99.32	36.88
4	9.87	82.22	-181.5	8.32
5	0.746			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.3%
I.P.R.G.	Nitzanim dunes 111000/124642 (+15.5m)
Data Set: HS506 / Date: 23-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS508

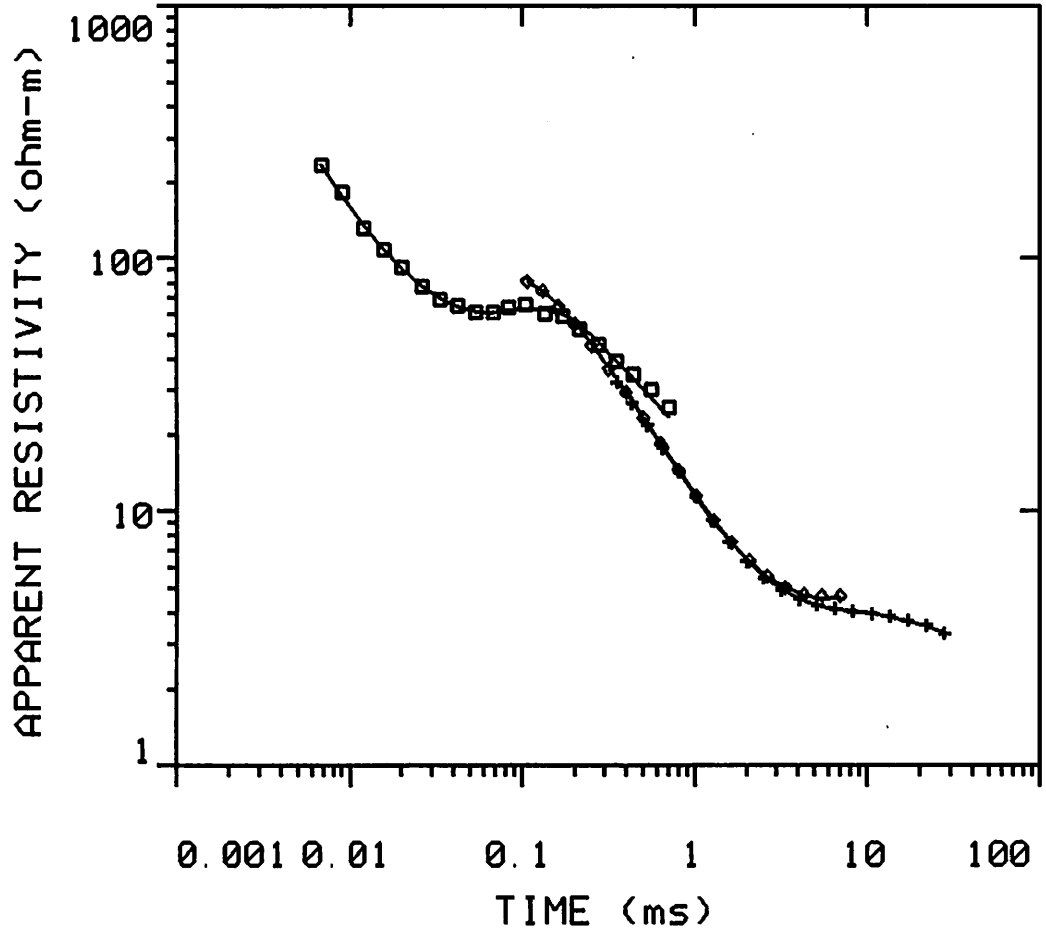
CLIENT: 100x100m DATE: 24-Nov-99
LOCATION: Nitzanim dunes SOUNDING: z
COUNTY: 112445/128120 (+4.6m) ELEVATION: 4.60 m
PROJECT: FITTING ERROR: 2.5% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 92.0000 SLOPE: NONE

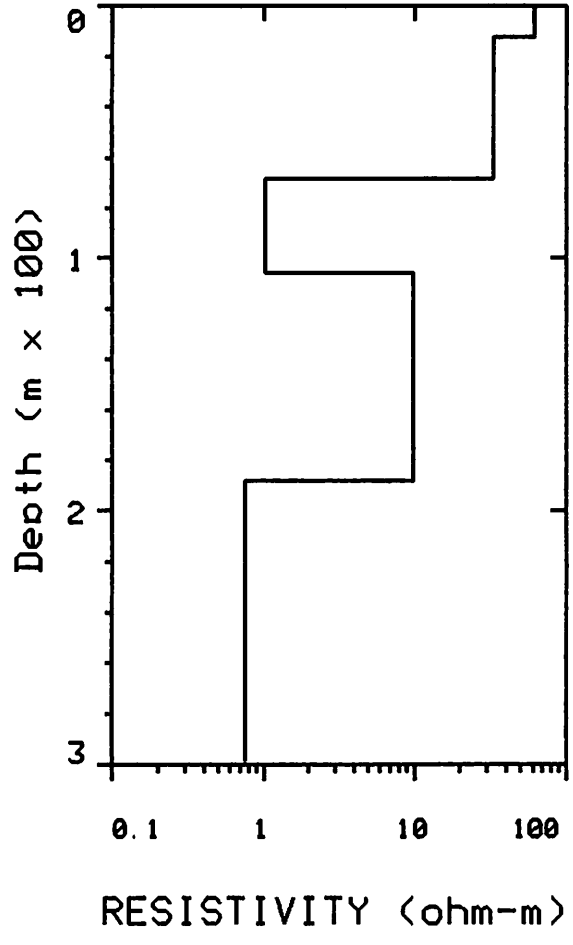
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.573 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.60	
1	30.29	52.73	-48.13	1.74
2	1.46	66.75	-114.8	45.71
3	8.57	47.72	-162.6	5.56
4	1.03			

ALL PARAMETERS ARE FREE





100×100m	FITTING ERROR: 3.2%
I.P.R.G.	Hof Nitzanim 112871/128064 (+6.8m)
Data Set: HS507 / Date: 23-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS509

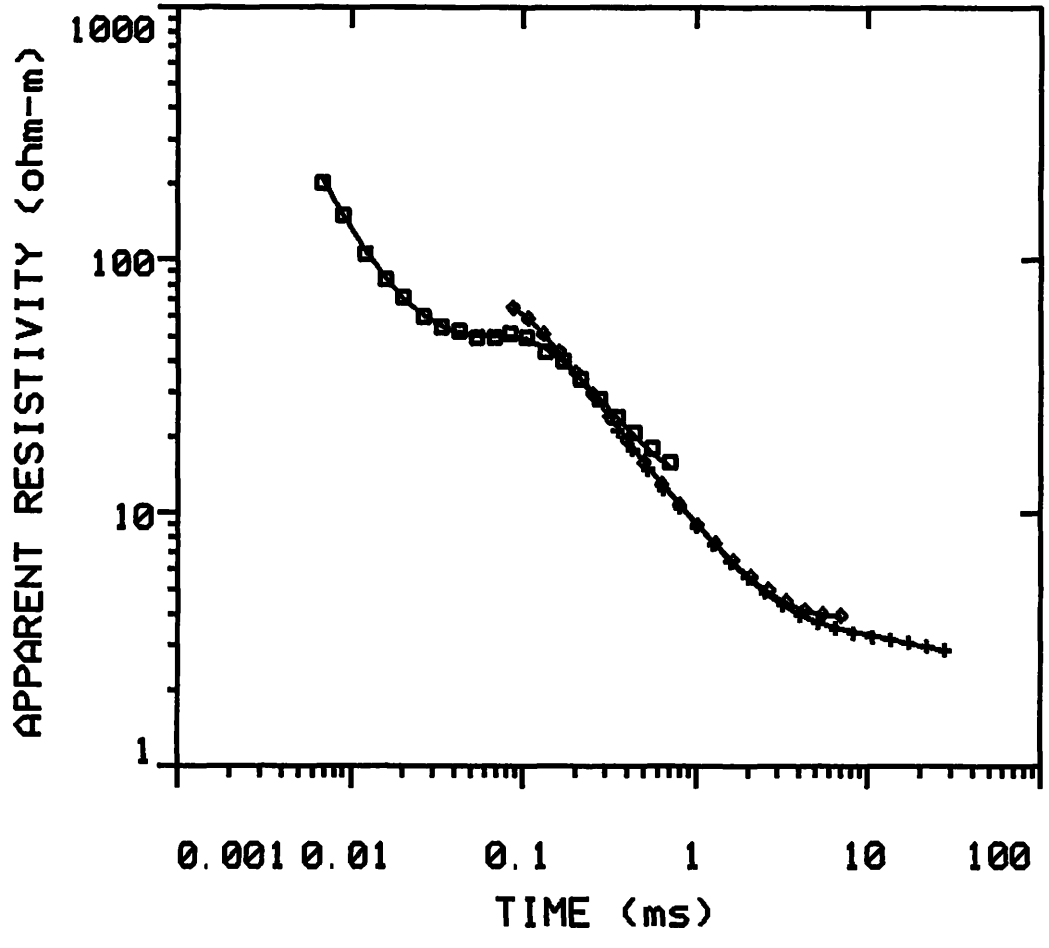
CLIENT: 100x100m DATE: 24-Nov-99
LOCATION: Ashdod South SOUNDING: z
COUNTY: 114330/130891 (+17.4) ELEVATION: 17.40 m
PROJECT: FITTING ERROR: 3.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 93.0000 SLOPE: NONE

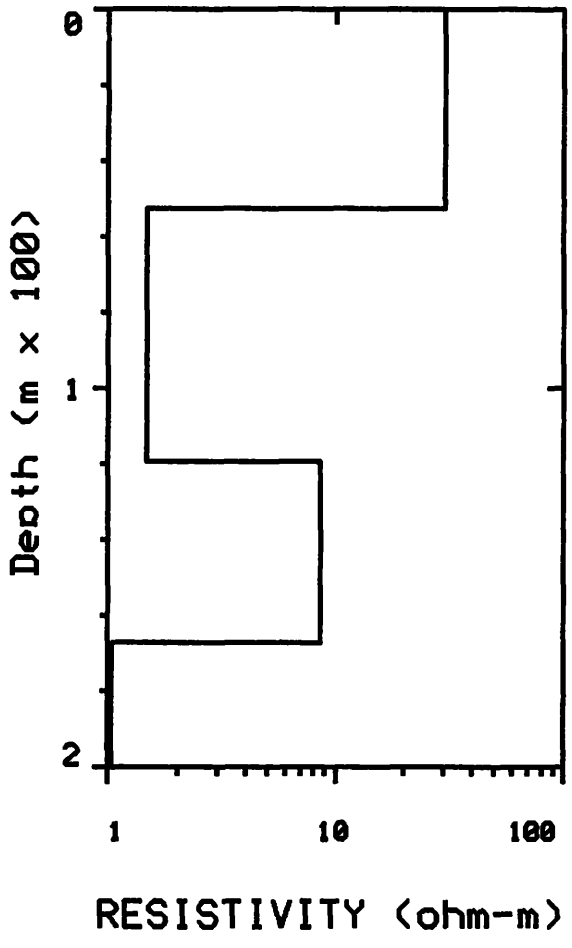
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.661 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			17.40	
1	39.20	15.75	1.64	0.401
2	272.7	53.91	-52.26	0.197
3	1.90	50.67	-102.9	26.54
4	40.44	53.22	-156.1	1.31
5	1.52			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.5%
I.P.R.G.	Nitzanim dunes 112445/128120 (+4.6m)
Data Set: HS508 / Date: 24-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS510

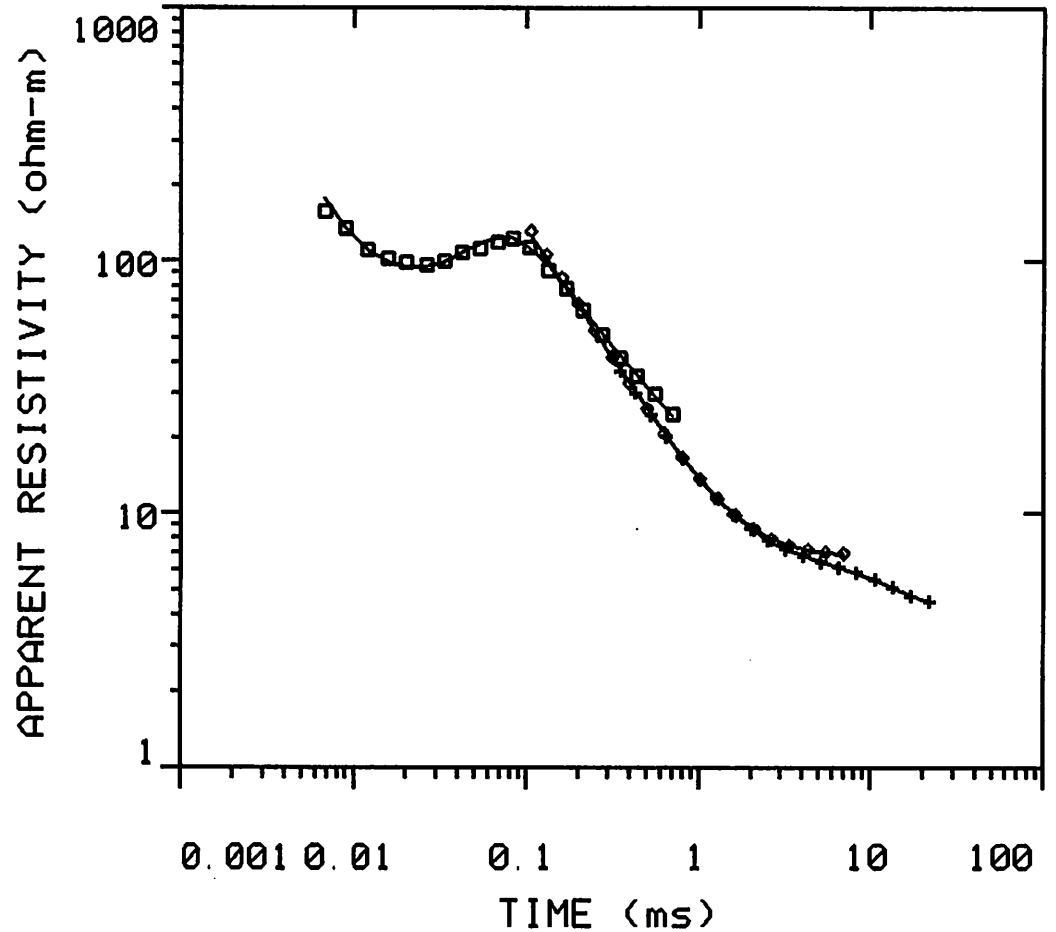
CLIENT: 100x100m DATE: 24-Nov-99
LOCATION: Hof Rishon SOUNDING: z
COUNTY: 124978/156167 (+4.2) ELEVATION: 4.20 m
PROJECT: FITTING ERROR: 2.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 94.0000 SLOPE: NONE

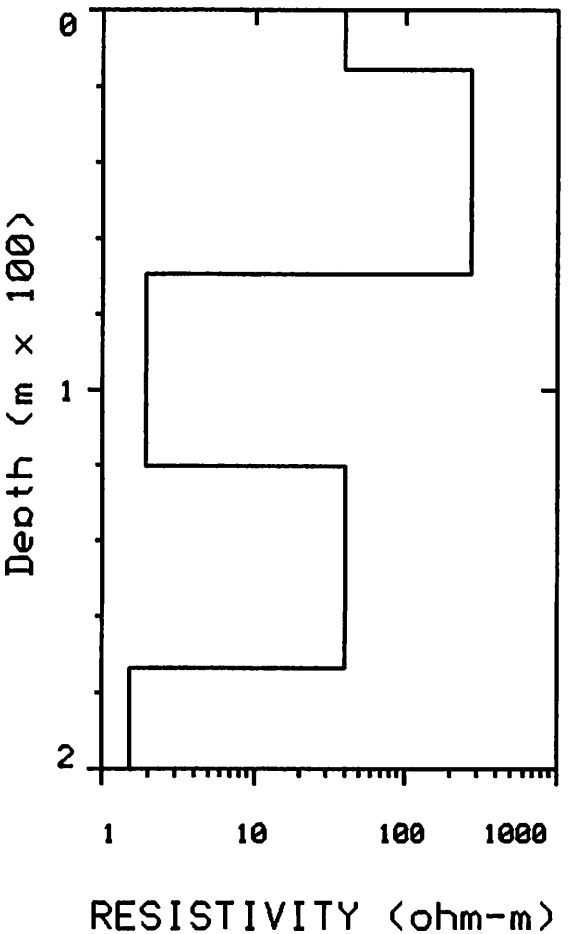
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.728 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.20	
1	6.23	14.63	-10.43	2.34
2	94.73	29.96	-40.39	0.316
3	3.92	24.39	-64.79	6.22
4	22.28	54.75	-119.5	2.45
5	3.64			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.7%
I.P.R.G.	Ashdod South 114330/130891 (+17.4)
Data Set: HS509 / Date: 24-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS511

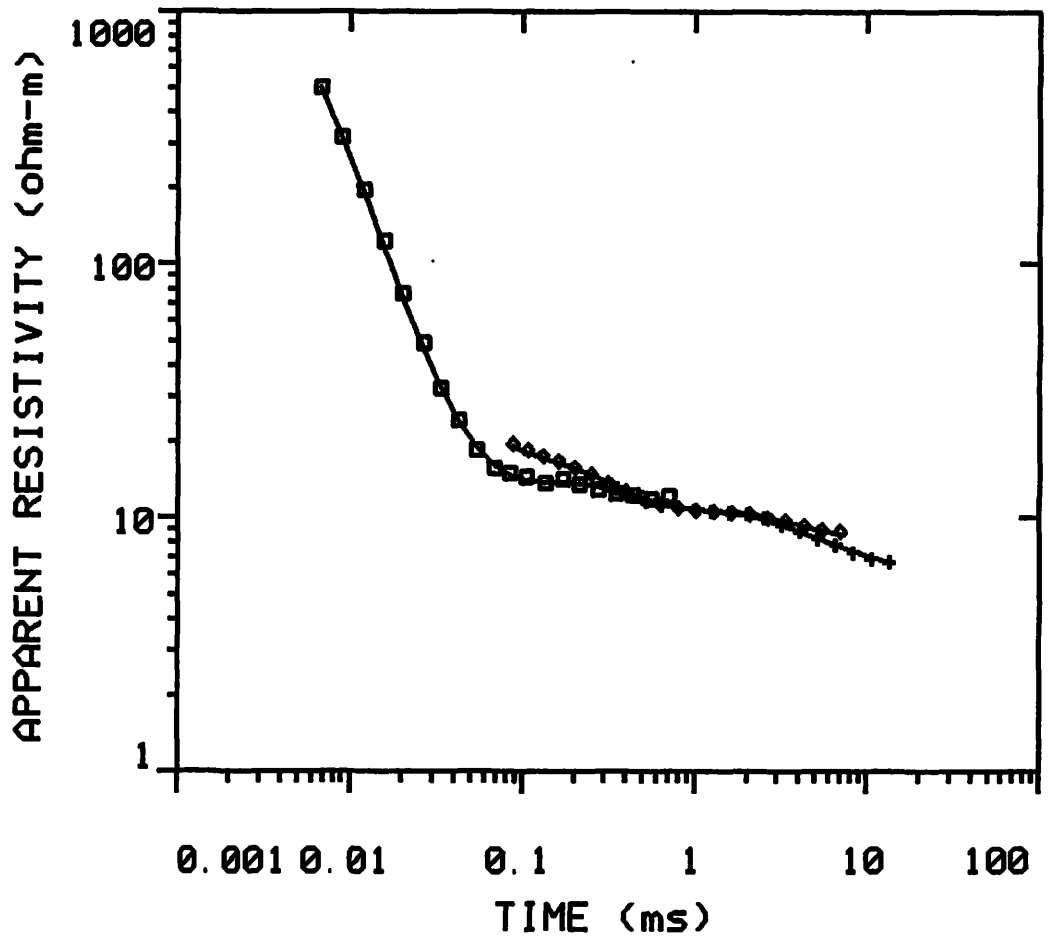
CLIENT: 100x100m DATE: 24-Nov-99
LOCATION: Herzliya SOUNDING: z
COUNTY: 131690/174374 (+18.3) ELEVATION: 18.30 m
PROJECT: FITTING ERROR: 8.4% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 95.0000 SLOPE: NONE

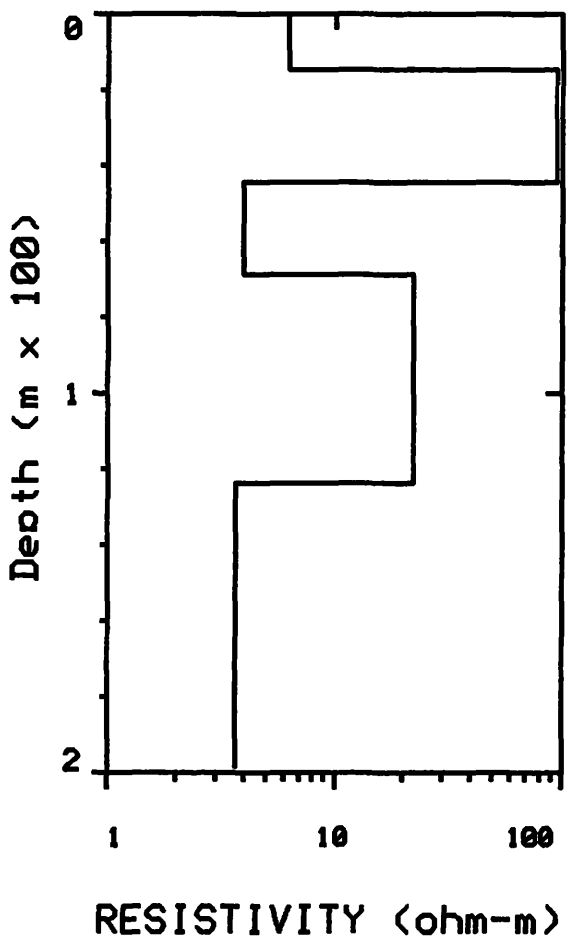
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 8.369 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			18.30	
1	62.70	25.87	-7.57	0.412
2	8.70	20.72	-28.30	2.38
3	0.347	42.07	-70.37	120.9
4	646.1			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.7%
I.P.R.G.	Hof Rishon 124978/156167 (+4.2)
Data Set: HS518 / Date: 24-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS512

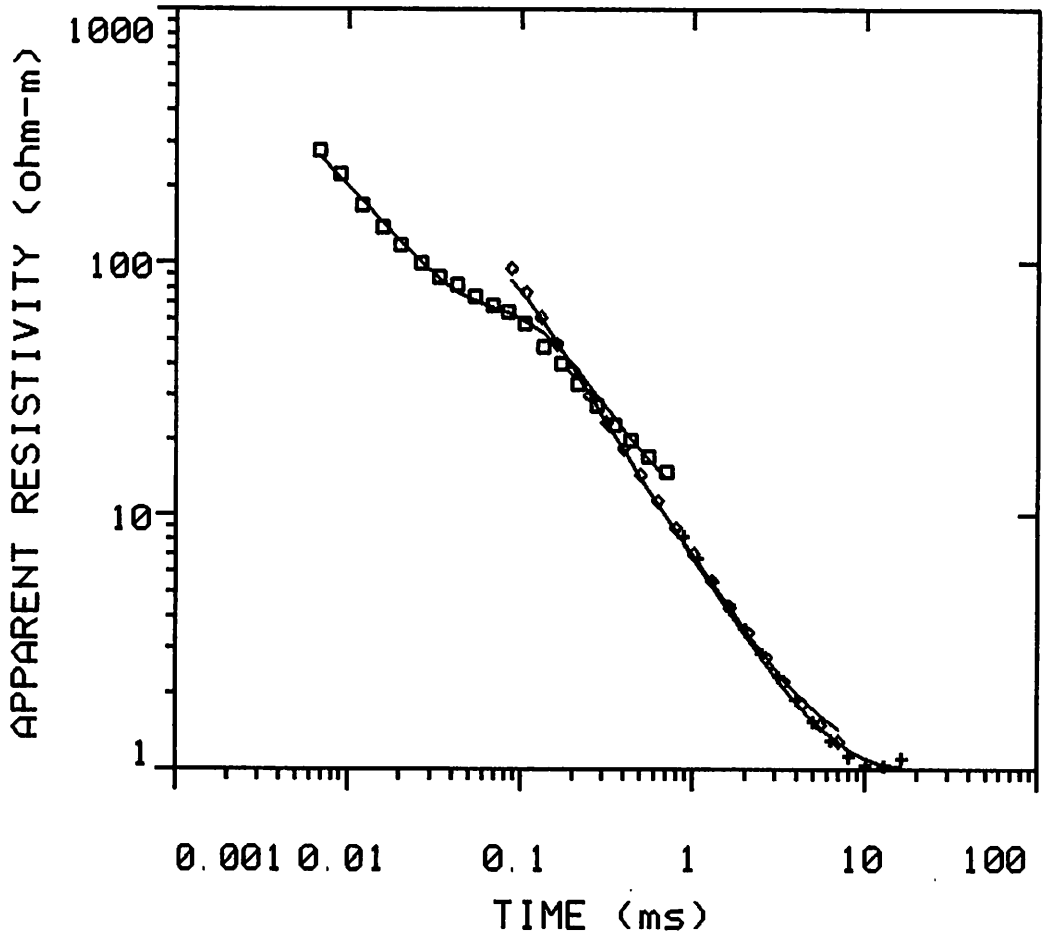
CLIENT: 100x100m DATE: 02-Dec-99
LOCATION: Shefaim SOUNDING: z
COUNTY: 133906/181147 (+28.2) ELEVATION: 28.20 m
PROJECT: FITTING ERROR: 2.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 105.0000 SLOPE: NONE

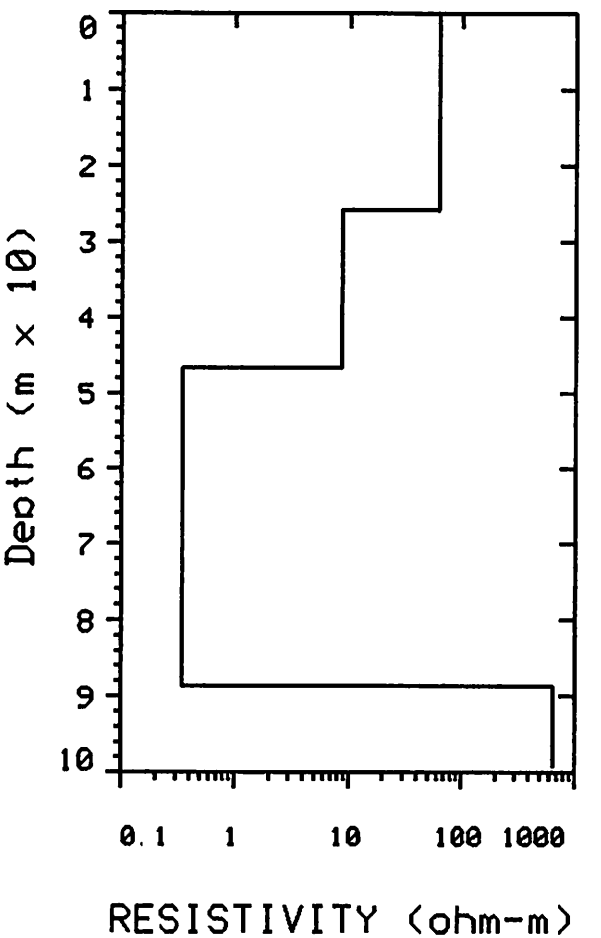
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.269 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			28.20	
1	42.19	72.73	-44.53	1.72
2	5.31	19.32	-63.86	3.63
3	1.57	19.14	-83.00	12.12
4	39.14	72.33	-155.3	1.84
5	1.56			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 8.4%
I.P.R.G.	Herzliya 131690/174374 (+18.3)
Data Set: HSS11 / Date: 24-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS513

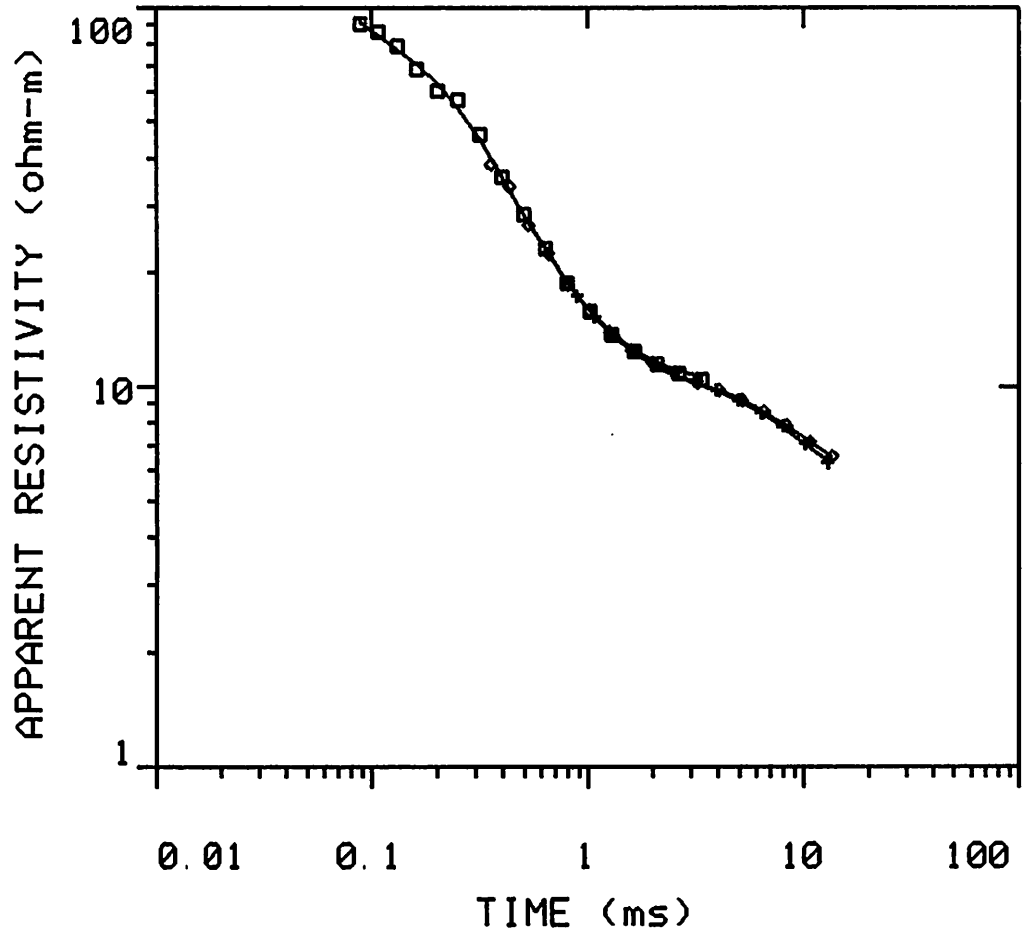
CLIENT: 100x100m DATE: 25-Nov-99
LOCATION: Yaqum SOUNDING: z
COUNTY: 133954/184130 (+41.2) ELEVATION: 41.20 m
PROJECT: FITTING ERROR: 4.9% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 97.0000 SLOPE: NONE

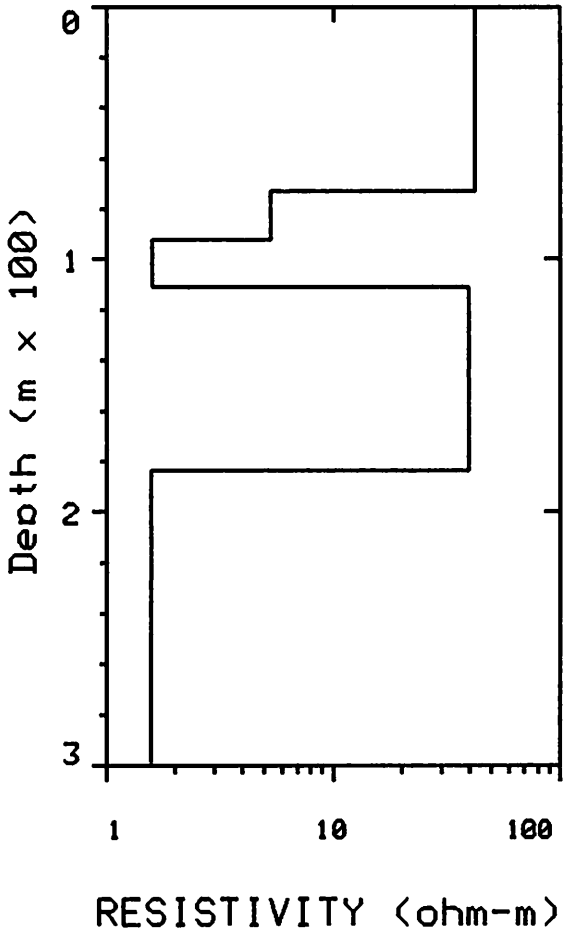
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.902 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			41.20	
1	89.23	50.15	-8.95	0.561
2	16.49	23.09	-32.04	1.39
3	1.92	21.58	-53.62	11.24
4	1.13	36.44	-90.06	32.15
5	5.27	48.28	-138.3	9.15
6	1.96			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.3%
I.P.R.G.	Shefaim 133906/181147 (+28.2)
Data Set: HS512 / Date: 02-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS514

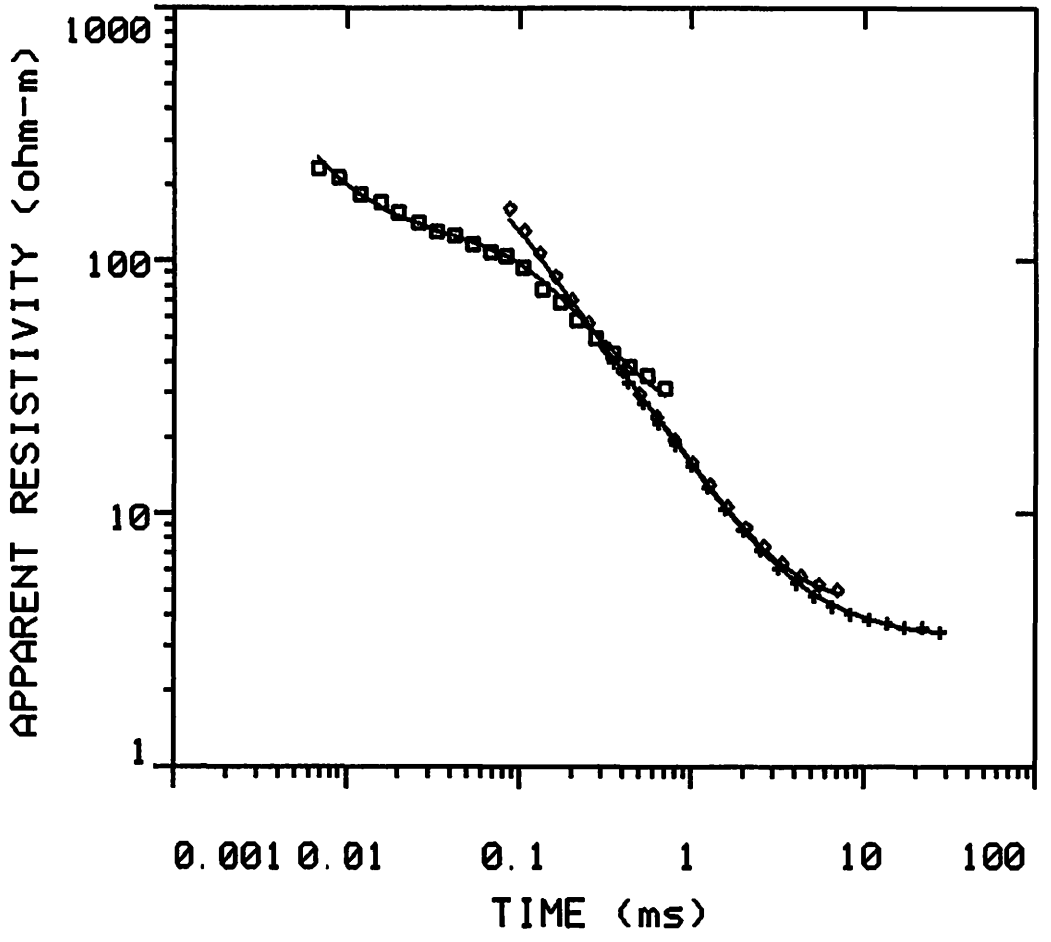
CLIENT: 100x100m DATE: 25-Nov-99
LOCATION: Yaqum SOUNDING: z
COUNTY: 134582/184231 (+16.5) ELEVATION: 16.50 m
PROJECT: FITTING ERROR: 6.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 98.0000 SLOPE: NONE

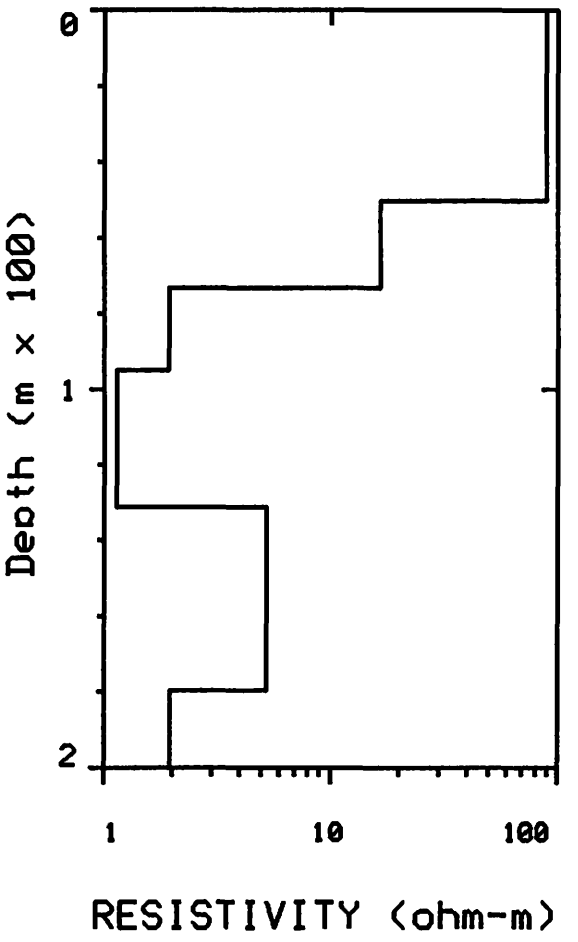
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 6.189 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			16.50	
1	7.08	7.37	9.12	1.04
2	114.7	40.64	-31.52	0.354
3	12.00	18.83	-50.35	1.56
4	1.49	24.03	-74.39	16.12
5	11.36	78.60	-152.9	6.91
6	2.13			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 4.9%
I.P.R.G.	Yaqum 133954/184130 (+41.2)
Data Set: HS513 / Date: 25-Nov-99	
Equipment: Geonics PROTEM / Sounding: Z	Azimuth:

DATA SET: HS515

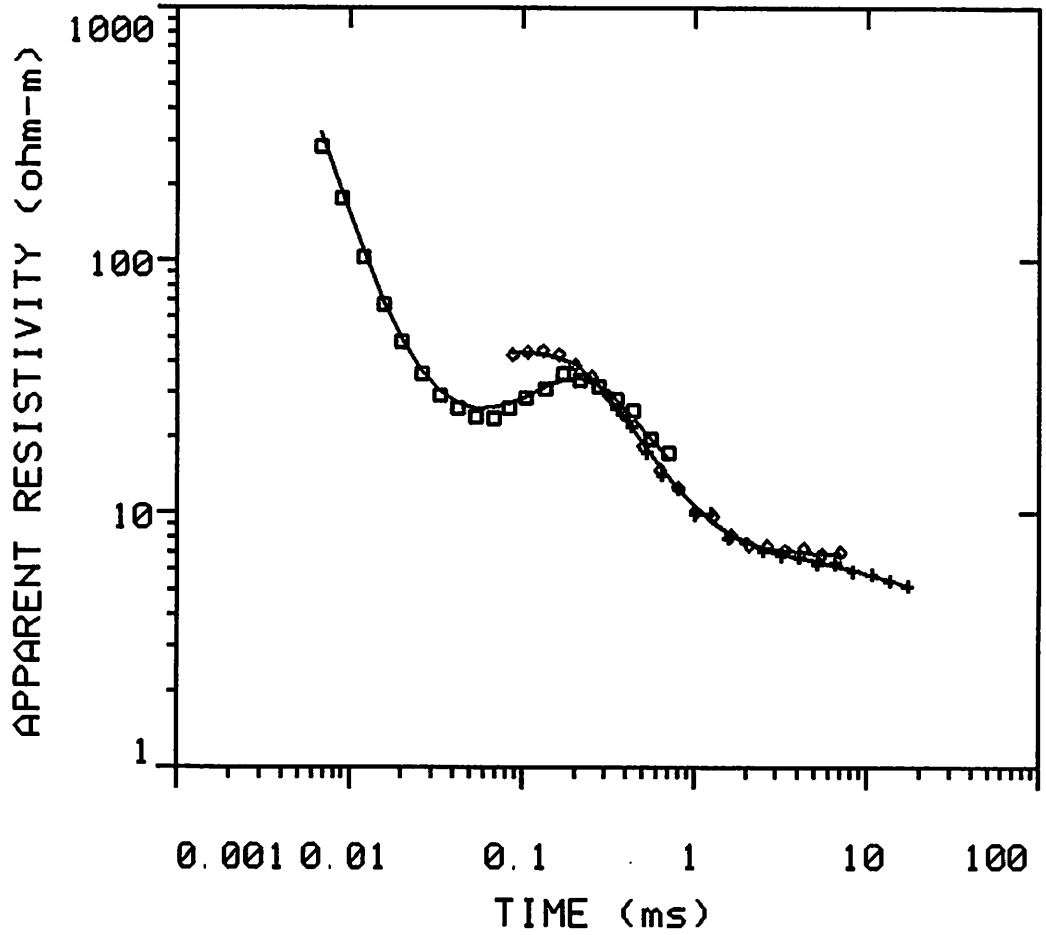
CLIENT: 100x100m DATE: 01-Dec-99
LOCATION: Poleg SOUNDING: z
COUNTY: 135212/186986 (+14.2) ELEVATION: 14.20 m
PROJECT: FITTING ERROR: 1.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 99.0000 SLOPE: NONE

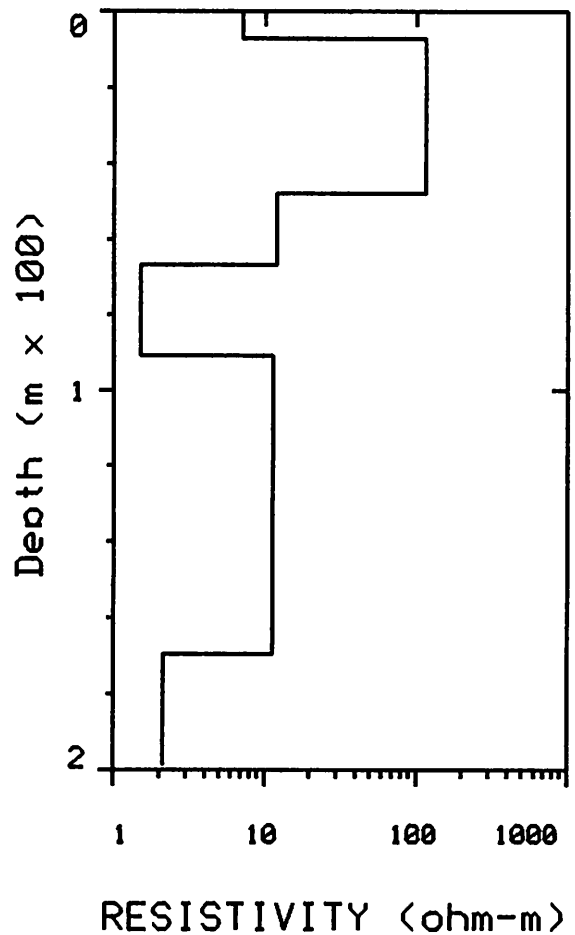
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.072 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			14.20	
1	311.4	48.23	-34.03	0.154
2	10.01	21.03	-55.07	2.10
3	2.54	21.99	-77.06	8.63
4	11.36	68.23	-145.3	6.00
5	1.95			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 6.2%
I.P.R.G.	Yaqum 134582/184231 (+16.5)
Data Set: HS514 / Date: 25-Nov-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS516

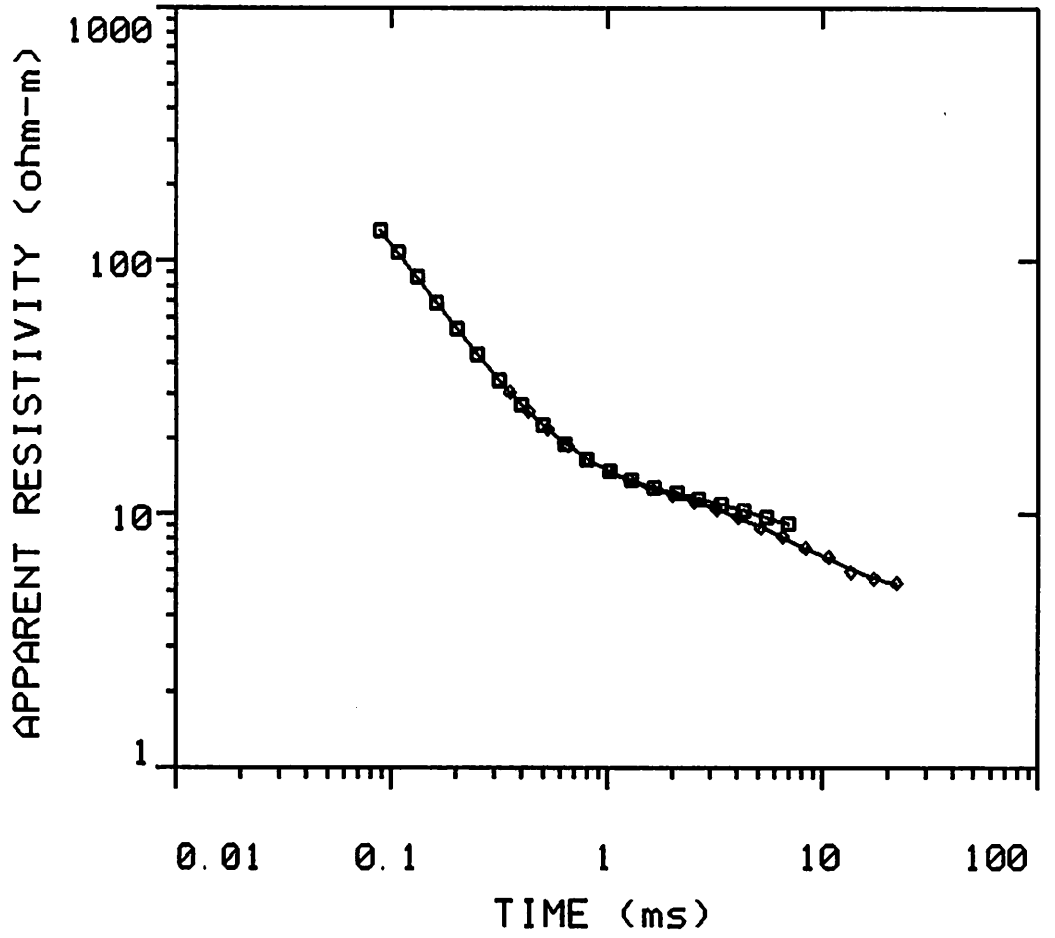
CLIENT: 100x100m DATE: 01-Dec-99
LOCATION: Hofit SOUNDING: z
COUNTY: 138625/199356 (+10.2) ELEVATION: 10.20 m
PROJECT: FITTING ERROR: 4.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 100.0000 SLOPE: NONE

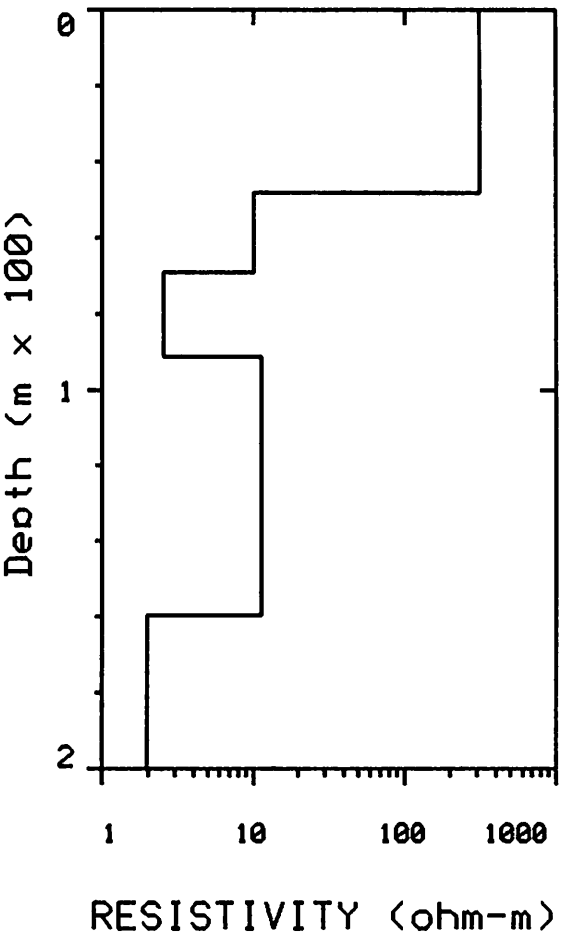
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.188 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			10.20	
1	95.04	9.98	0.211	0.105
2	29.07	46.38	-46.16	1.59
3	3.18	100.1	-146.2	31.47
4	1.05			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 1.1%
I.P.R.G.	Poleg 135212/186986 (+14.2)
Data Set: HS515 / Date: 01-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS517

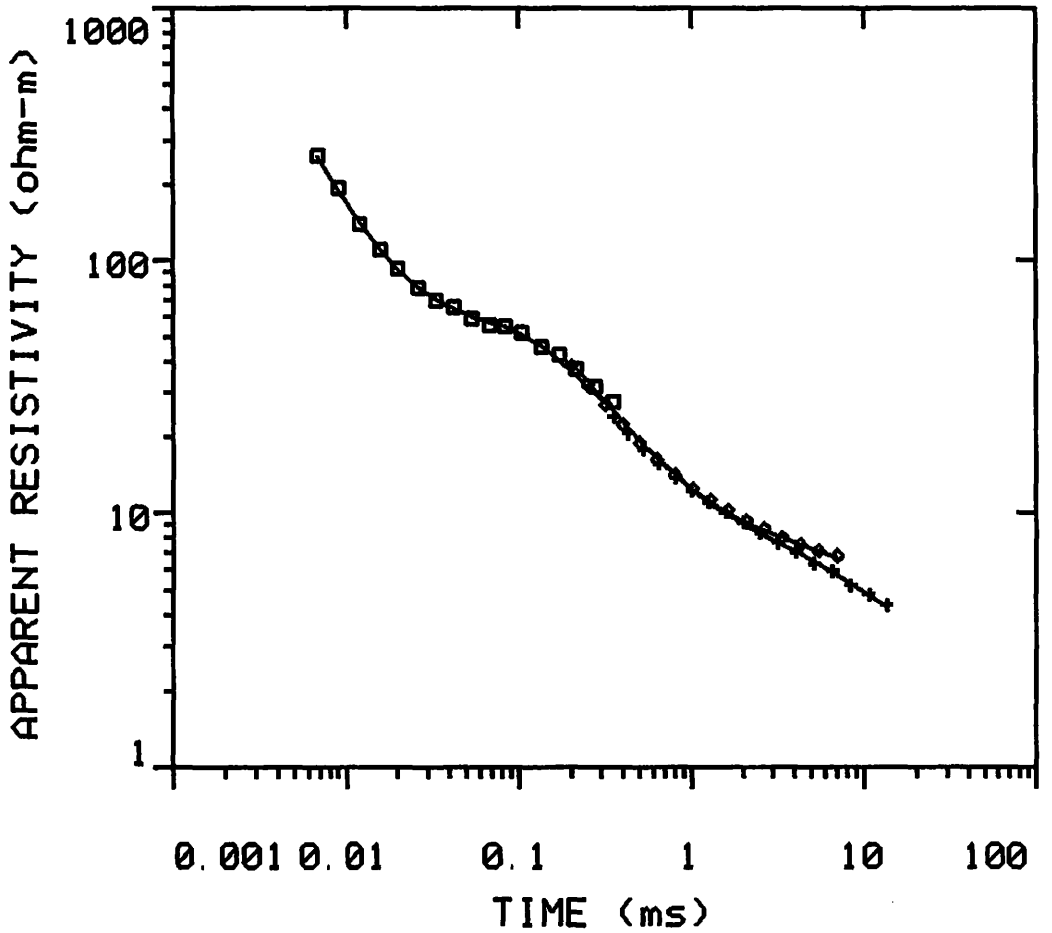
CLIENT: 100x100m DATE: 01-Dec-99
LOCATION: Mikhmoret East SOUNDING: z
COUNTY: 139479/202392 (+21.0) ELEVATION: 21.00 m
PROJECT: FITTING ERROR: 4.4% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 101.0000 SLOPE: NONE

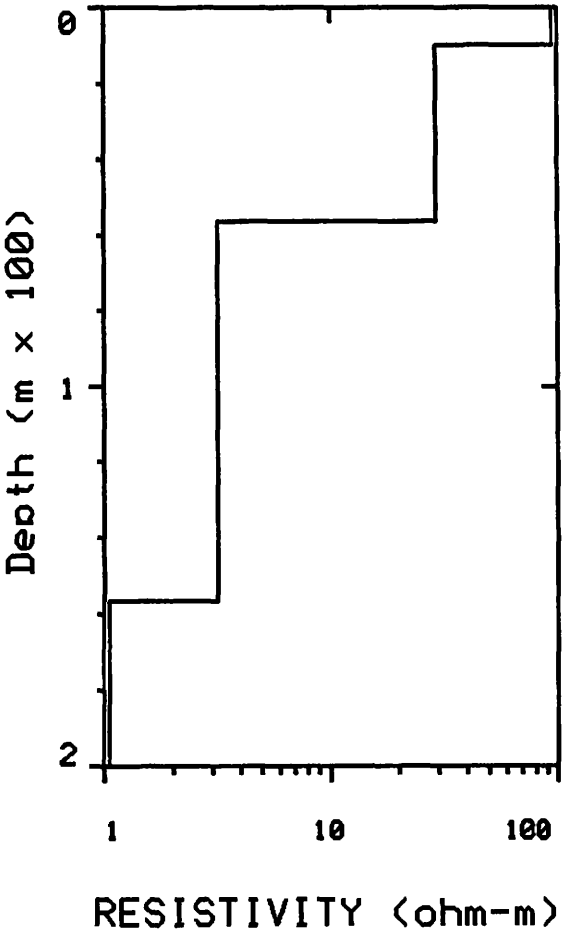
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.364 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			21.00	
1	283.3	15.18	5.81	0.0535
2	35.00	52.74	-46.93	1.50
3	2.86	13.33	-60.27	4.65
4	90.84	62.16	-122.4	0.684
5	1.76			

ALL PARAMETERS ARE FREE

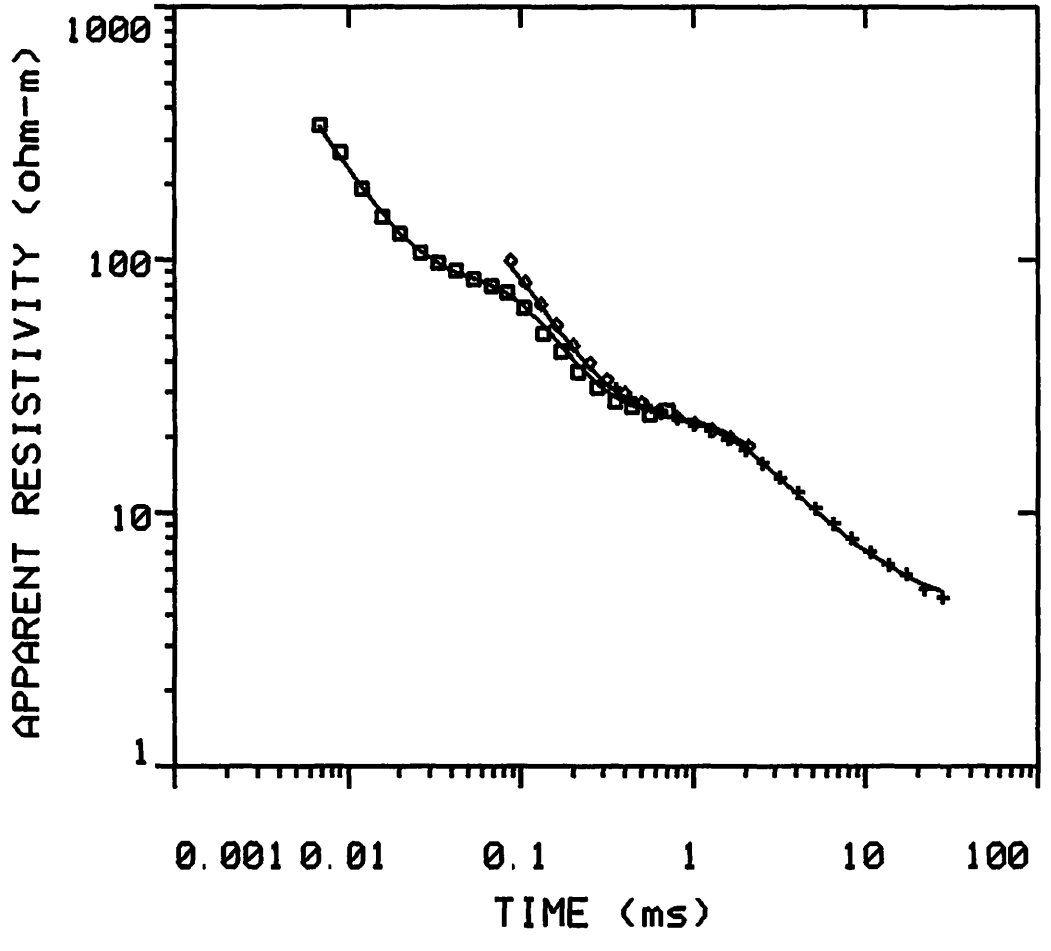


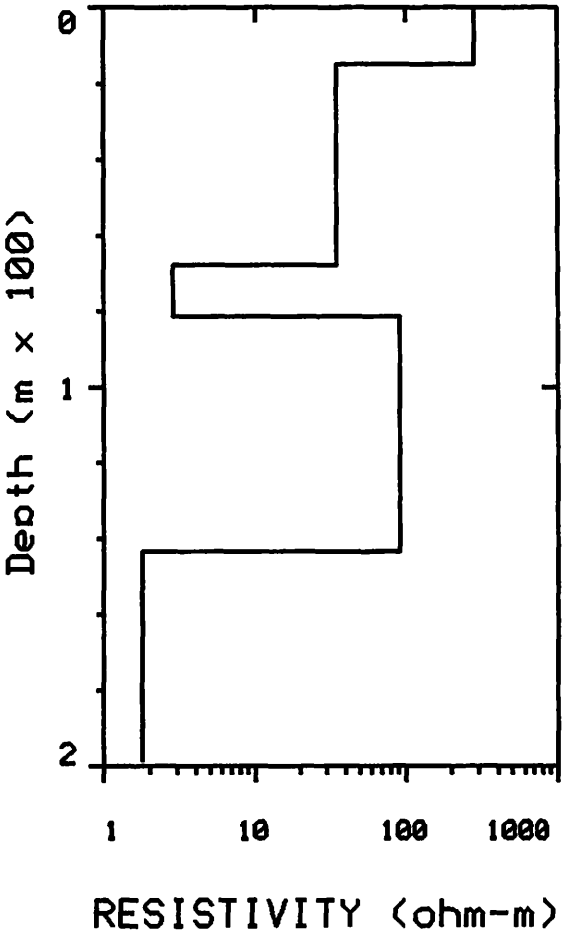


100x100m	FITTING ERROR: 4.2%
I.P.R.G.	Hofit 138625/199356 (+10.2)
Data Set: HS516 / Date: 01-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

HS518

Not interpreted





100x100m	FITTING ERROR: 4.4%
I.P.R.G.	Mikhmoret East 139479/202392 (+21.0)
Data Set: HSS17 / Date: 01-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS519

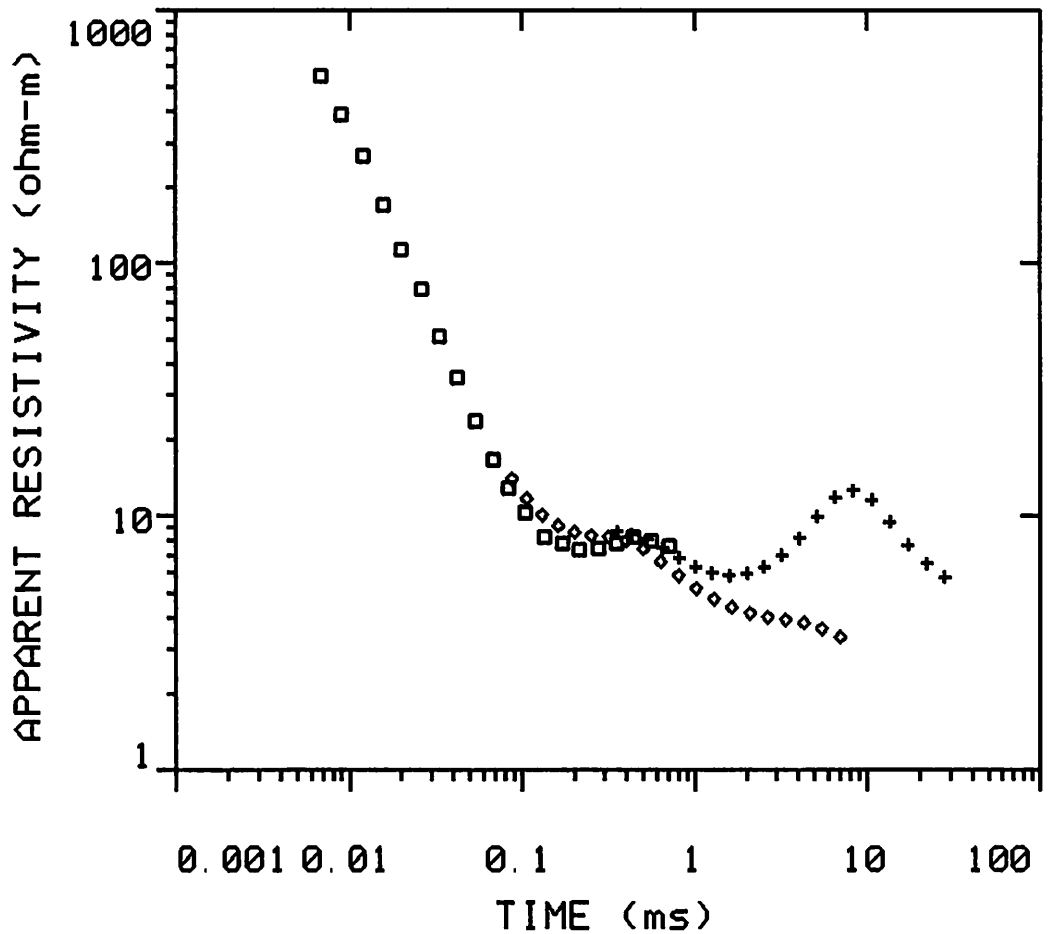
CLIENT: 100x100m DATE: 01-Dec-99
LOCATION: Giv'at Ol'ga SOUNDING: z
COUNTY: 139295/205860 (+10.6) ELEVATION: 10.60 m
PROJECT: FITTING ERROR: 2.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 103.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.772 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			10.60	
1	17.13	1.27	9.32	0.0744
2	39.09	37.49	-28.17	0.959
3	4.56	15.04	-43.21	3.29
4	1.68	36.50	-79.71	21.60
5	18.40	40.66	-120.3	2.20
6	1.13			

ALL PARAMETERS ARE FREE



100x100m	Not interpreted
I.P.R.G.	Hadera waste site 139756/203934 (+12.1)
Data Set: HS518	Date: 01-Dec-99
Equipment: Geonics PROTEM	Sounding: z
	Azimuth:

DATA SET: HS520

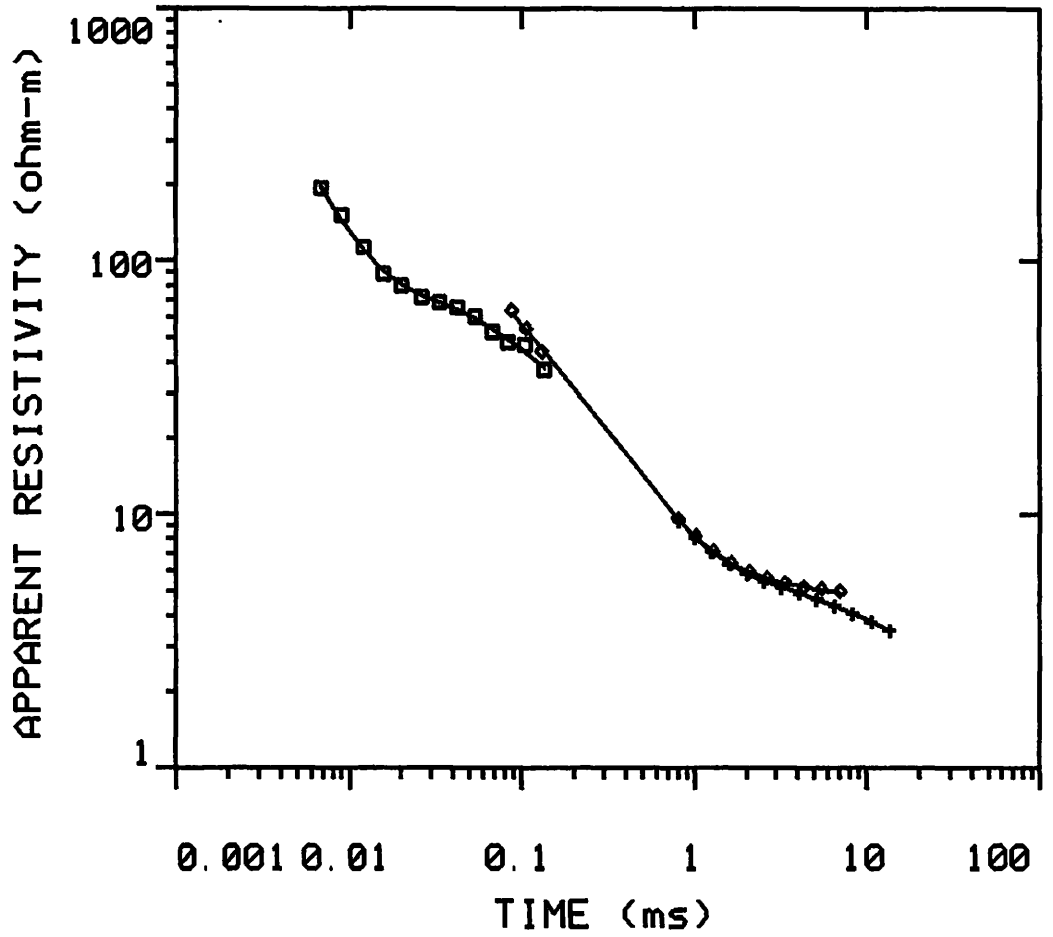
CLIENT: 100x100m DATE: 01-Dec-99
LOCATION: Giv'at Ol'ga SOUNDING: z
COUNTY: 139929/205986 (+23.8) ELEVATION: 10.60 m
PROJECT: FITTING ERROR: 3.6% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 104.0000 SLOPE: NONE

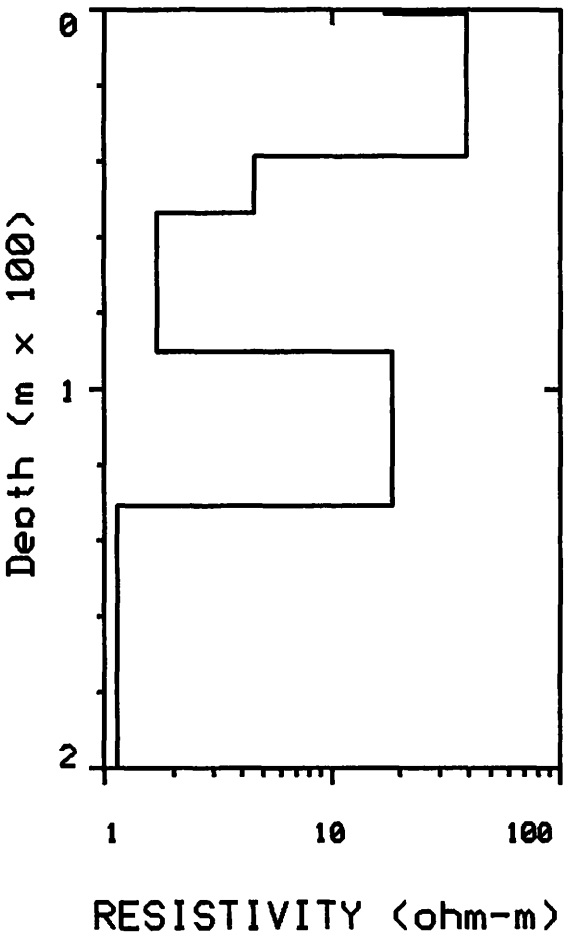
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.620 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			10.60	
1	96.59	22.51	-11.91	0.233
2	40.14	41.29	-53.21	1.02
3	1.19	6.96	-60.18	5.80
4	7.69	81.63	-141.8	10.61
5	0.749			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.7%
I.P.R.G.	Giv'at Ol'ga 139295/205860 (+10.6)
Data Set: HS519 / Date: 01-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS521

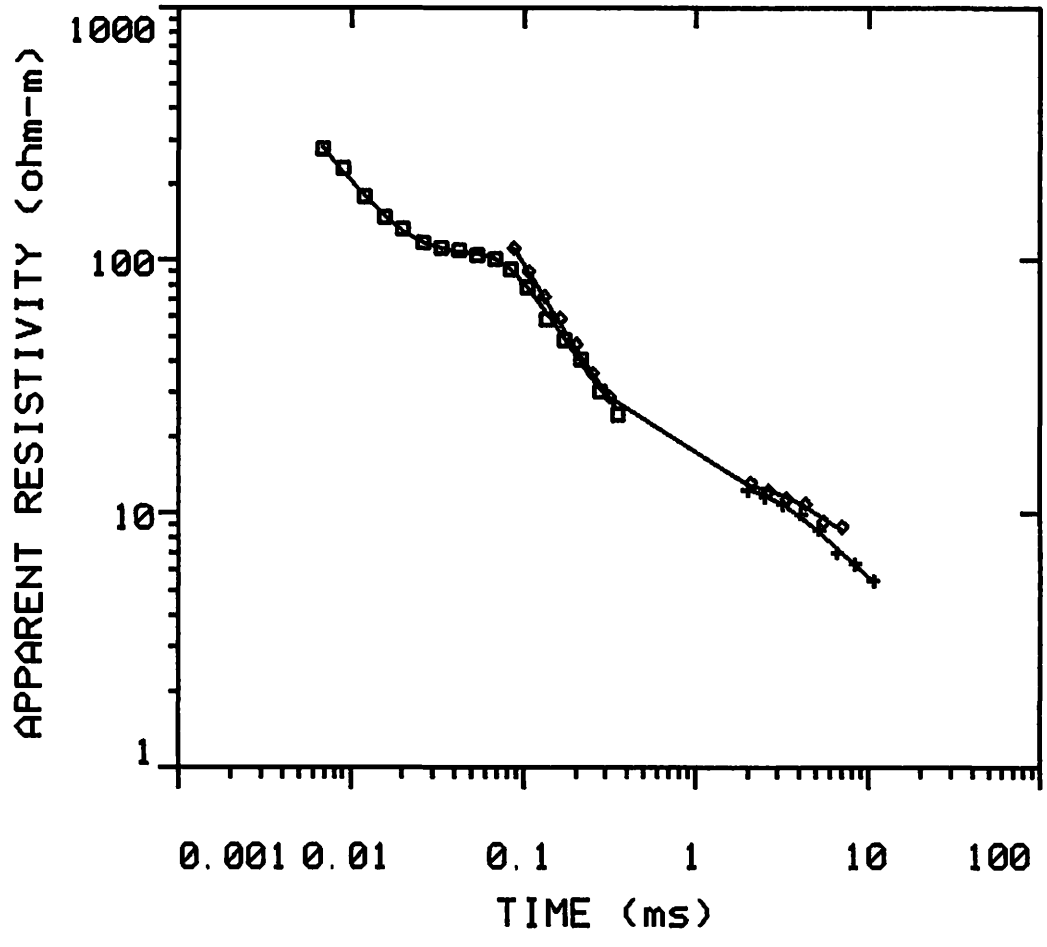
CLIENT: 50x50m DATE: 02-Dec-99
LOCATION: Sdot Yam SOUNDING: z
COUNTY: 139750/209851 (+2.1) ELEVATION: 2.10 m
PROJECT: FITTING ERROR: 2.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 106.0000 SLOPE: NONE

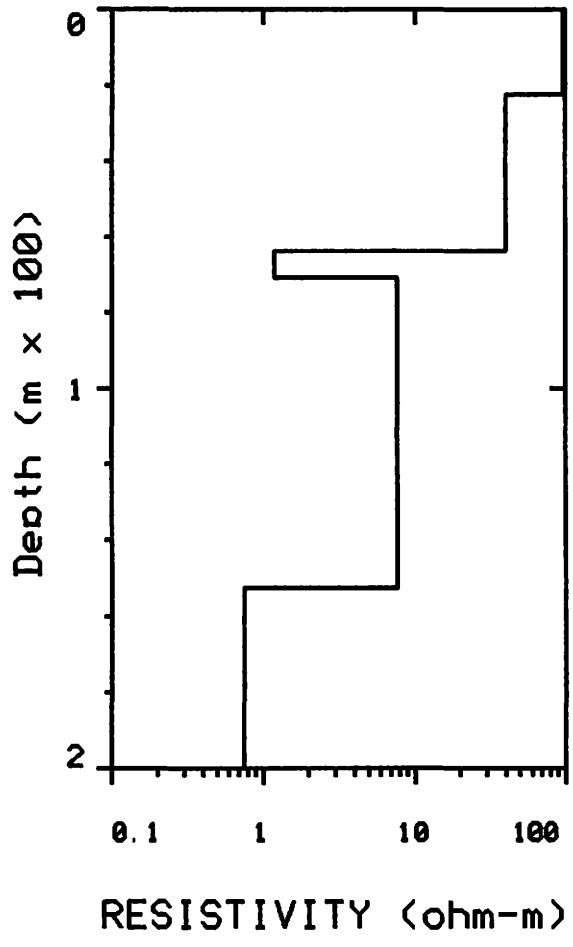
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.739 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			2.10	
1	5.69	3.71	-1.61	0.652
2	42.99	5.76	-7.38	0.134
3	6.07	8.48	-15.87	1.39
4	32.48	13.44	-29.31	0.413
5	1.91	22.38	-51.70	11.70
6	1.36	52.70	-104.4	38.58
7	2.10			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.6%
I.P.R.G.	Giv'at Ol'ga 139929/205986 (+23.8)
Data Set: HS520 / Date: 01-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS522

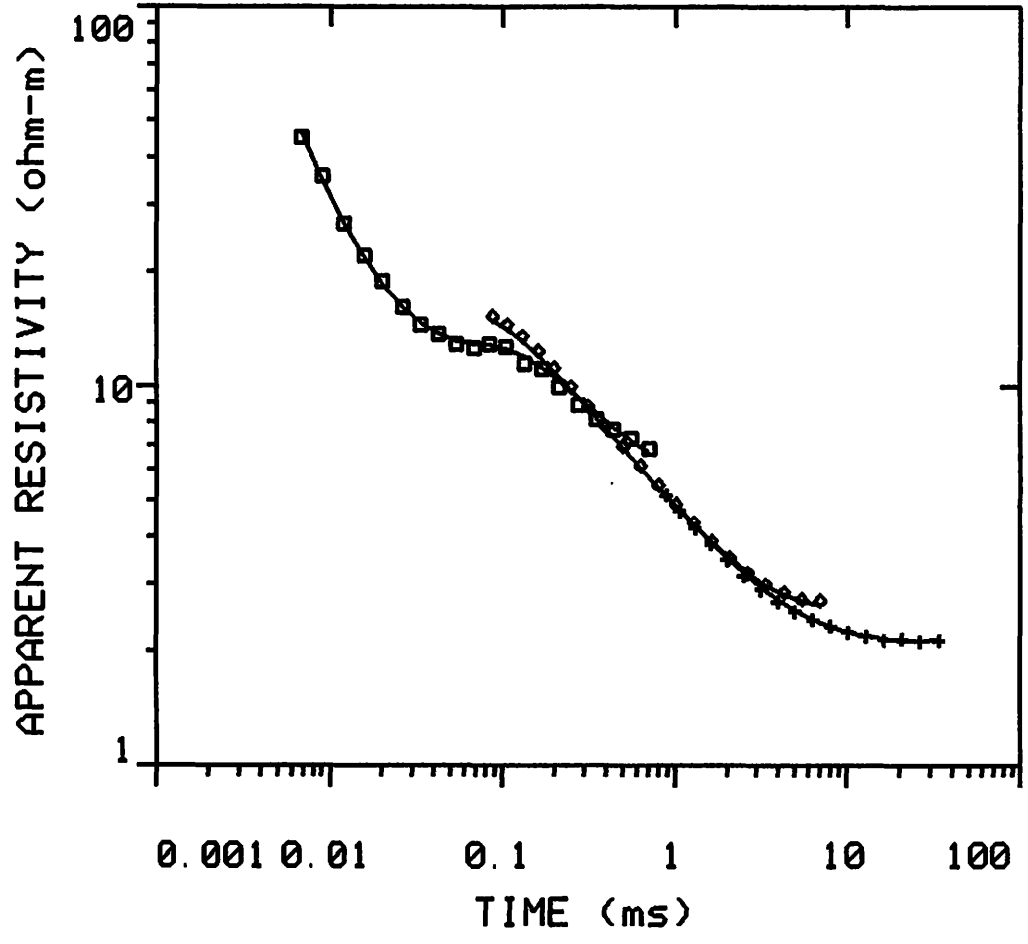
CLIENT: 100x100m DATE: 02-Dec-99
LOCATION: Caesarea SOUNDING: z
COUNTY: 140271/210199 (+5.5) ELEVATION: 5.50 m
PROJECT: FITTING ERROR: 3.9% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 107.0000 SLOPE: NONE

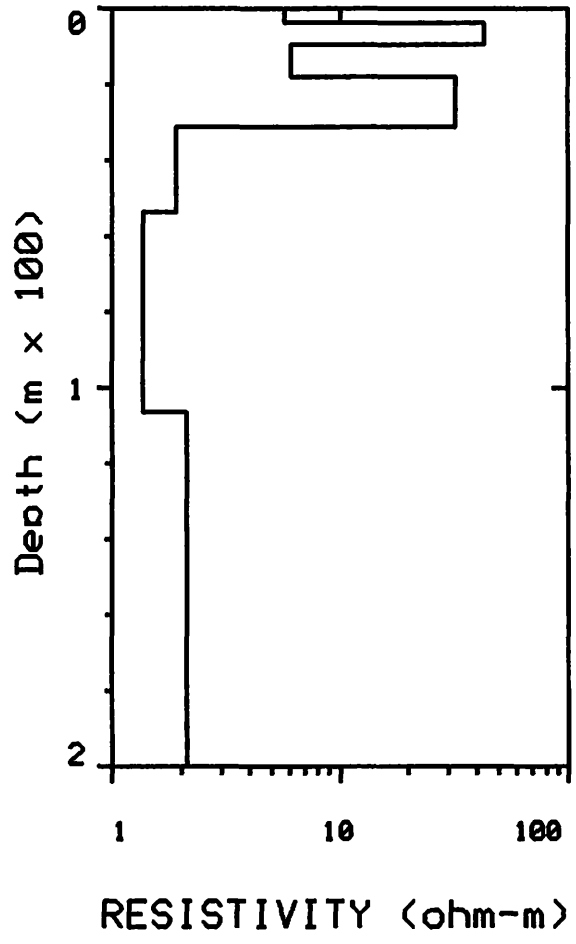
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.869 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			5.50	
1	55.24	42.68	-37.18	0.772
2	25.80	14.01	-51.20	0.543
3	3.12			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 2.7%
I.P.R.G.	Sdot Yam 139750/209851 (+2.1)
Data Set: HS521 / Date: 02-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS523

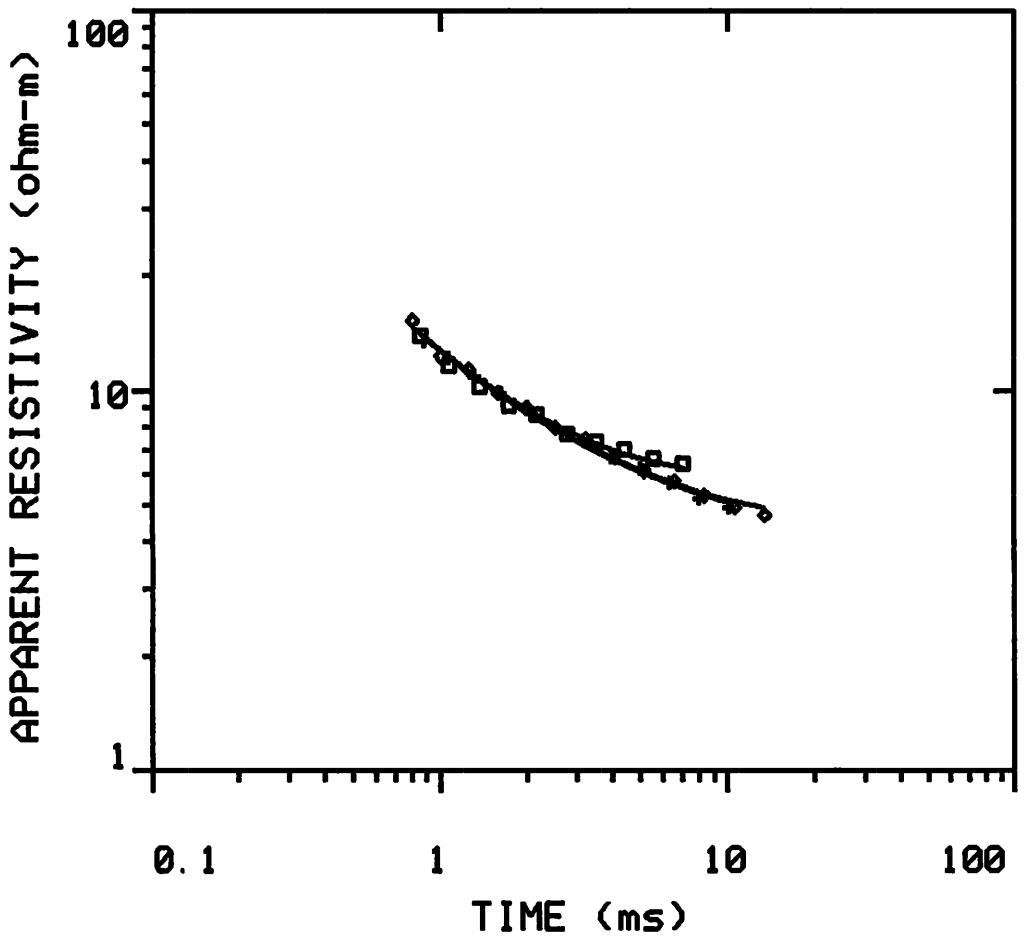
CLIENT: 100x100m DATE: 02-Dec-99
LOCATION: Caesarea SOUNDING: z
COUNTY: 140933/213671 (+5.7) ELEVATION: 5.70 m
PROJECT: FITTING ERROR: 3.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 108.0000 SLOPE: NONE

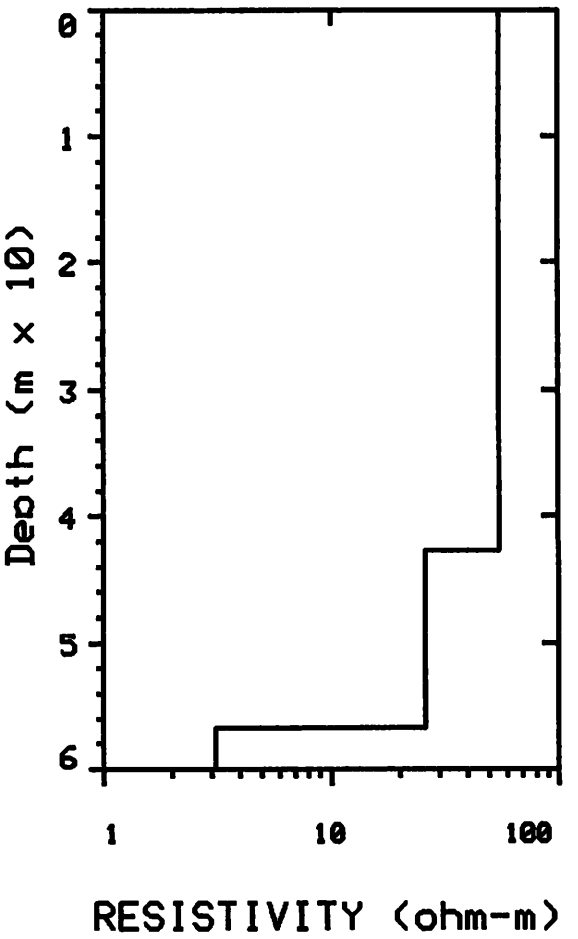
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.092 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			5.70	
1	22.21	33.40	-27.70	1.50
2	6.69	125.8	-153.5	18.78
3	2.26			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.9%
I.P.R.G.	Caesarea 140271/210199 (+5.5)
Data Set: HS522 / Date: 02-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS524

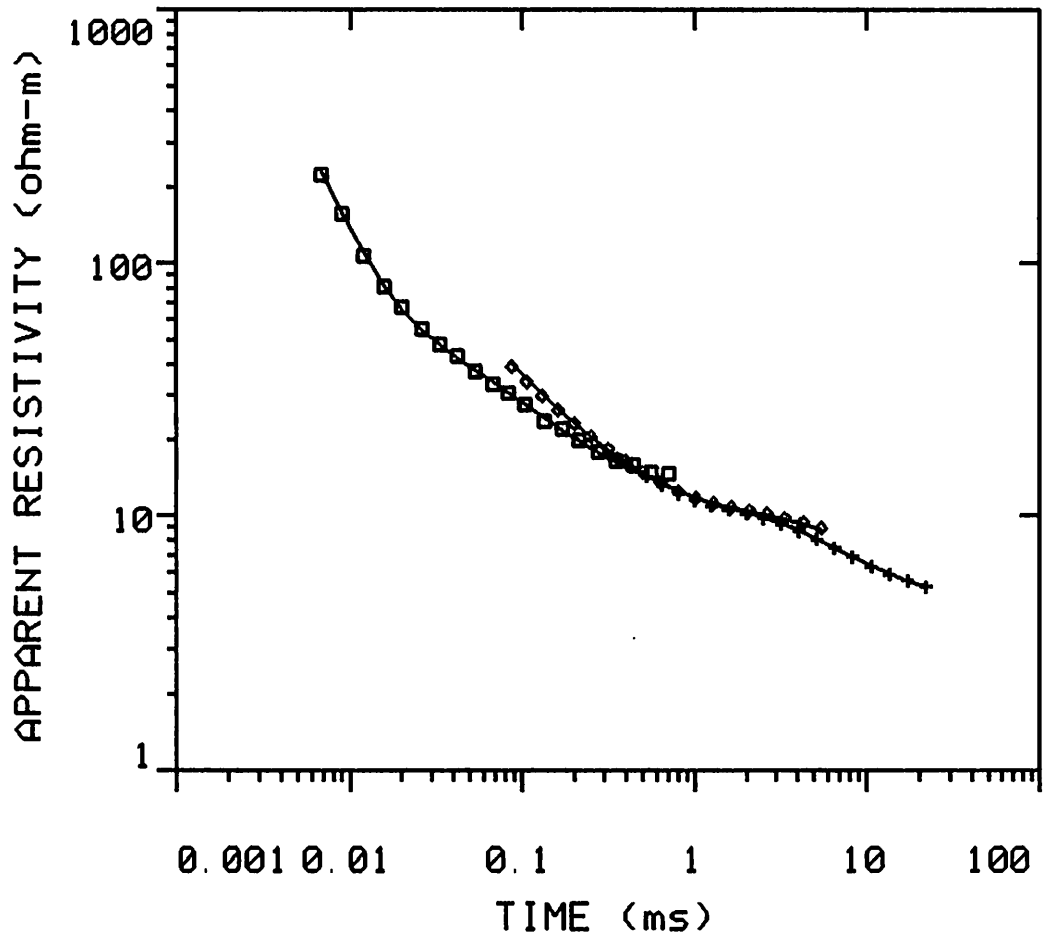
CLIENT: 50x50m
DATE: 02-Dec-99
LOCATION: Caesarea
SOUNDING: z
COUNTY: 140632/213363 (+4.4)
ELEVATION: 4.40 m
PROJECT: FITTING ERROR: 2.9%
EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m
AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y)
TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 109.0000
SLOPE: NONE

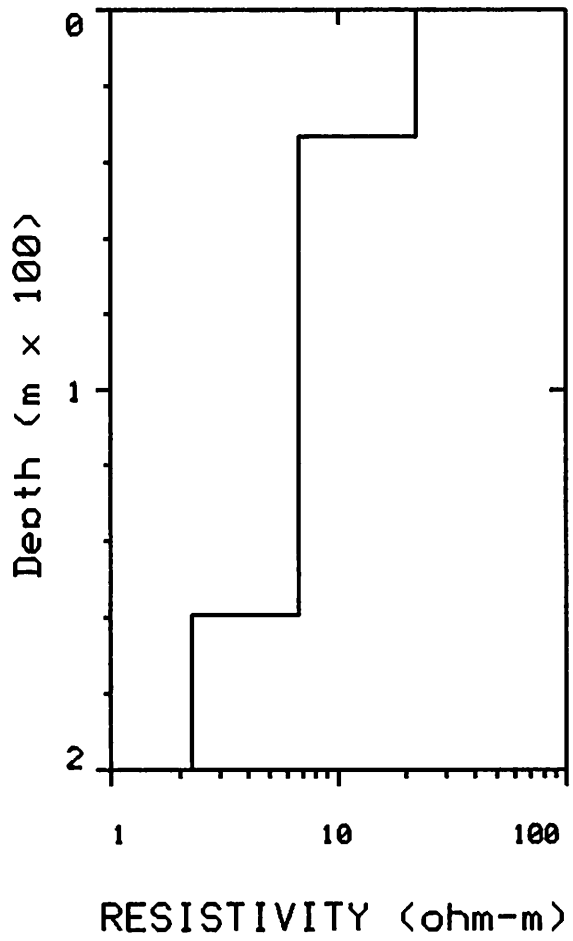
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.930 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.40	
1	7.64	3.01	1.38	0.394
2	24.60	46.43	-45.04	1.88
3	2.95			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.1%
I.P.R.G.	Caesarea 140933/213671 (+5.7)
Data Set: HS523 / Date: 02-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS525

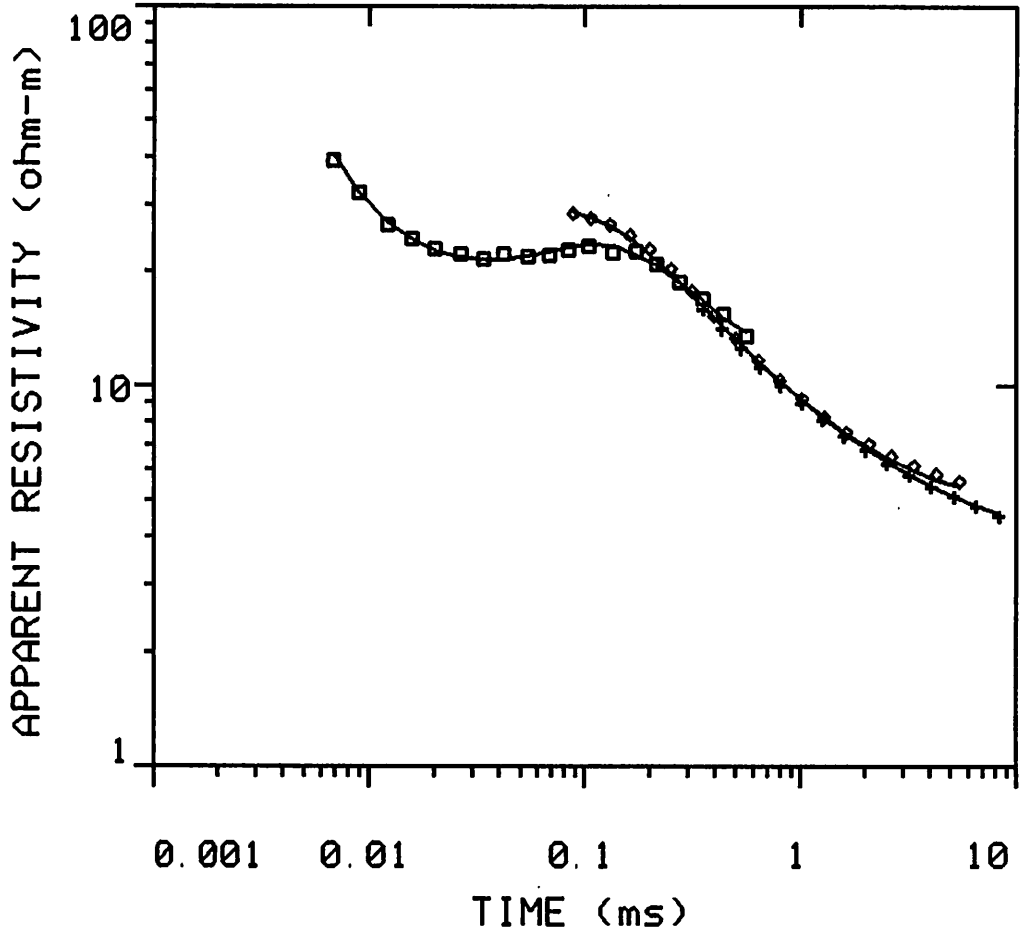
CLIENT: 50x50m DATE: 06-Dec-99
LOCATION: Hof Nachsholim SOUNDING: z
COUNTY: 142888/225303 (+3.7) ELEVATION: 3.70 m
PROJECT: FITTING ERROR: 4.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 110.0000 SLOPE: NONE

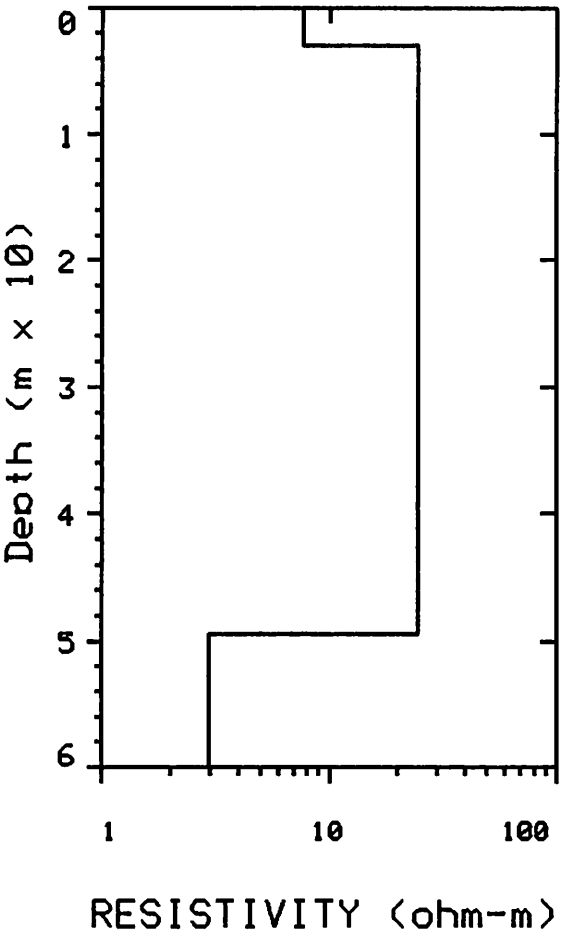
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.091 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			3.70	
1	11.70	3.92	-0.224	0.335
2	20.43	14.57	-14.80	0.713
3	2.33	8.11	-22.91	3.47
4	18.47	44.66	-67.58	2.41
5	12.55	72.84	-140.4	5.80
6	20.00			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 2.9%
I.P.R.G.	Caesarea 140632/213363 (+4.4)
Data Set: HS524 / Date: 02-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS526

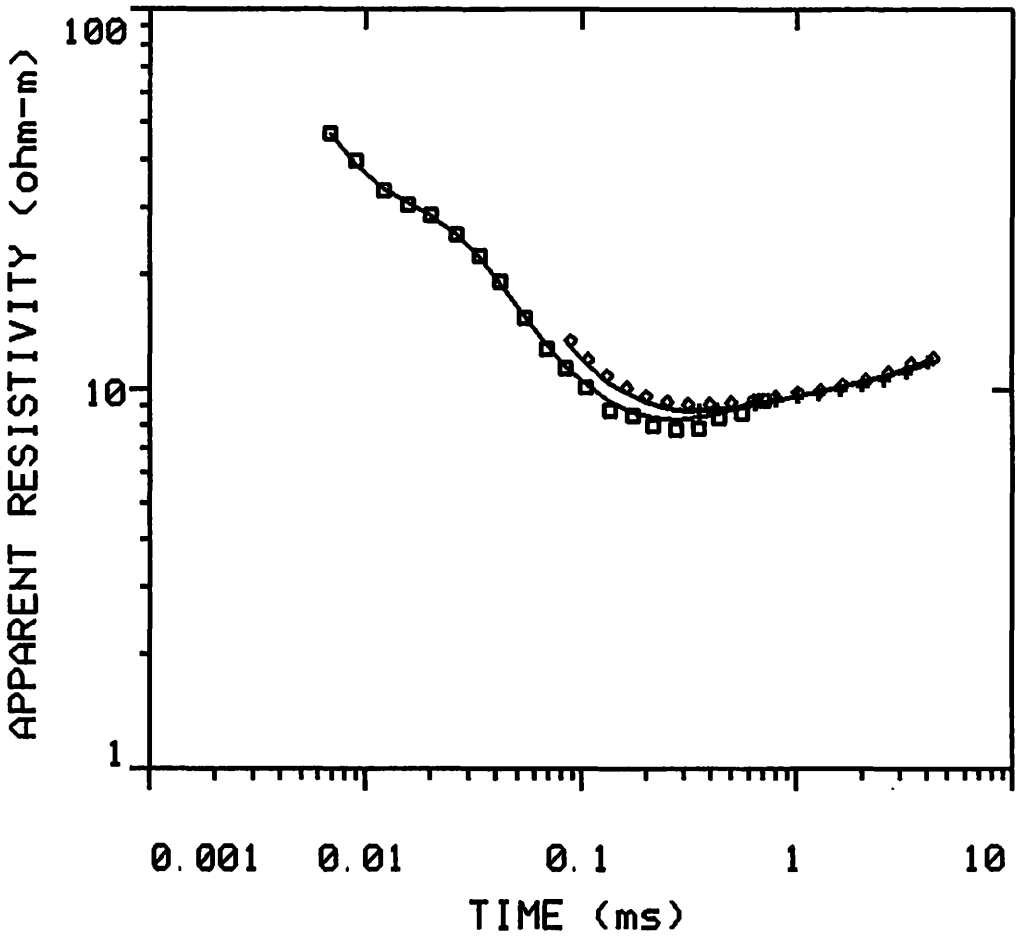
CLIENT: 100x100m DATE: 06-Dec-99
LOCATION: Newe-Yam SOUNDING: z
COUNTY: 143540/229343 (+4.8) ELEVATION: 4.80 m
PROJECT: FITTING ERROR: 1.5% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 111.0000 SLOPE: NONE

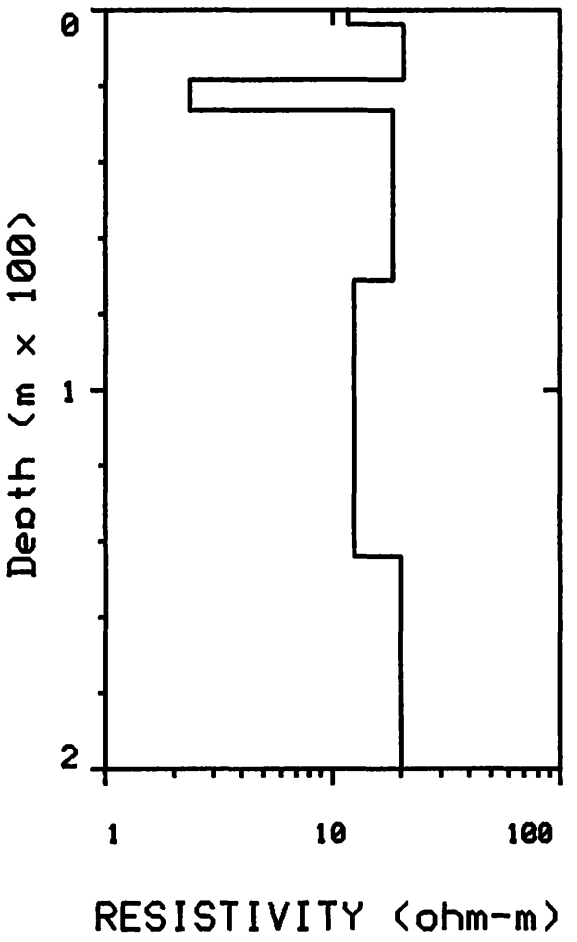
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.542 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.80	
1	4.25	25.10	-20.30	5.90
2	2.76	24.54	-44.85	8.87
3	1.51	64.54	-109.4	42.61
4	10.16			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 4.1%
I.P.R.G.	Hof Nachsholim 142888/225303 (+3.7)
Data Set: HS525 / Date: 06-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS527

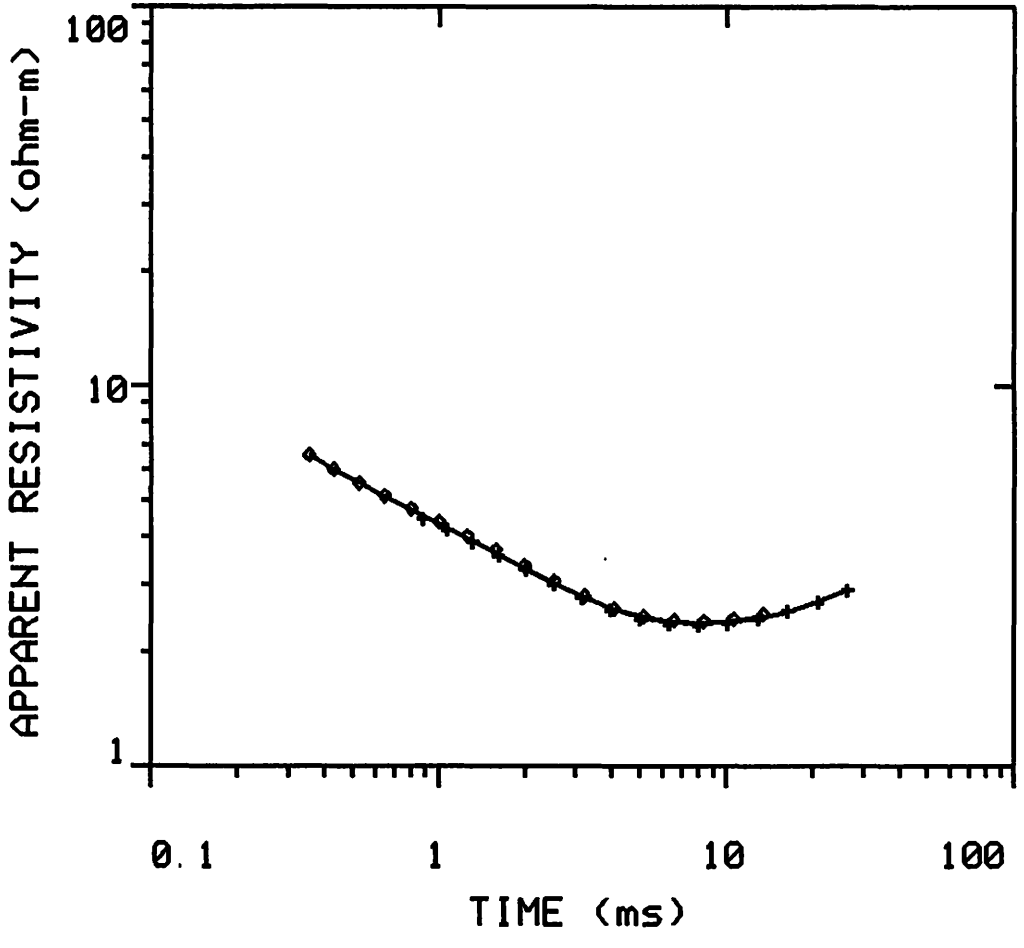
CLIENT: 100x100m DATE: 06-Dec-99
LOCATION: Newe-Yam SOUNDING: z
COUNTY: 143990/229260 (+5.9) ELEVATION: 5.90 m
PROJECT: FITTING ERROR: 3.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 112.0000 SLOPE: NONE

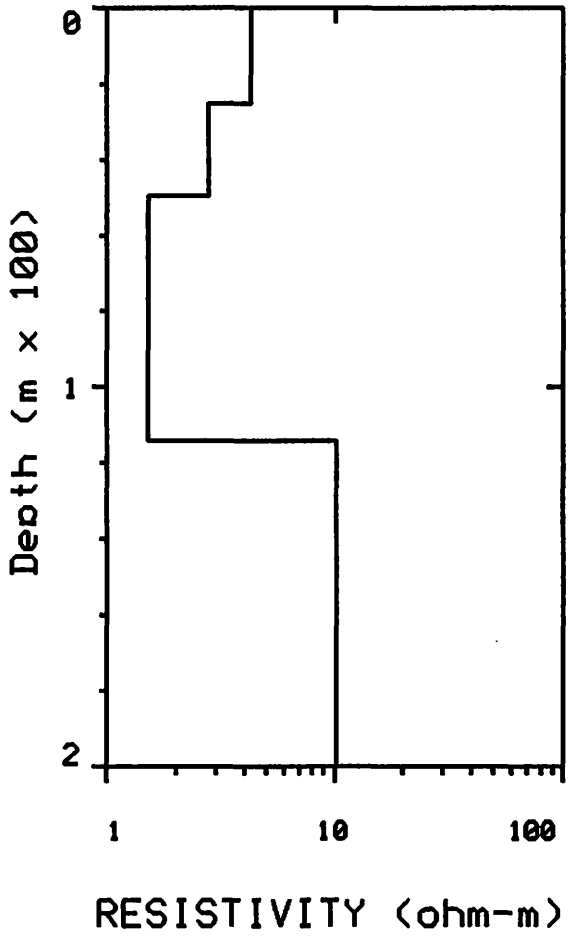
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.103 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			5.90	
1	13.11	1.35	4.54	0.103
2	14.96	22.92	-18.37	1.53
3	3.54	41.90	-60.28	11.81
4	5.90	115.5	-175.8	19.57
5	18.88			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 1.5%
I.P.R.G.	Newe-Yam 143540/229343 (+4.8)
Data Set: HS526 / Date: 06-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS528

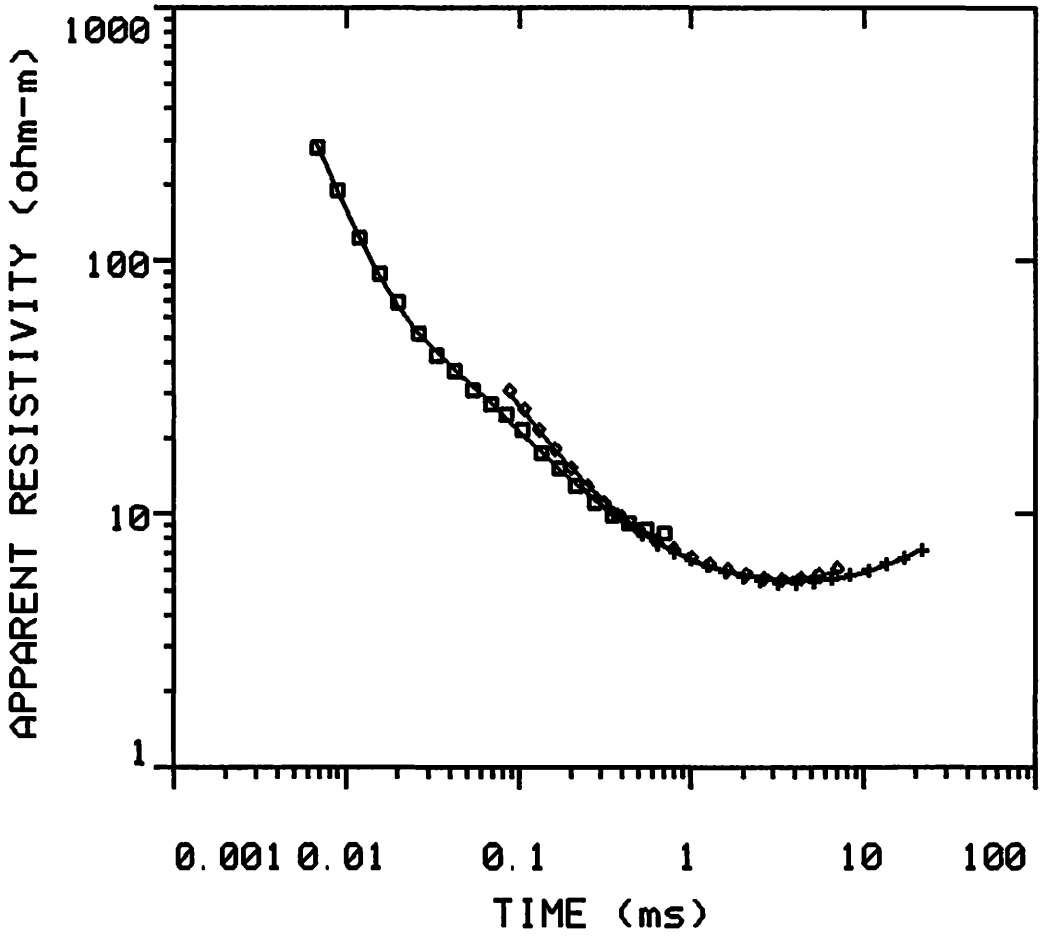
CLIENT: 50x50m DATE: 06-Dec-99
LOCATION: Newe-Yam SOUNDING: z
COUNTY: 143707/231164 (+4.3) ELEVATION: 4.30 m
PROJECT: FITTING ERROR: 4.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 113.0000 SLOPE: NONE

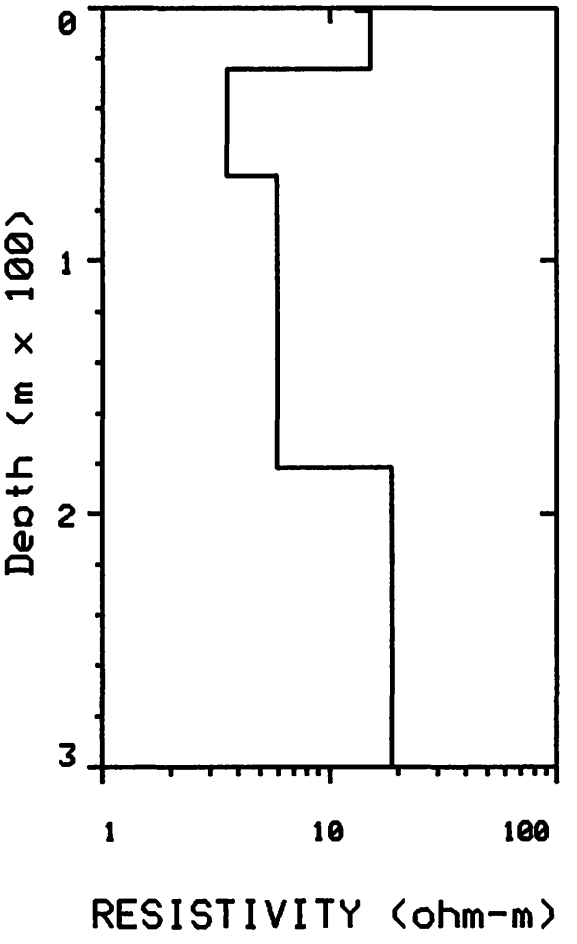
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.134 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.30	
1	3.02	4.55	-0.251	1.50
2	105.4	18.20	-18.45	0.172
3	3.30	38.86	-57.31	11.76
4	17.21	63.58	-120.9	3.69
5	6.67			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.1%
I.P.R.G.	Newe-Yam 143990/229260 (+5.9)
Data Set: HS527 / Date: 06-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS529

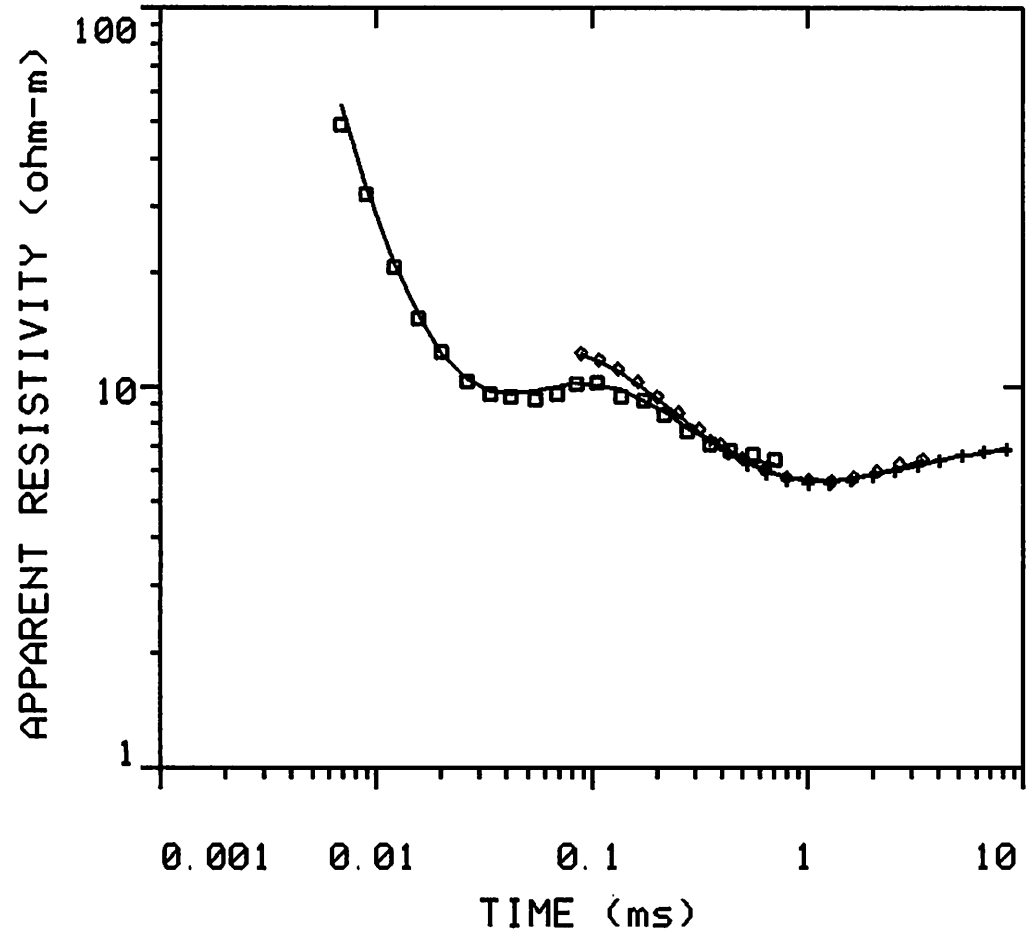
CLIENT: 50x50m DATE: 07-Dec-99
LOCATION: Atlit SOUNDING: z
COUNTY: 145185/236104 (+4.4) ELEVATION: 4.40 m
PROJECT: FITTING ERROR: 2.9% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 115.0000 SLOPE: NONE

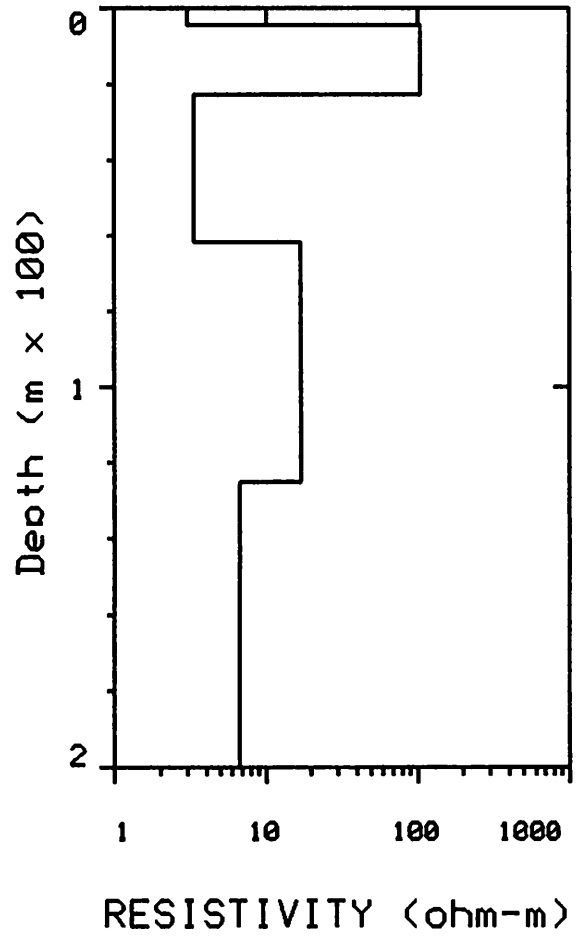
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.862 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.40	
1	8.62	5.52	-1.12	0.640
2	100.1	8.97	-10.09	0.0896
3	5.26	36.16	-46.26	6.86
4	7.59	33.88	-80.15	4.46
5	5.01			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 4.1%
I.P.R.G.	Newe-Yam 143707/231164 (+4.3)
Data Set: HS528 / Date: 06-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS530

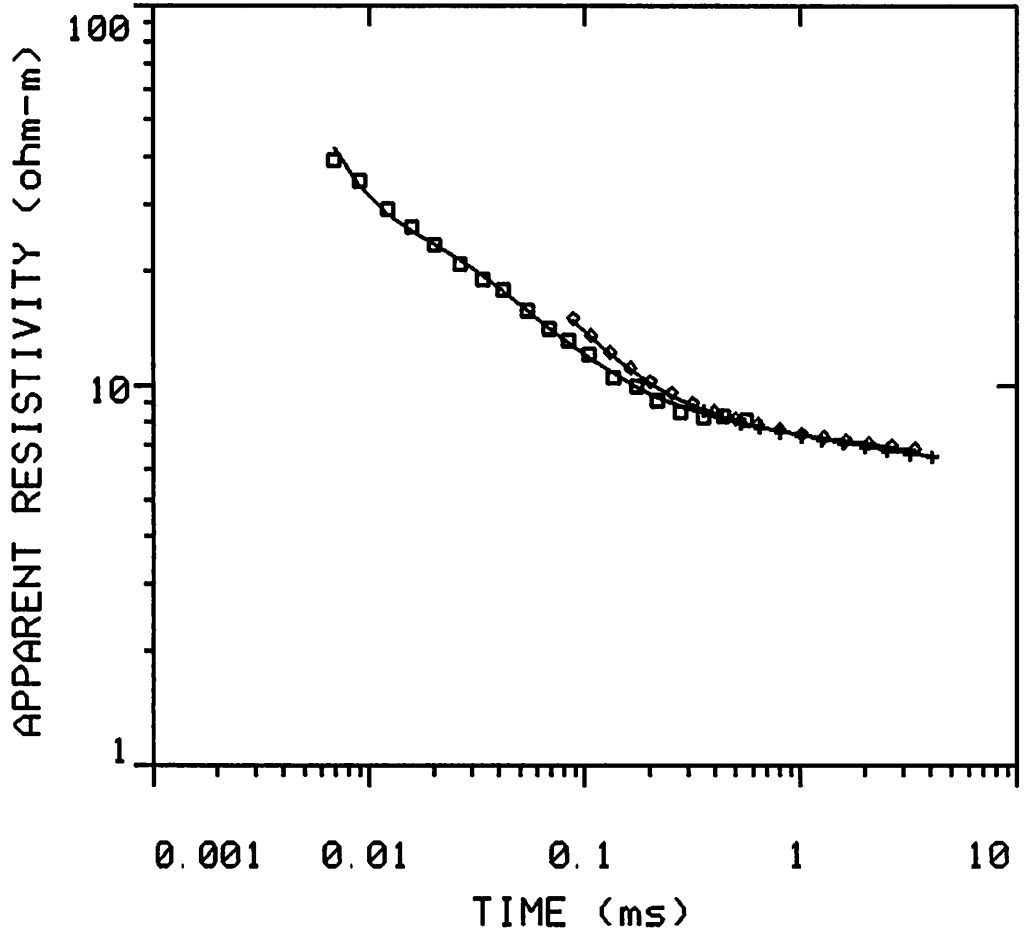
CLIENT: 100x100m
DATE: 07-Dec-99
LOCATION: Kfar Galim
SOUNDING: z
COUNTY: 146051/240800 (+15.1)
ELEVATION: 15.10 m
PROJECT: FITTING ERROR: 4.6%
EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m
AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y)
TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 116.0000
SLOPE: NONE

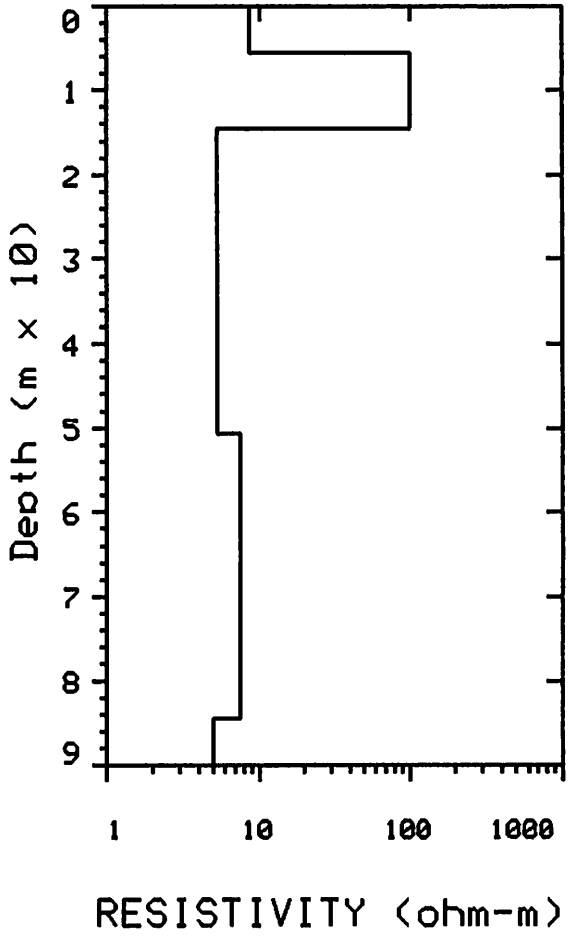
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.574 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			15.10	
1	4.37	13.34	1.75	3.04
2	36.21	26.73	-24.97	0.738
3	6.68	11.14	-36.11	1.66
4	2.33	54.98	-91.10	23.50
5	15.57			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 2.9%
I.P.R.G.	Atlit 145185/236104 (+4.4)
Data Set: HS529 / Date: 07-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS531

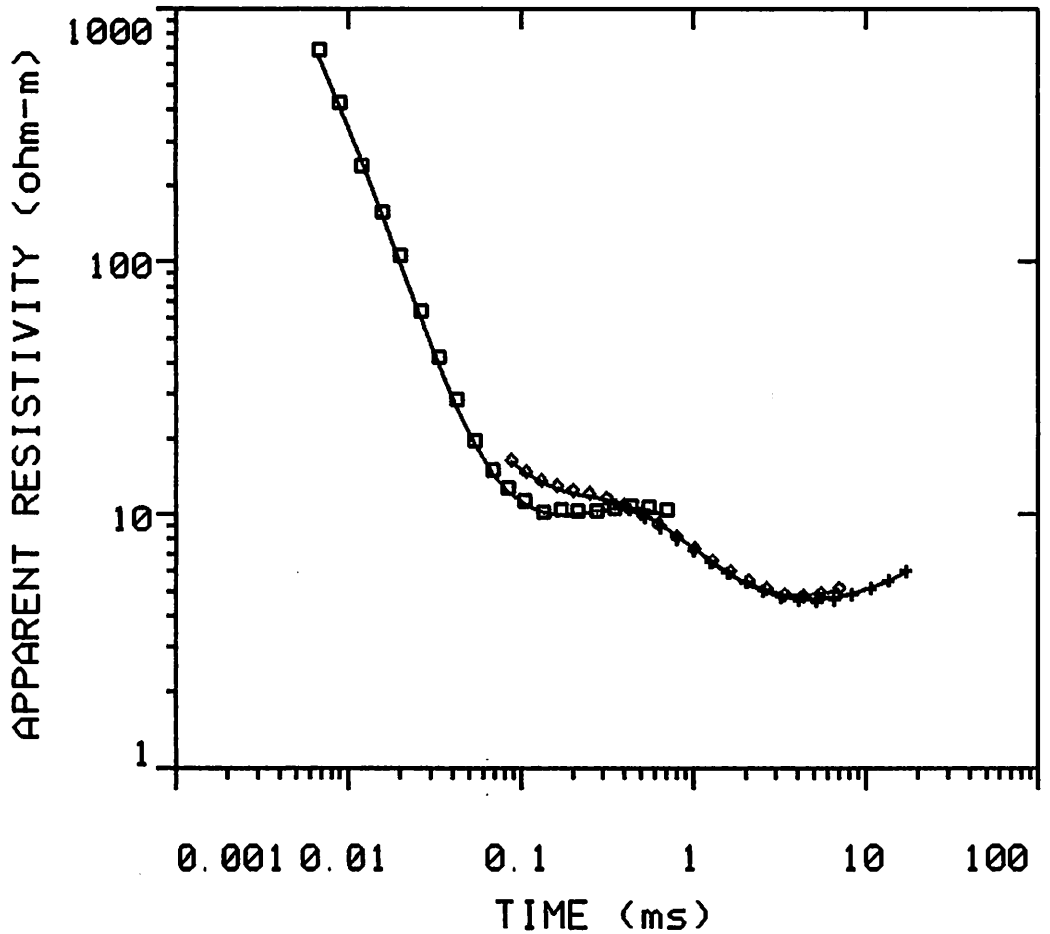
CLIENT: 100x100m DATE: 07-Dec-99
LOCATION: Kfar Shamir SOUNDING: z
COUNTY: 146525/244091 (+15.6) ELEVATION: 15.60 m
PROJECT: FITTING ERROR: 4.0% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 117.0000 SLOPE: NONE

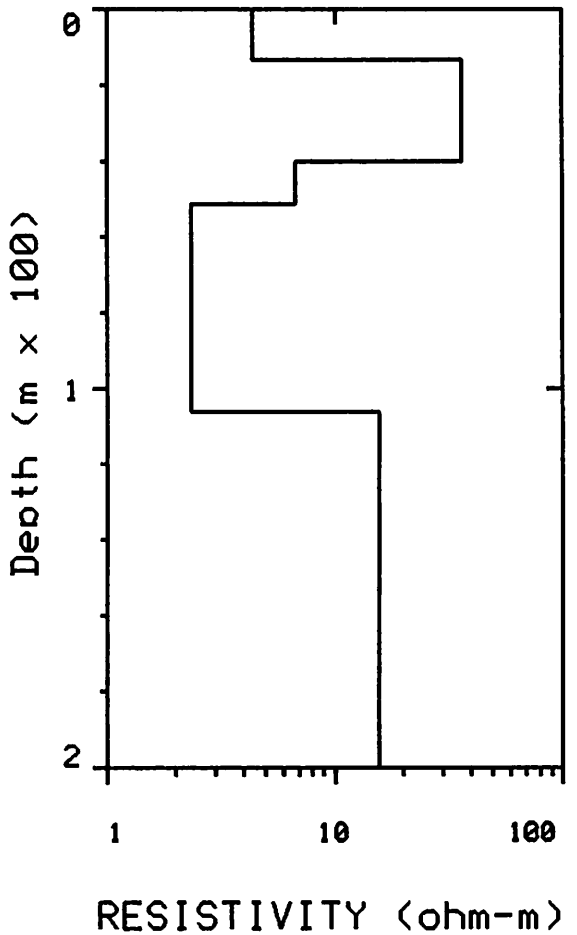
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.046 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			15.60	
1	23.54	5.84	9.75	0.248
2	1.42	5.74	4.00	4.02
3	31.93	41.13	-37.12	1.28
4	8.30	111.2	-148.3	13.39
5	2.46			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 4.6%
I.P.R.G.	Kfar Galim 146051/240800 (+15.1)
Data Set: HS530 / Date: 07-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS532

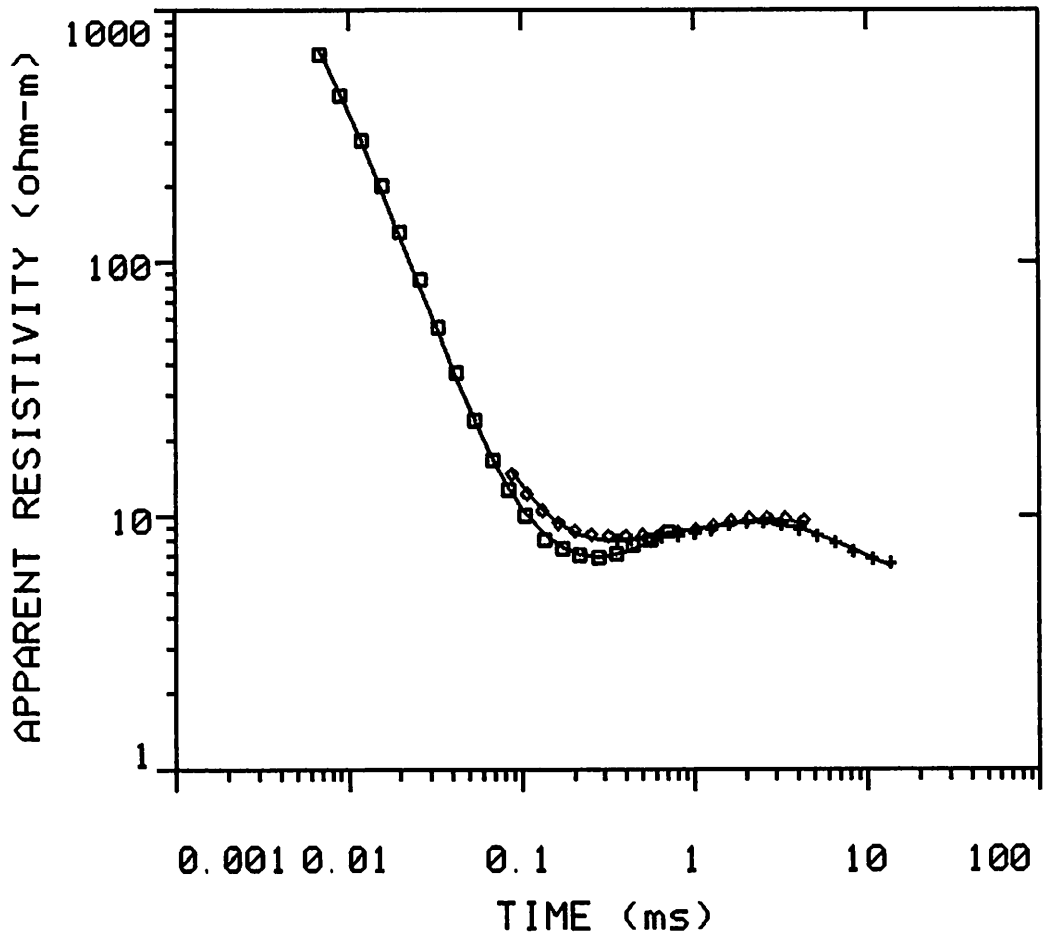
CLIENT: 100x100m DATE: 07-Dec-99
LOCATION: Hakhotrim SOUNDING: z
COUNTY: 145966/239222 (+9.9) ELEVATION: 9.90 m
PROJECT: FITTING ERROR: 2.6% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 118.0000 SLOPE: NONE

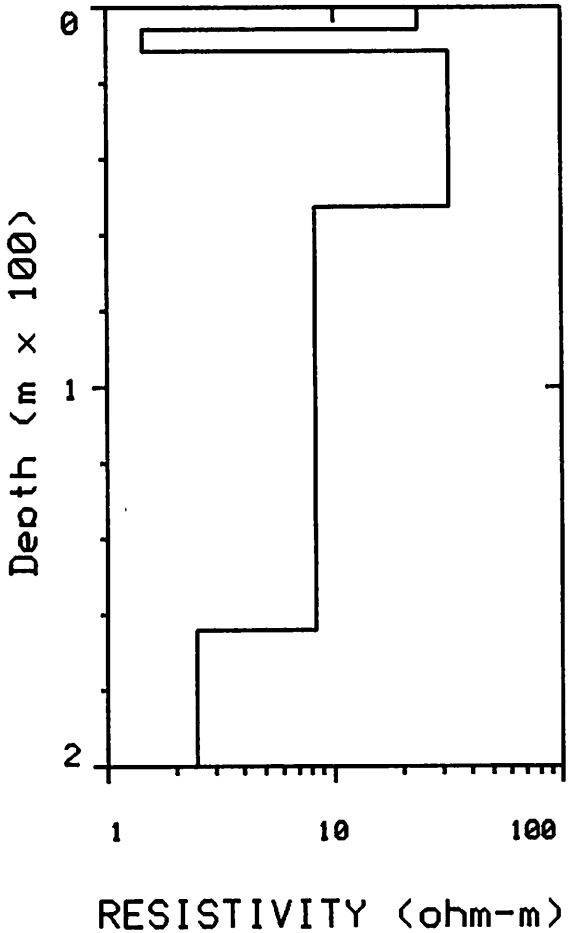
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.609 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			9.90	
1	7.98	6.61	3.28	0.828
2	2.24	8.28	-5.00	3.69
3	7.48	57.79	-62.79	7.71
4	4.55	51.58	-114.3	11.33
5	10.24			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 4.0%
I.P.R.G.	Kfar Shamir 146525/244091 (+15.6)
Data Set: HS531 / Date: 07-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS533

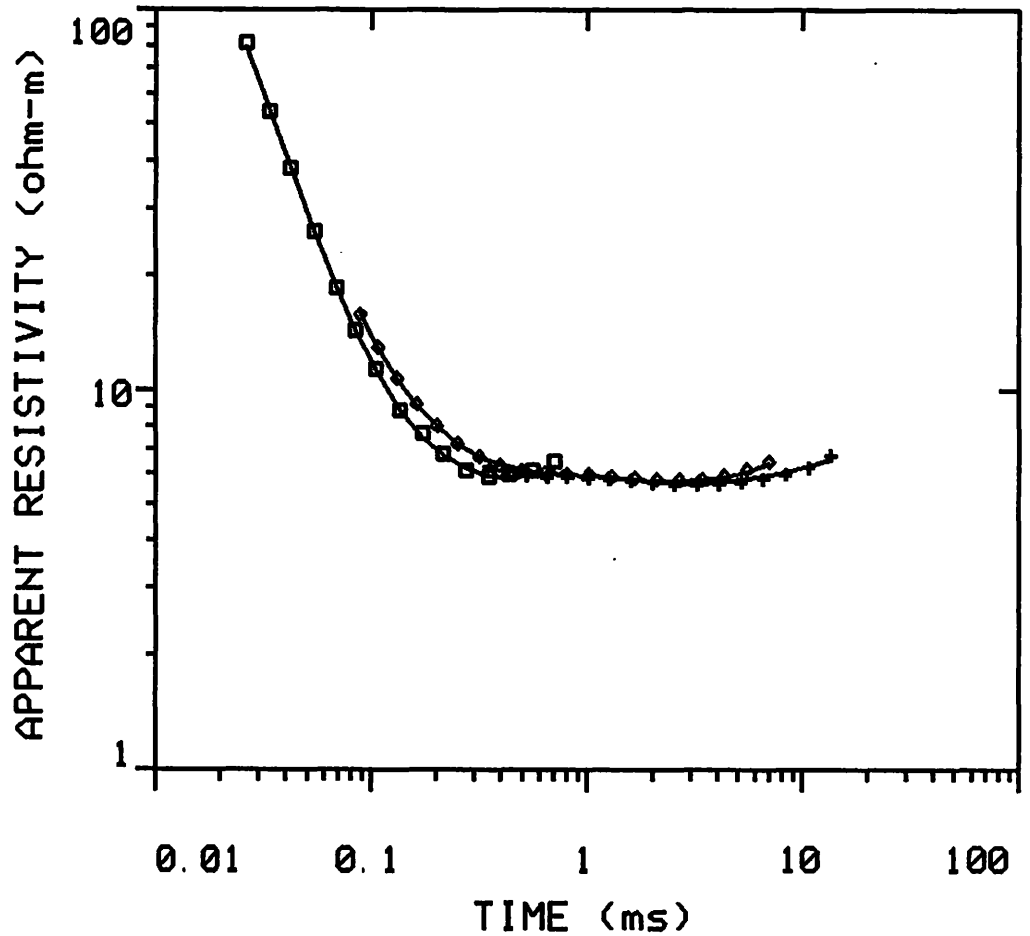
CLIENT: 50x50m DATE: 07-Dec-99
LOCATION: Habonim SOUNDING: z
COUNTY: 143098/227199 (+3.9) ELEVATION: 3.90 m
PROJECT: FITTING ERROR: 2.6% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 119.0000 SLOPE: NONE

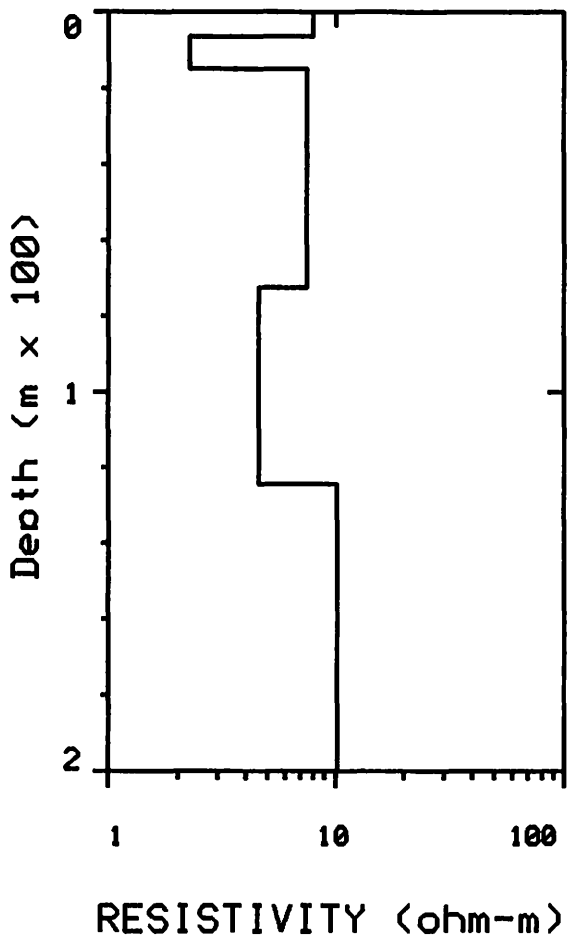
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.621 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			3.90	
1	4.32	5.84	-1.94	1.35
2	16.42	8.39	-10.34	0.511
3	30.31	14.78	-25.12	0.487
4	3.79	57.76	-82.89	15.22
5	9.88			

ALL PARAMETERS ARE FREE

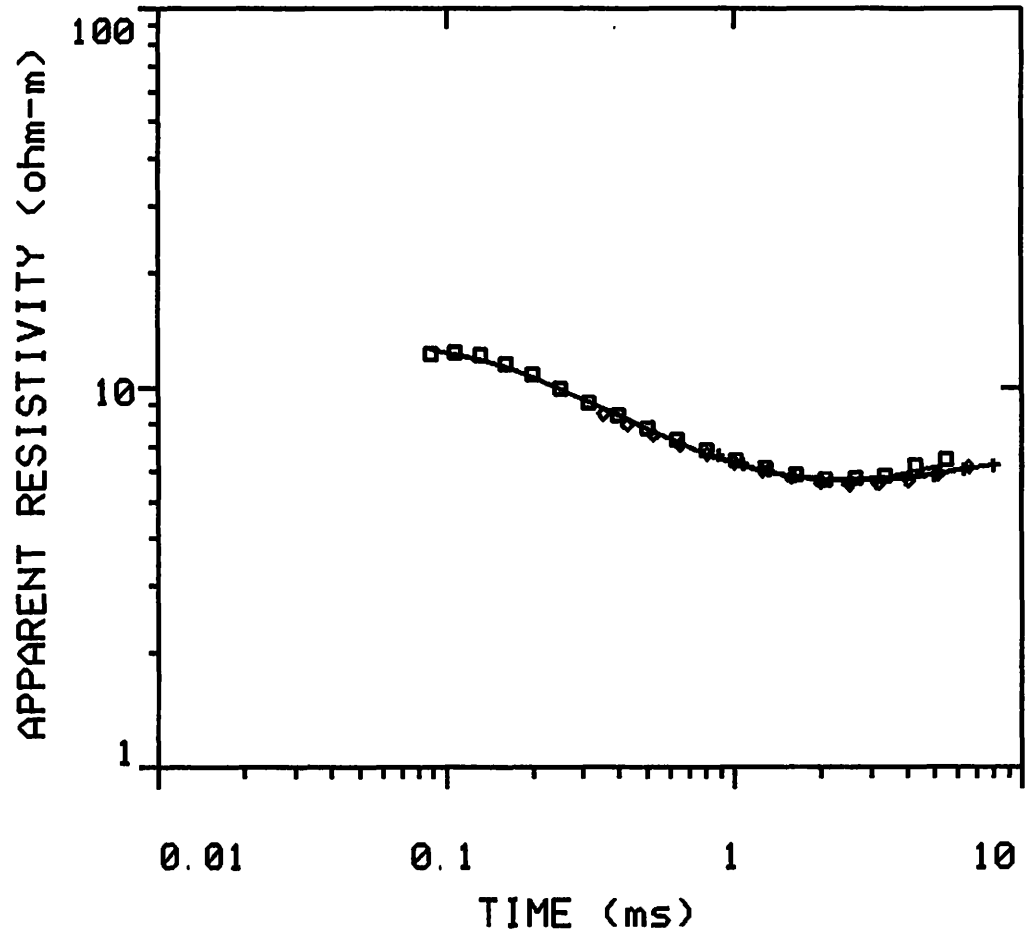


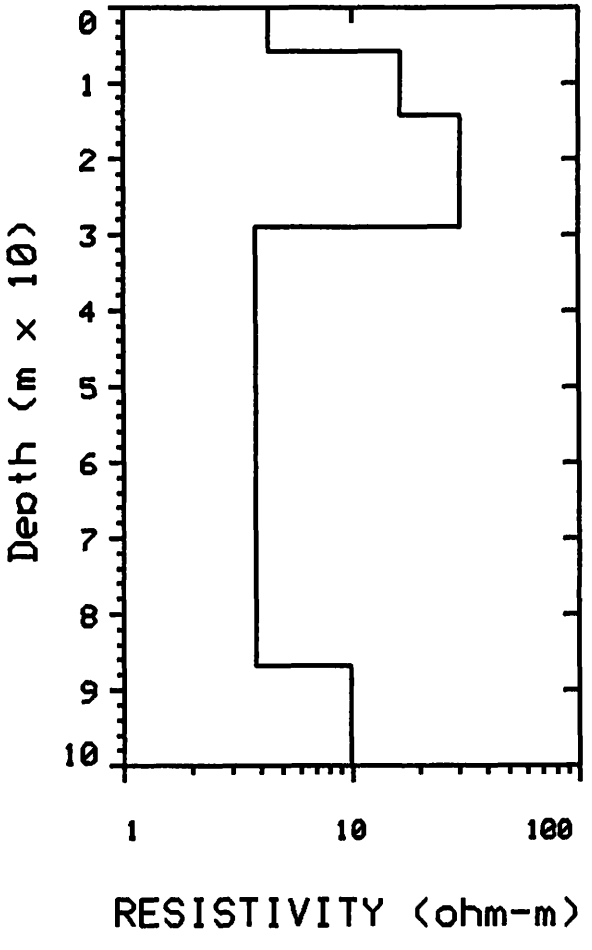


100×100m	FITTING ERROR: 2.6%
I.P.R.G.	Hakhotrim 145966/239222 (+9.9)
Data Set: HS532 / Date: 07-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

HS534

Not interpreted





50x50m	FITTING ERROR: 2.6%
I.P.R.G.	Habonim 143098/227199 (+3.9)
Data Set: HS533 / Date: 07-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS535

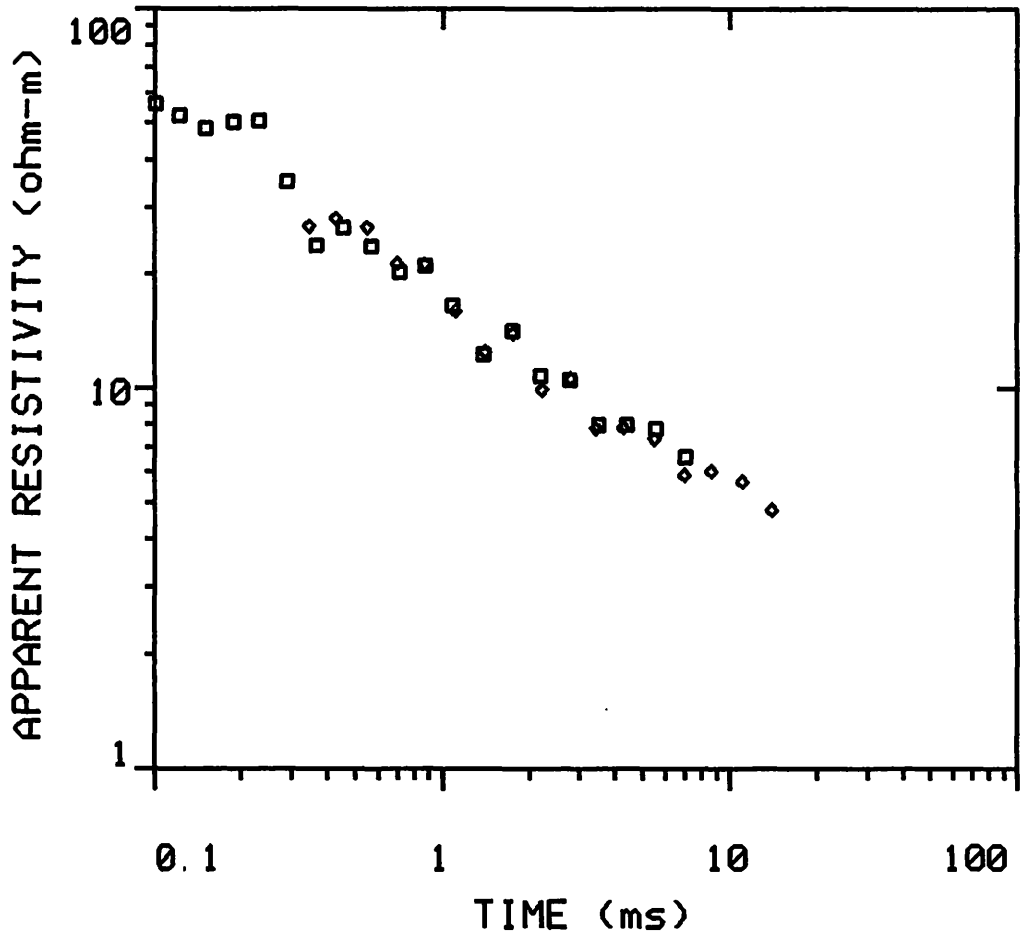
CLIENT: 200x200m DATE: 08-Dec-99
LOCATION: Caesarea SOUNDING: z
COUNTY: 140302/211513 (+7.0) ELEVATION: 7.00 m
PROJECT: FITTING ERROR: 3.9% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 200.000 m by 200.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 120.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.903 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			7.00	
1	32.74	50.46	-43.46	1.54
2	1.34	33.05	-76.52	24.65
3	6.14	20.55	-97.07	3.34
4	2.22	116.2	-213.3	52.36
5	9.77			

ALL PARAMETERS ARE FREE



100x100m	Not interpreted
I.P.R.G.	Ga'ash 134329/182796 (+23.0)
Data Set: HG534 / Date: 08-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS536

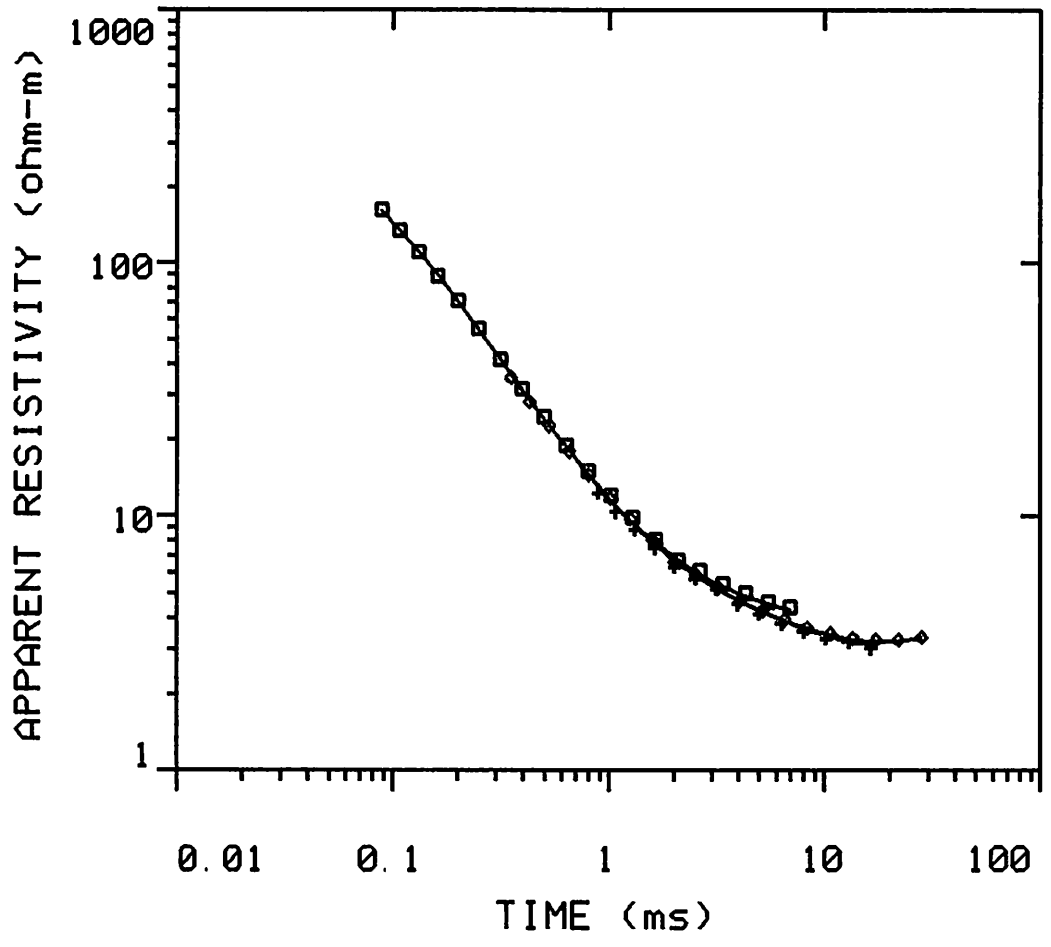
CLIENT: 100x100m DATE: 08-Dec-99
LOCATION: Jisr-Al-Zarka SOUNDING: z
COUNTY: 141390/215940 ELEVATION: 0.00 m
PROJECT: FITTING ERROR: 2.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 121.0000 SLOPE: NONE

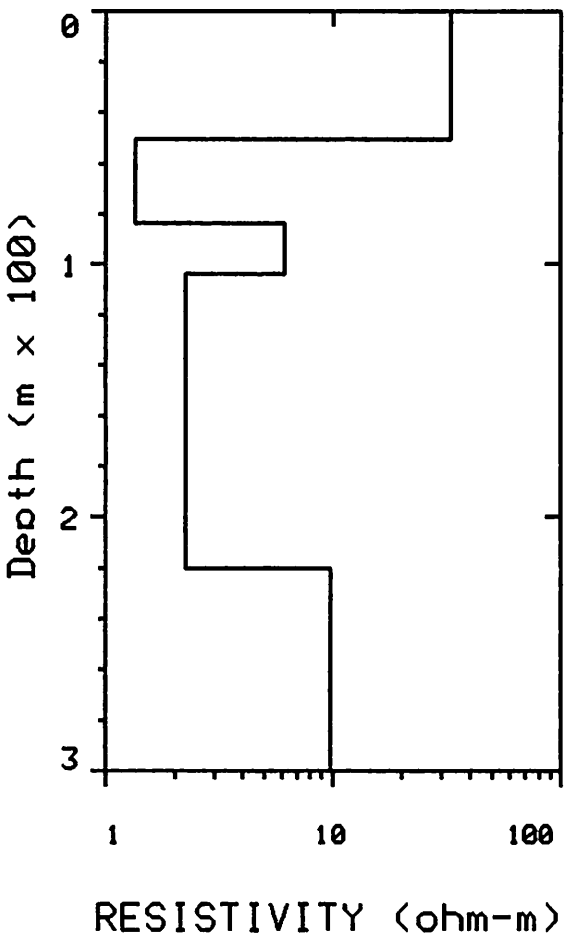
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.316 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	9.02	22.67	-22.67	2.51
2	16.31	17.16	-39.84	1.05
3	10.32	18.84	-58.68	1.82
4	14.72			

ALL PARAMETERS ARE FREE





200x200m	FITTING ERROR: 3.9%
I.P.R.G.	Caesarea 140302/211513 (+7.0)
Data Set: HS535 / Date: 08-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS537

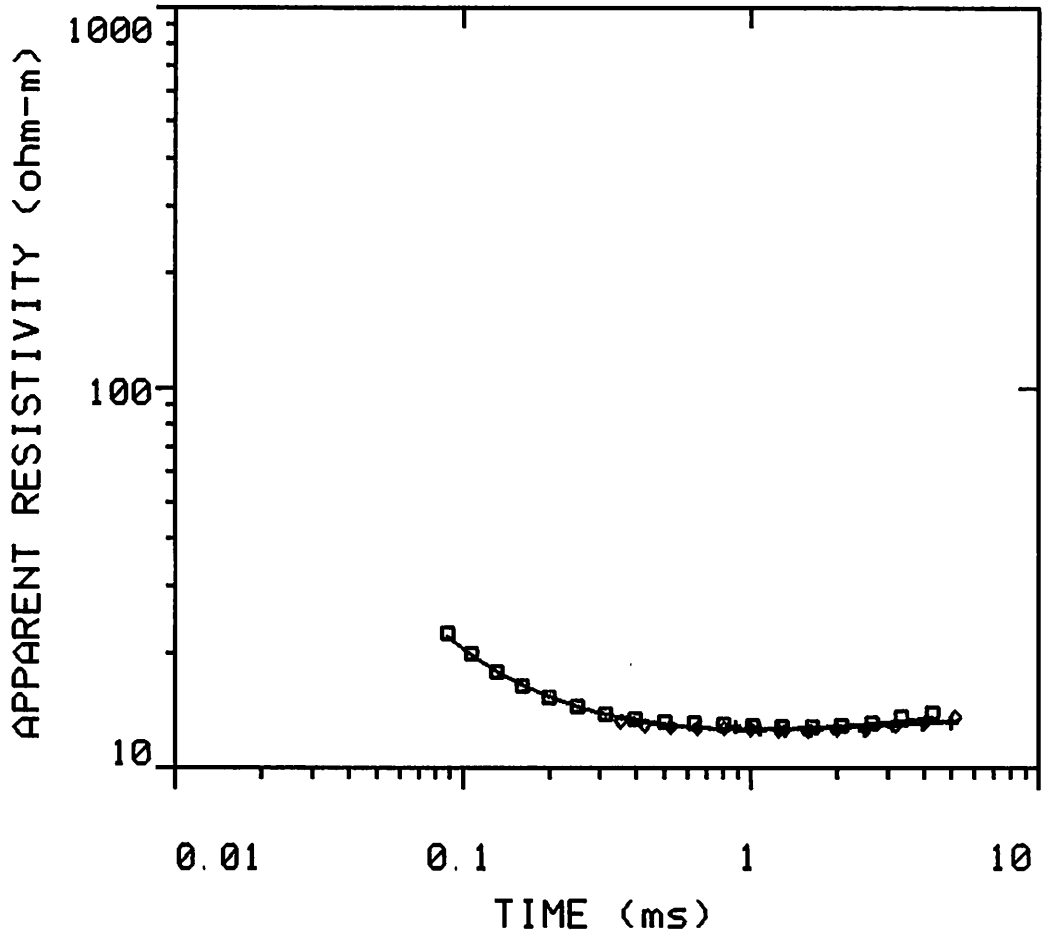
CLIENT: 200x200m DATE: 08-Dec-99
LOCATION: Jisr-Al-Zarka SOUNDING: z
COUNTY: 142550/216010 ELEVATION: 0.00 m
PROJECT: FITTING ERROR: 2.9% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 200.000 m by 200.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 122.0000 SLOPE: NONE

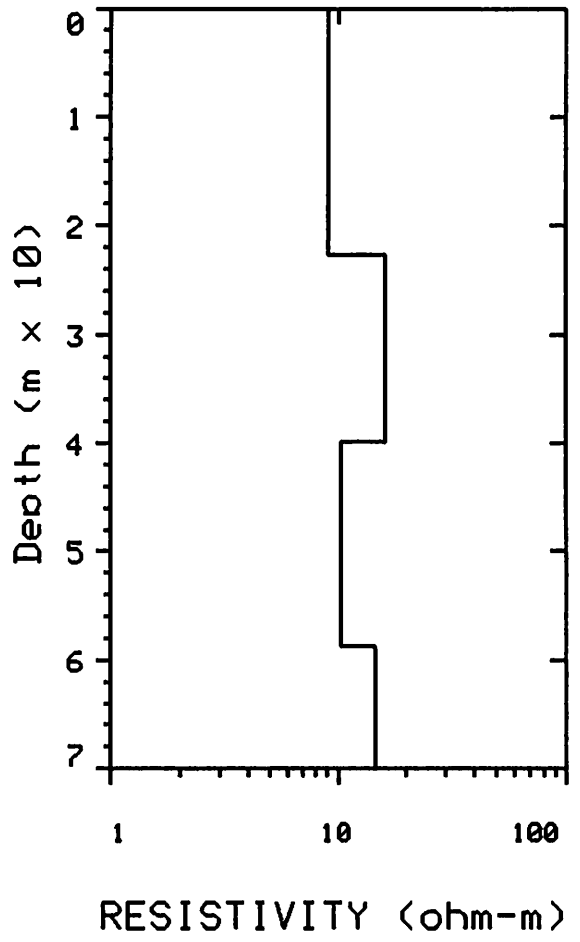
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.885 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	7.14	40.95	-40.95	5.72
2	26.06	18.91	-59.86	0.725
3	53.11	38.47	-98.33	0.724
4	13.68			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.3%
I.P.R.G.	Jisr-Al-Zarka 141390/215940
Data Set: HS536 / Date: 08-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS538

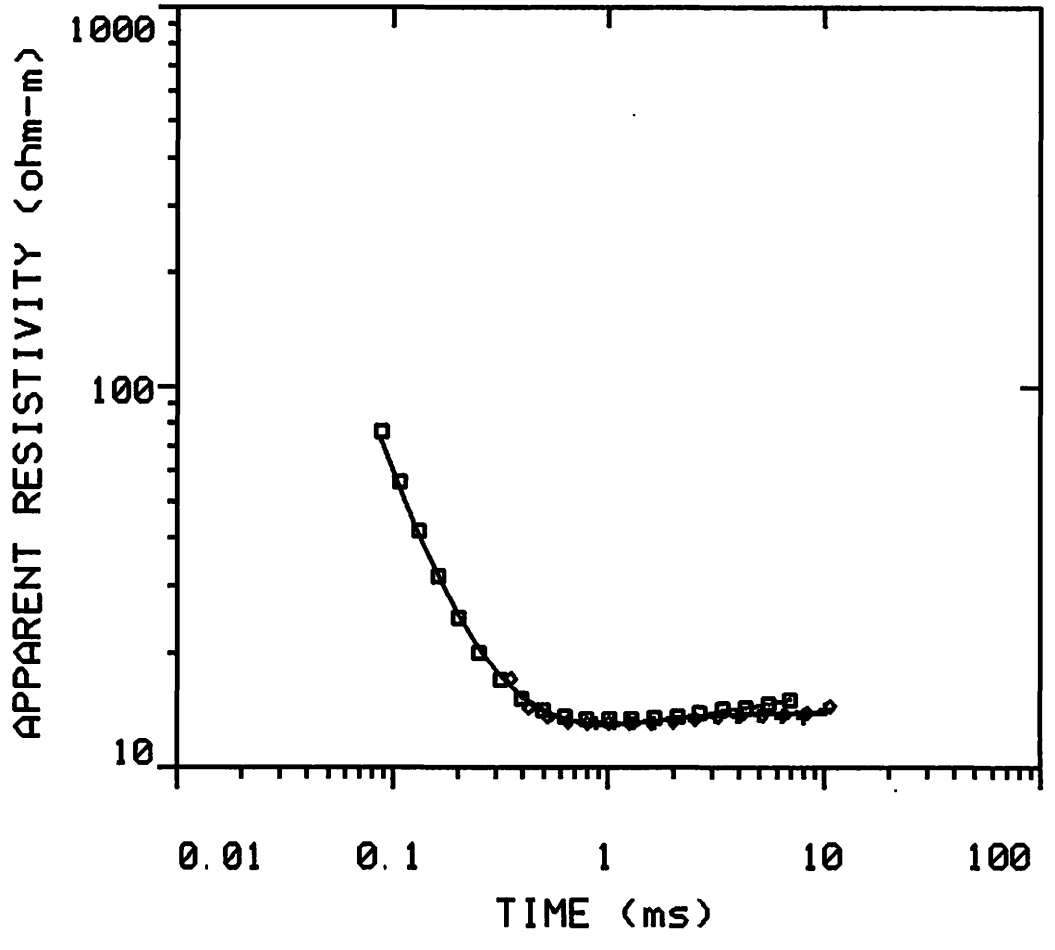
CLIENT: 100x100m DATE: 12-Dec-99
LOCATION: Rosh Ha-Nikra SOUNDING: z
COUNTY: 160794/276209 (+6.5m) ELEVATION: 6.50 m
PROJECT: FITTING ERROR: 3.6% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 123.0000 SLOPE: NONE

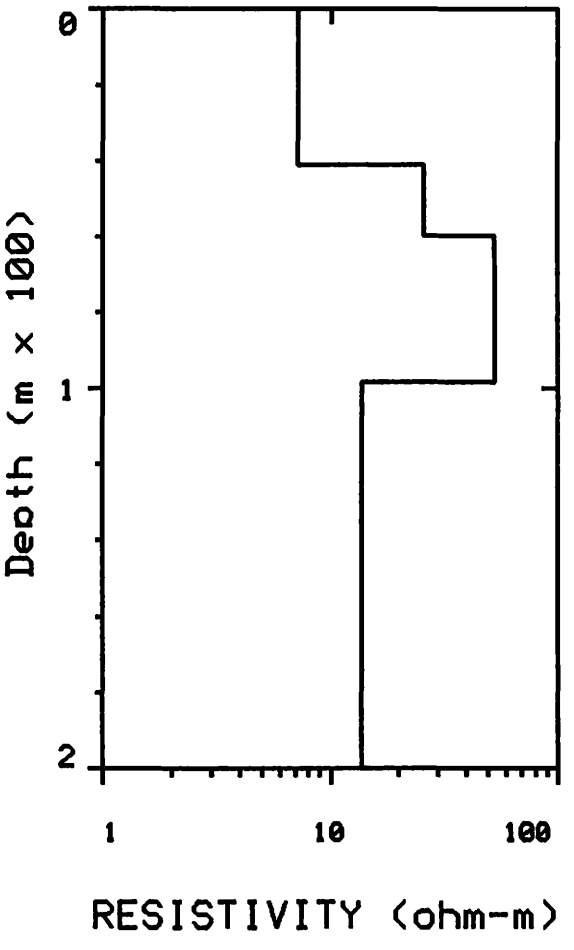
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.551 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			6.50	
1	4.35	16.92	-10.42	3.88
2	28.75	16.61	-27.03	0.577
3	3.90	51.06	-78.10	13.06
4	29.08			

ALL PARAMETERS ARE FREE





200x200m	FITTING ERROR: 2.9%
I.P.R.G.	Jisr-Al-Zarka 142550/216010
Data Set: HS537 / Date: 08-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS539

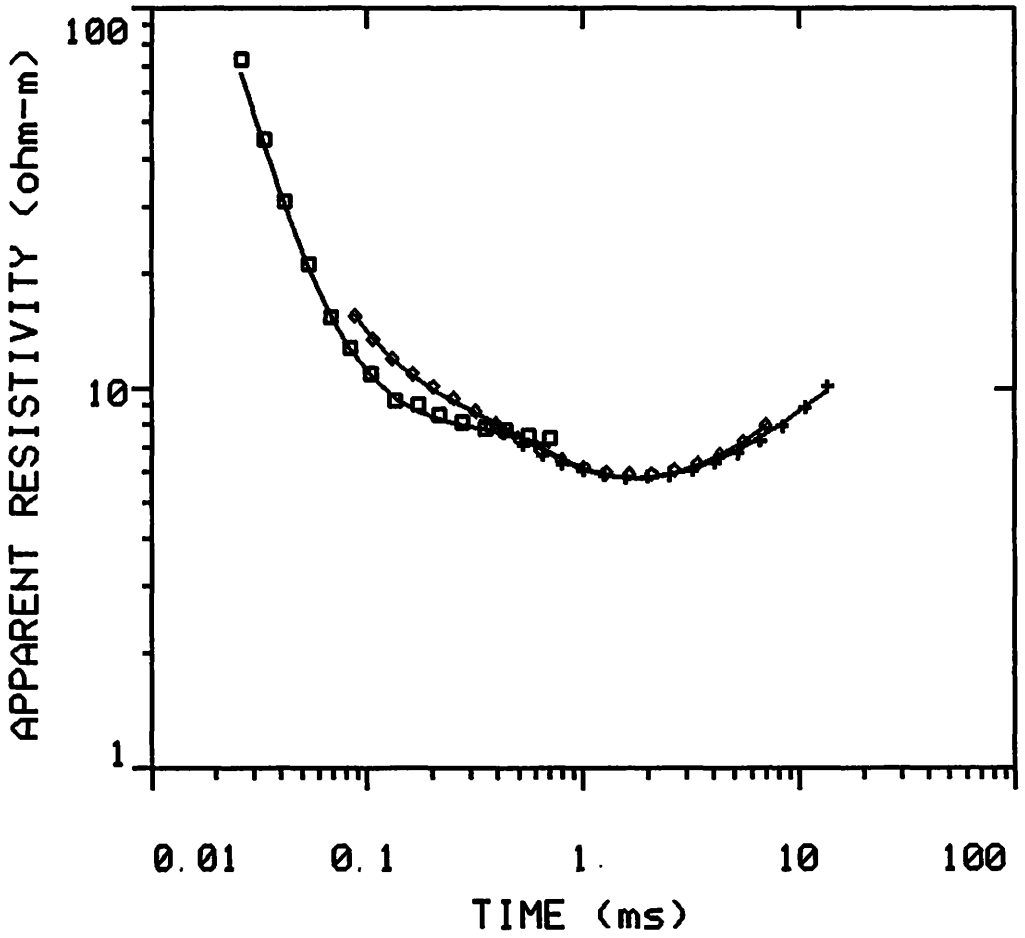
CLIENT: 50x50m DATE: 12-Dec-99
LOCATION: Betzet SOUNDING: z
COUNTY: 160707/275063 (+7.1m) ELEVATION: 7.10 m
PROJECT: FITTING ERROR: 1.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 124.0000 SLOPE: NONE

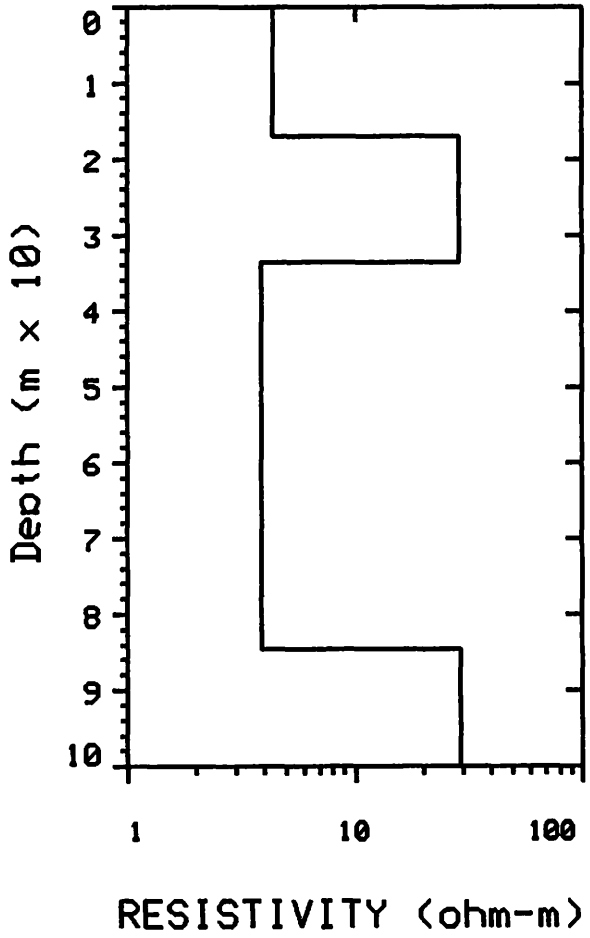
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.735 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			7.10	
1	5.77	8.30	-1.20	1.43
2	19.49	32.59	-33.79	1.67
3	1.80	99.02	-132.8	54.75
4	0.750			

ALL PARAMETERS ARE FREE

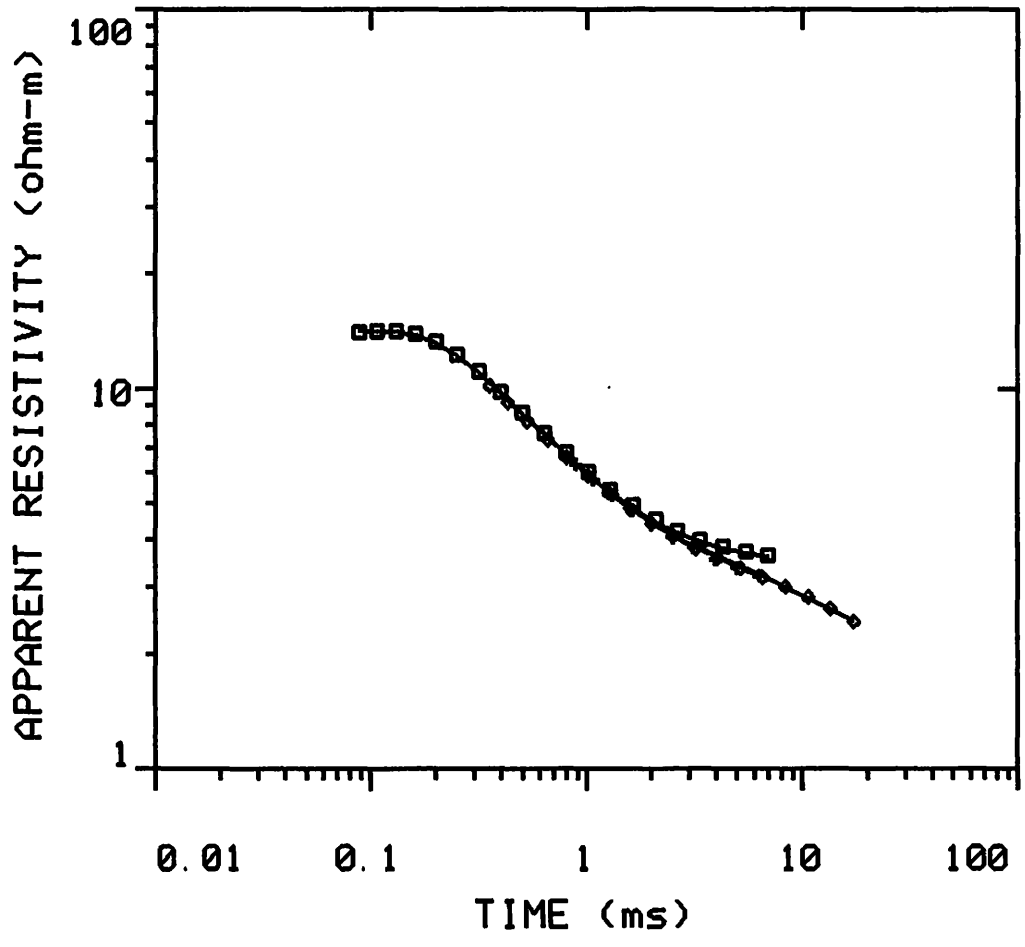


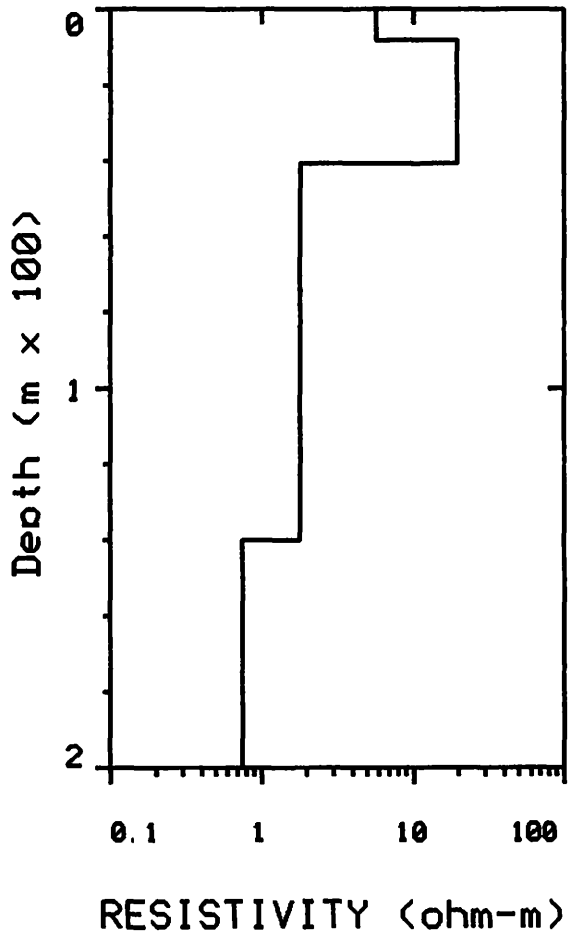


100x100m	FITTING ERROR: 3.6%
I.P.R.G.	Rosh Ha-Nikra 160794/276209 (+6.5m)
Data Set: HS538 / Date: 12-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

HS540

Not interpreted





50x50m	FITTING ERROR: 1.7%
I.P.R.G.	Betzet 160707/275063 (+7.1m)
Data Set: HS539 / Date: 12-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS540A

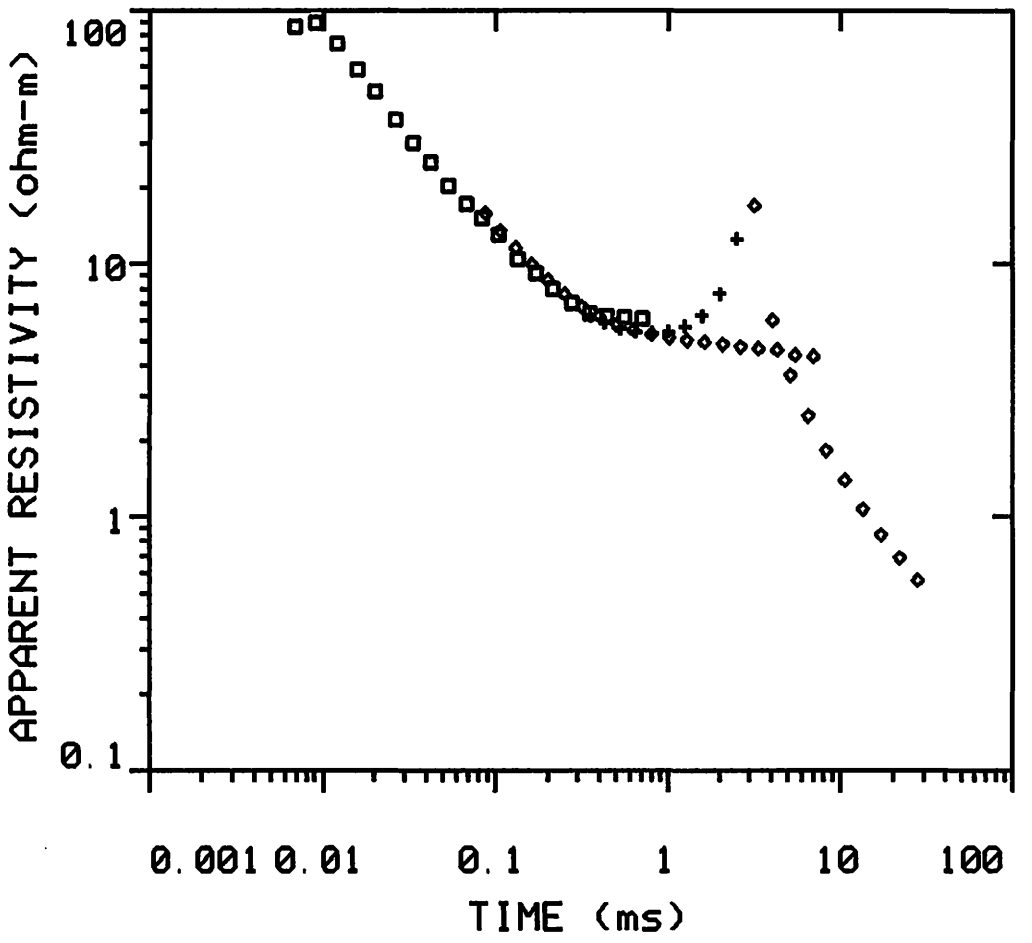
CLIENT: 50x50m DATE: 13-Dec-99
LOCATION: Ahziv SOUNDING: z
COUNTY: 159600/271200 ELEVATION: 0.00 m
PROJECT: FITTING ERROR: 4.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 127.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.716 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			0.0	
1	23.30	5.78	-5.78	0.248
2	317.4	11.18	-16.97	0.0352
3	2.57	122.4	-139.4	47.56
4	97.73			

ALL PARAMETERS ARE FREE



50x50m	Not interpreted
I.P.R.G.	Akhziv 159600/271200
Data Set: HS540 / Date: 13-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS541

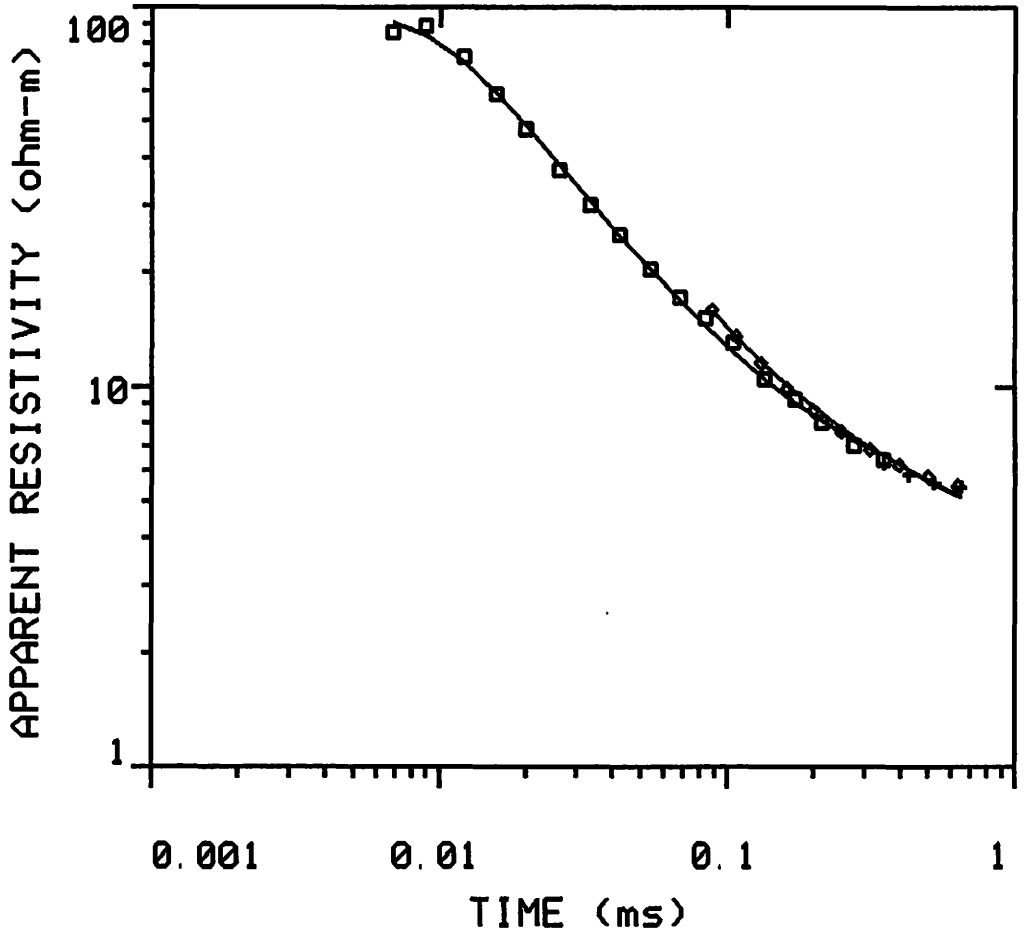
CLIENT: 100x100m DATE: 12-Dec-99
LOCATION: Nahariya South SOUNDING: z
COUNTY: 158134/265841 (+4.9m) ELEVATION: 4.90 m
PROJECT: FITTING ERROR: 2.6% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 125.0000 SLOPE: NONE

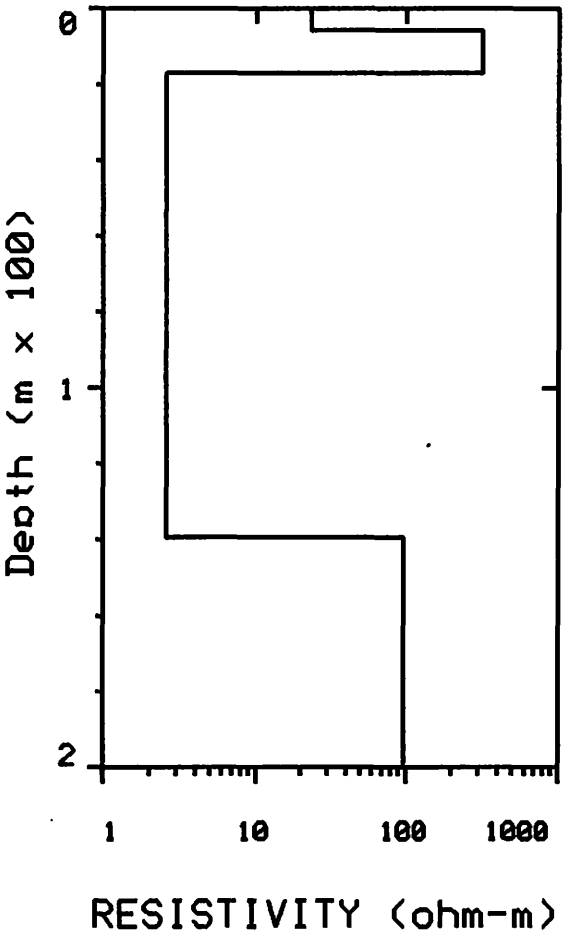
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.614 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.90	
1	9.61	21.70	-16.80	2.25
2	19.83	17.45	-34.26	0.880
3	3.25	64.20	-98.46	19.72
4	8.97			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 4.7%
I.P.R.G.	Ahziv 159600/271200
Data Set: HSS40A / Date: 13-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS542

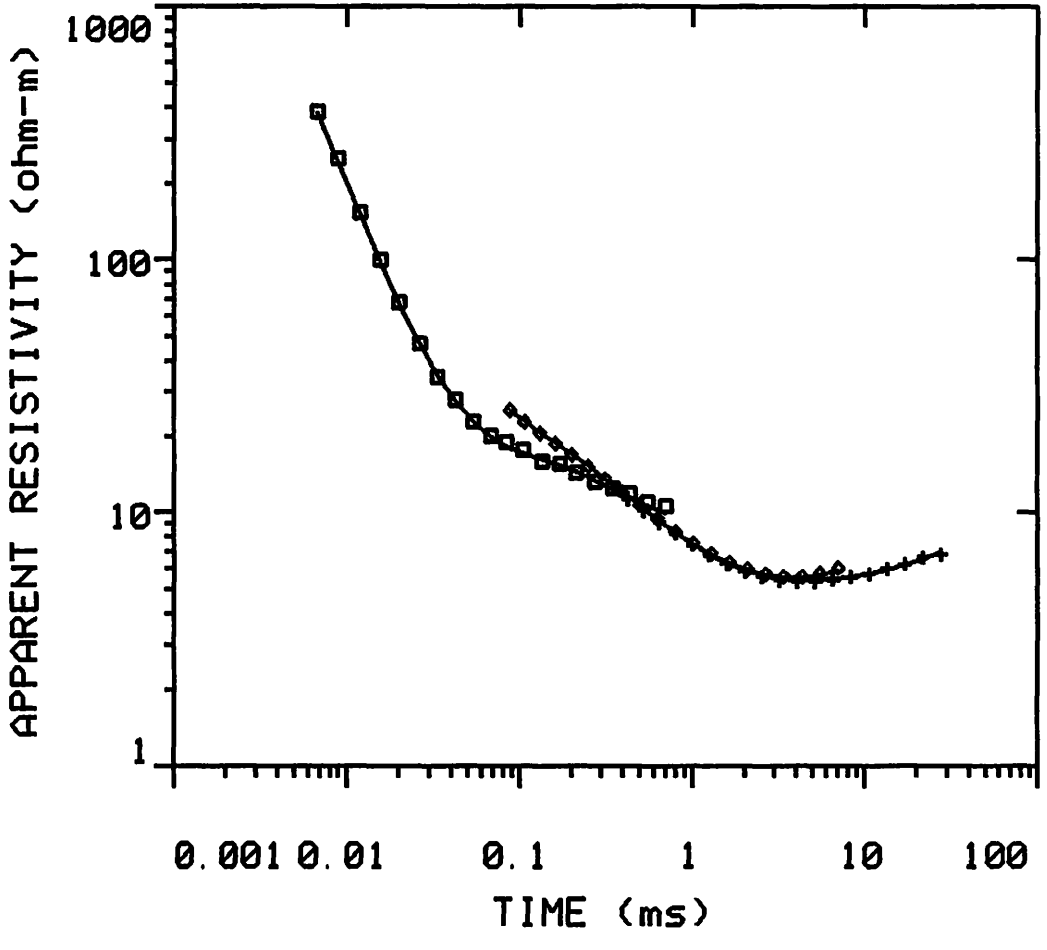
CLIENT: 100x100m DATE: 12-Dec-99
LOCATION: Nahariya South SOUNDING: z
COUNTY: 158520/265782 (+5.8m) ELEVATION: 5.80 m
PROJECT: FITTING ERROR: 3.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 126.0000 SLOPE: NONE

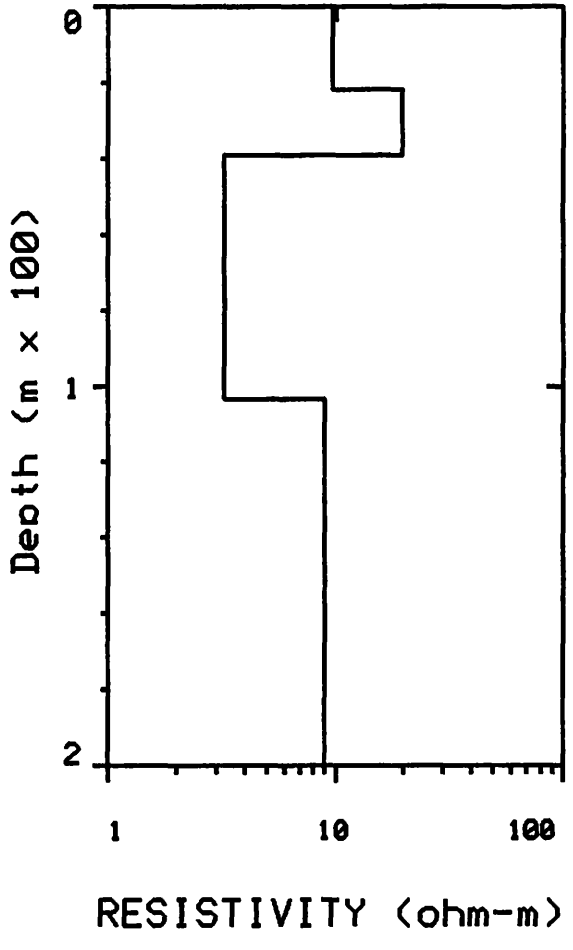
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.711 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			5.80	
1	6.45	7.00	-1.20	1.08
2	23.06	39.18	-40.38	1.69
3	5.43	22.77	-63.16	4.18
4	1.67			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.6%
I.P.R.G.	Nahariya South 158134/265841 (+4.9m)
Data Set: HS541 / Date: 12-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS543

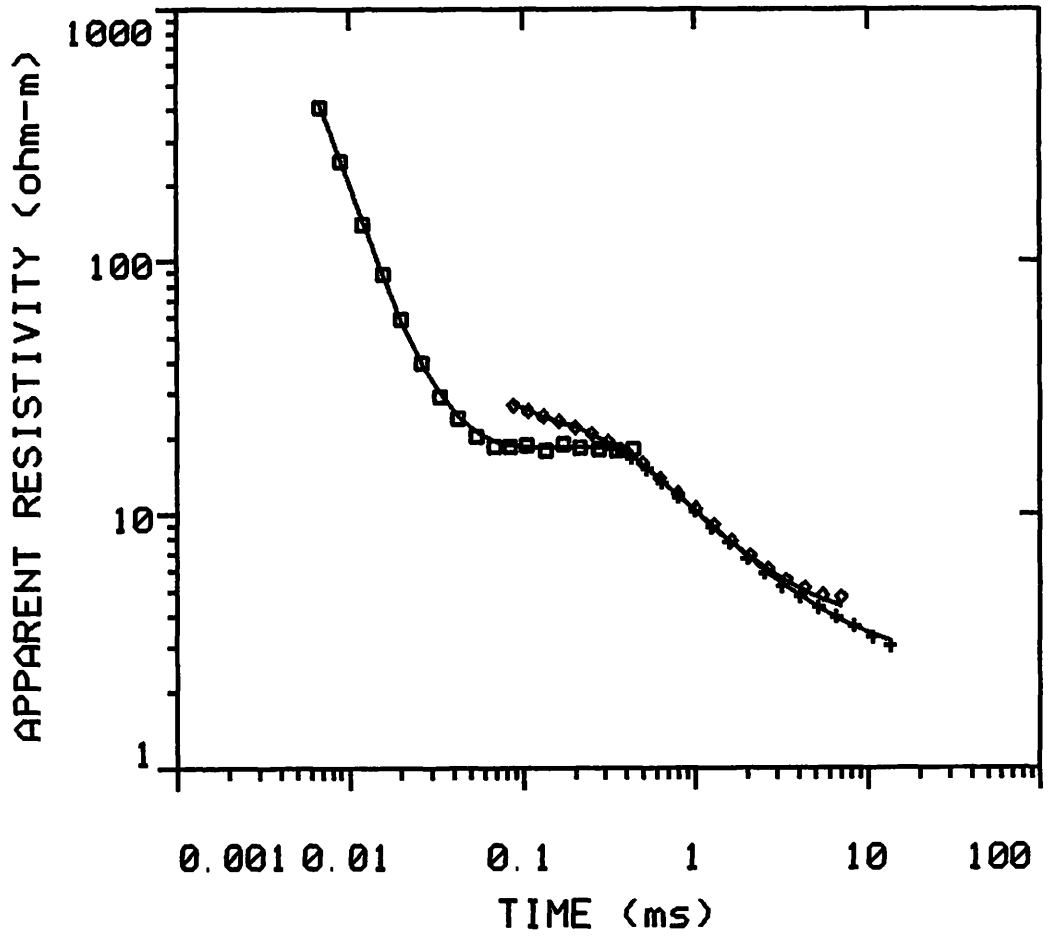
CLIENT: 50x50m DATE: 13-Dec-99
LOCATION: Khatzrot Yosef SOUNDING: z
COUNTY: 157491/262631 (+1.9m) ELEVATION: 1.90 m
PROJECT: FITTING ERROR: 3.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 128.0000 SLOPE: NONE

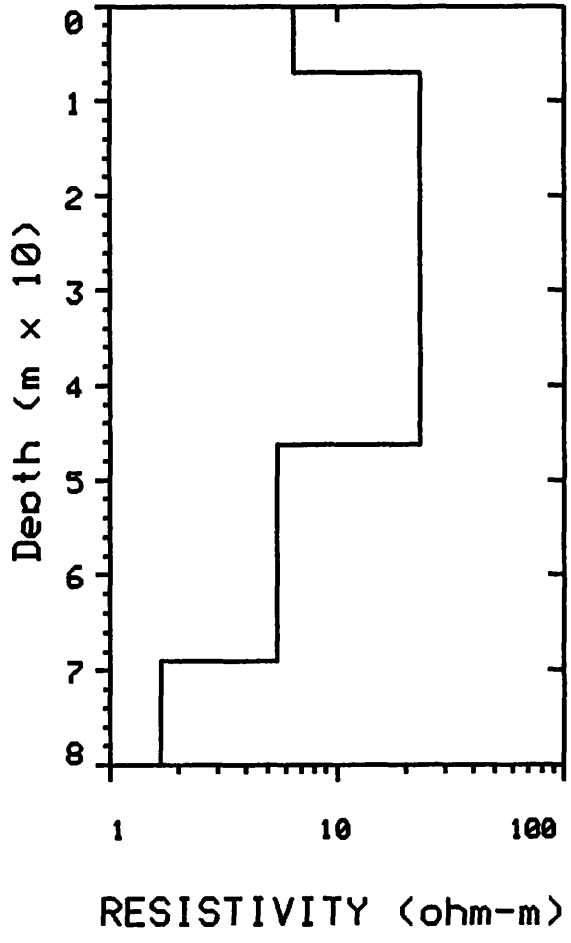
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.094 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			1.90	
1	3.29	14.81	-12.91	4.50
2	1.38	9.23	-22.15	6.68
3	2.33	43.83	-65.98	18.79
4	3.80			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.7%
I.P.R.G.	Nahariya South 158520/265782 (+5.8m)
Data Set: HS542 / Date: 12-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS544

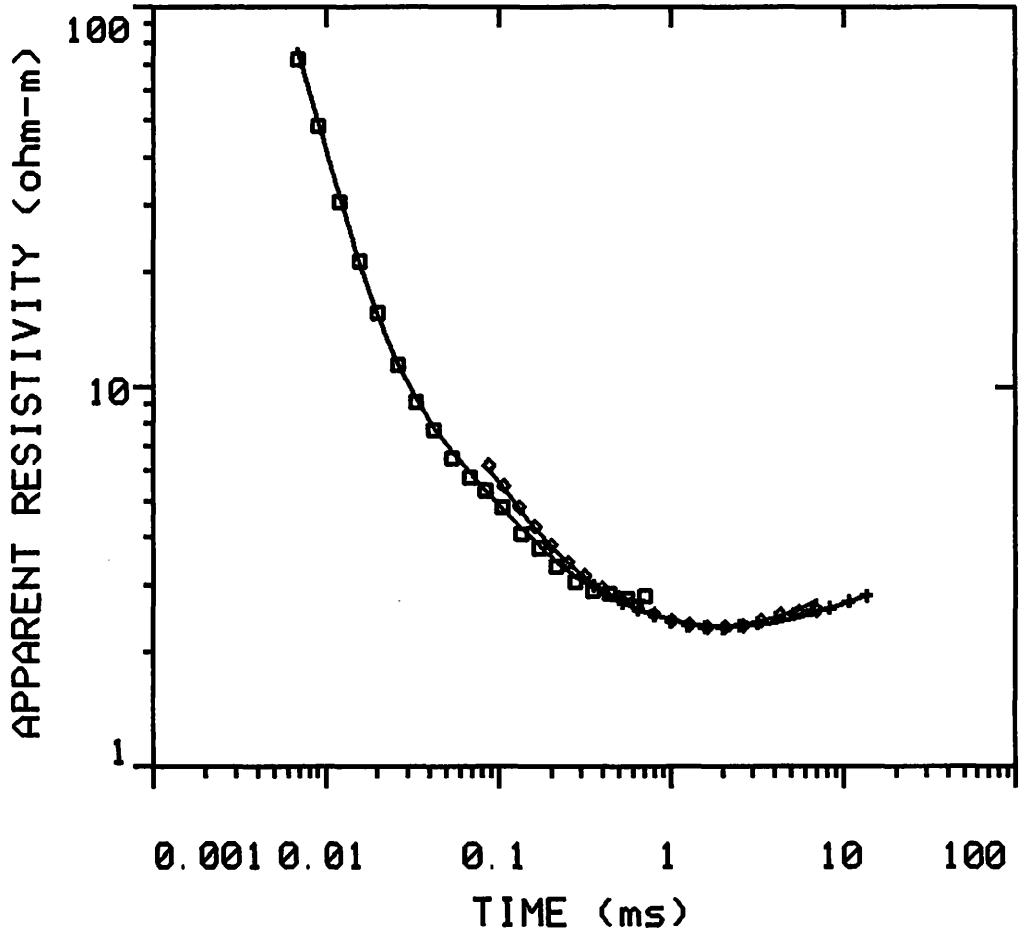
CLIENT: 100x100m DATE: 15-Dec-99
LOCATION: Geva Karmel SOUNDING: z
COUNTY: 146959/230378 (+35.1) ELEVATION: 35.10 m
PROJECT: FITTING ERROR: 7.9% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 129.0000 SLOPE: NONE

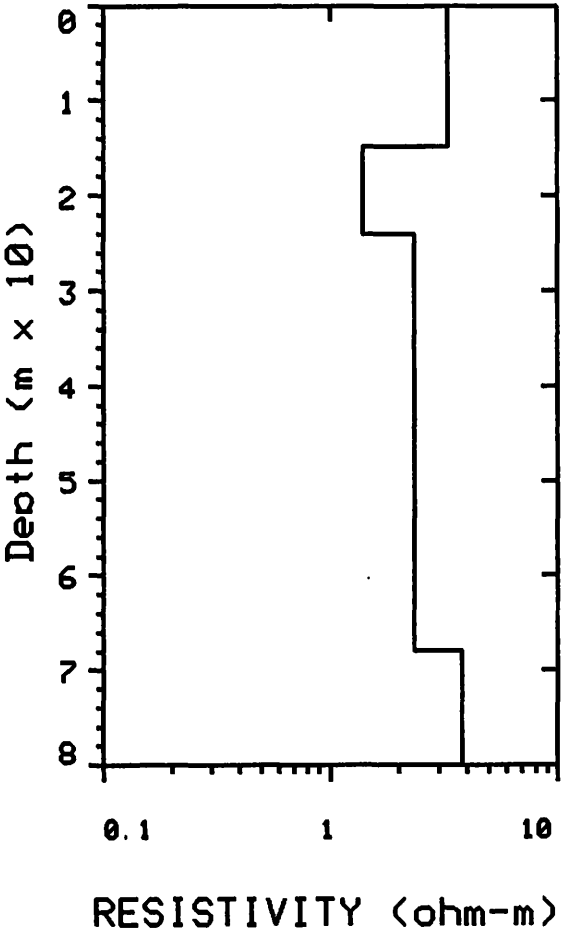
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 7.912 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			35.10	
1	9.41	13.74	21.35	1.45
2	334.4	66.35	-45.00	0.198
3	10.79	30.84	-75.85	2.85
4	249.2	187.7	-263.6	0.753
5	9.71	29.41	-293.0	3.02
6	0.979			

ALL PARAMETERS ARE FREE

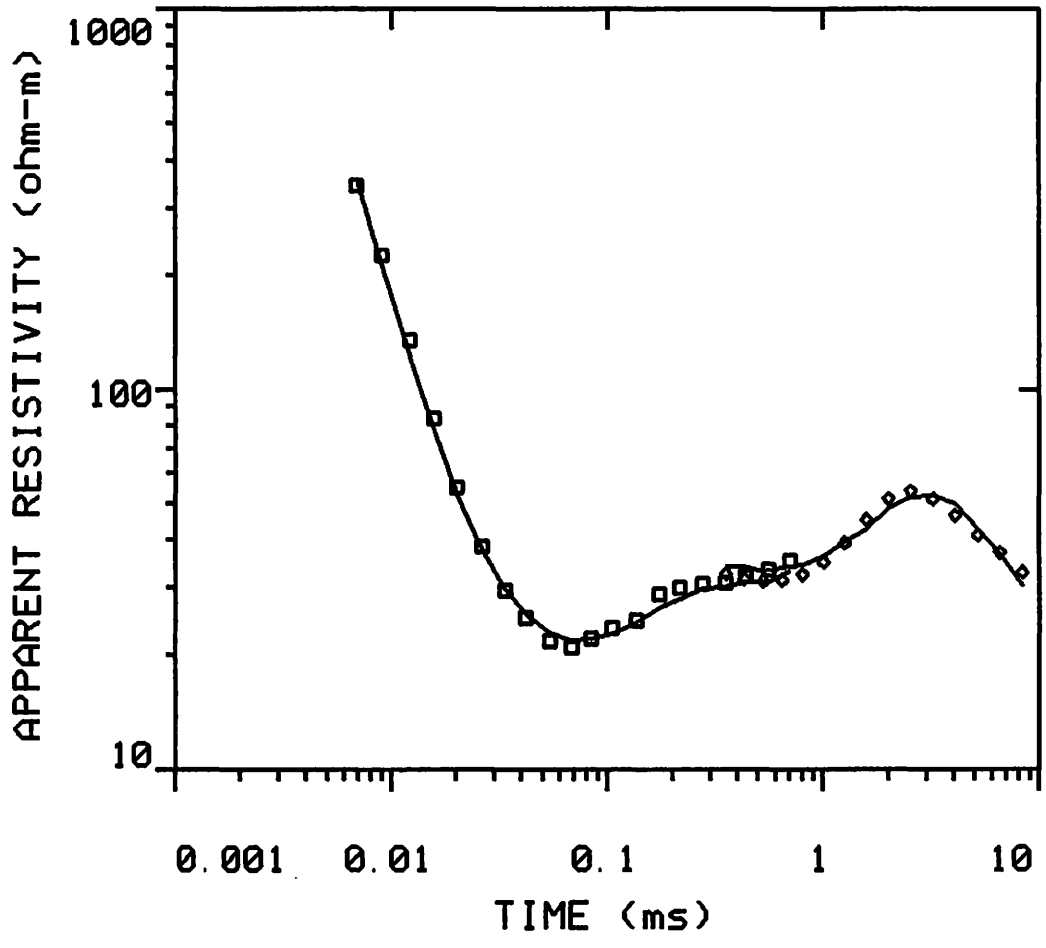


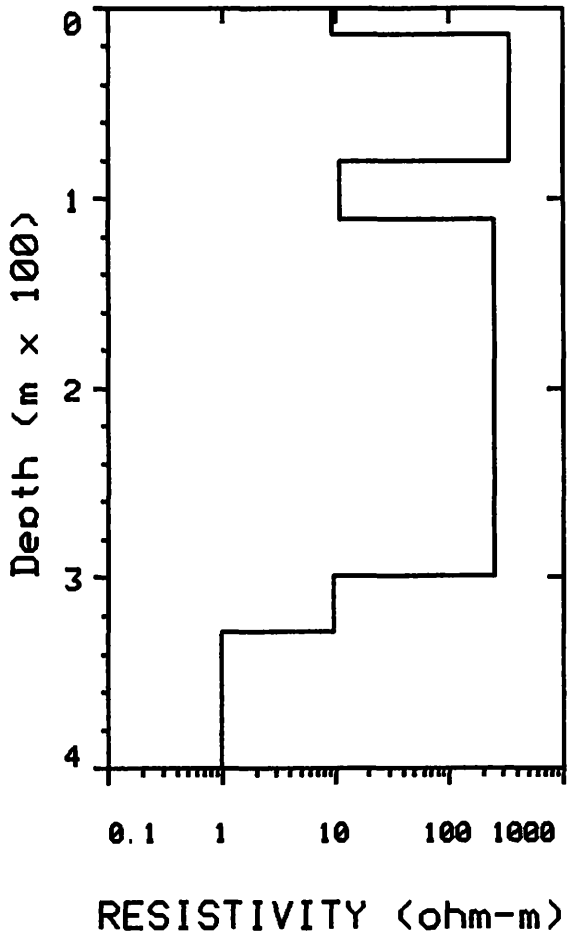


50x50m	FITTING ERROR: 3.1%
I.P.R.G.	Khatzrot Yosef 157491/262631 (+1.9m)
Data Set: HS543 / Date: 13-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

HS545

Not interpreted





100x100m	FITTING ERROR: 7.9%
I.P.R.G.	Geva Karmel 146959/230378 (+35.1)
Data Set: HS544 / Date: 15-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS546

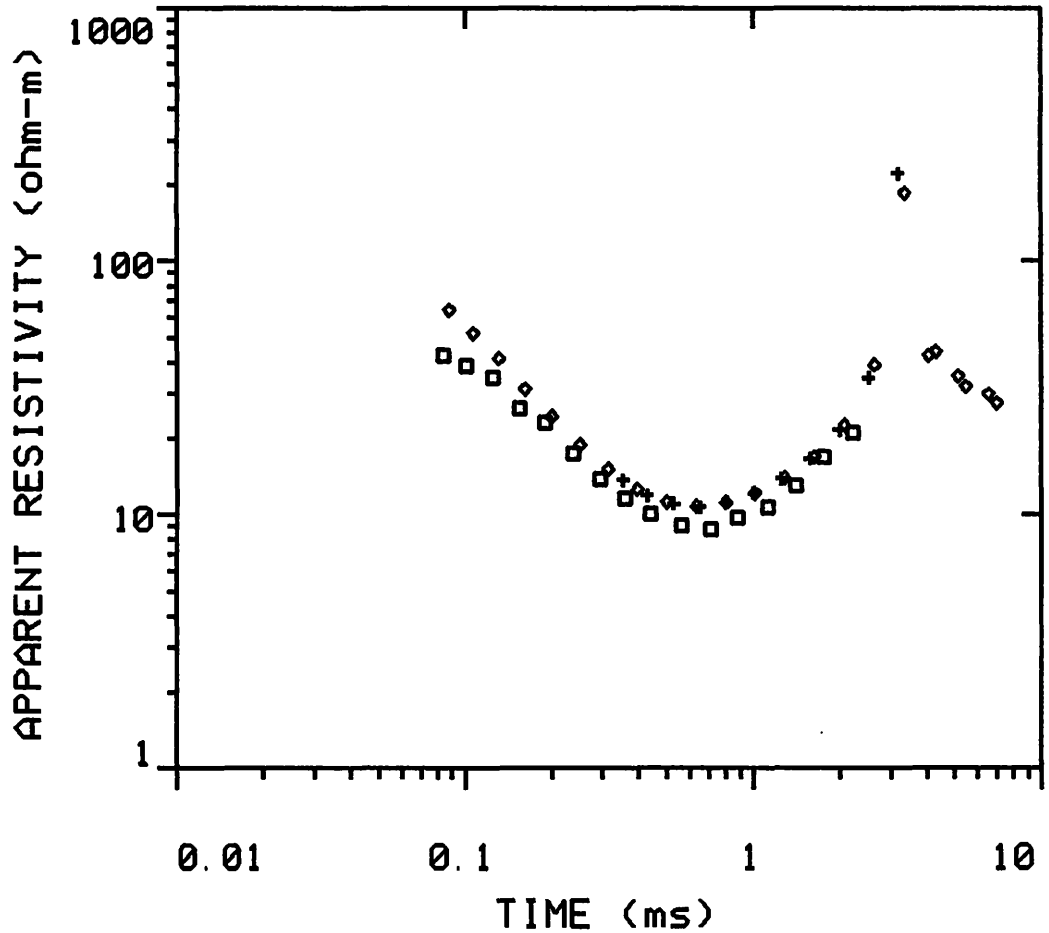
CLIENT: 50x50m DATE: 15-Dec-99
LOCATION: Shvei Zion SOUNDING: z
COUNTY: 157813/264233 (+2.6) ELEVATION: 2.60 m
PROJECT: FITTING ERROR: 3.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 131.0000 SLOPE: NONE

Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.214 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			2.60	
1	4.32	9.30	-6.70	2.15
2	8.17	11.64	-18.35	1.42
3	1.32	11.55	-29.91	8.73
4	4.81	119.7	-149.6	24.89
5	3.04			

ALL PARAMETERS ARE FREE



100x100m	Not interpreted
I.P.R.G.	Park Ha-Karmel 148323/235546 (+56.9)
Data Set: HS545 / Date: 15-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS547

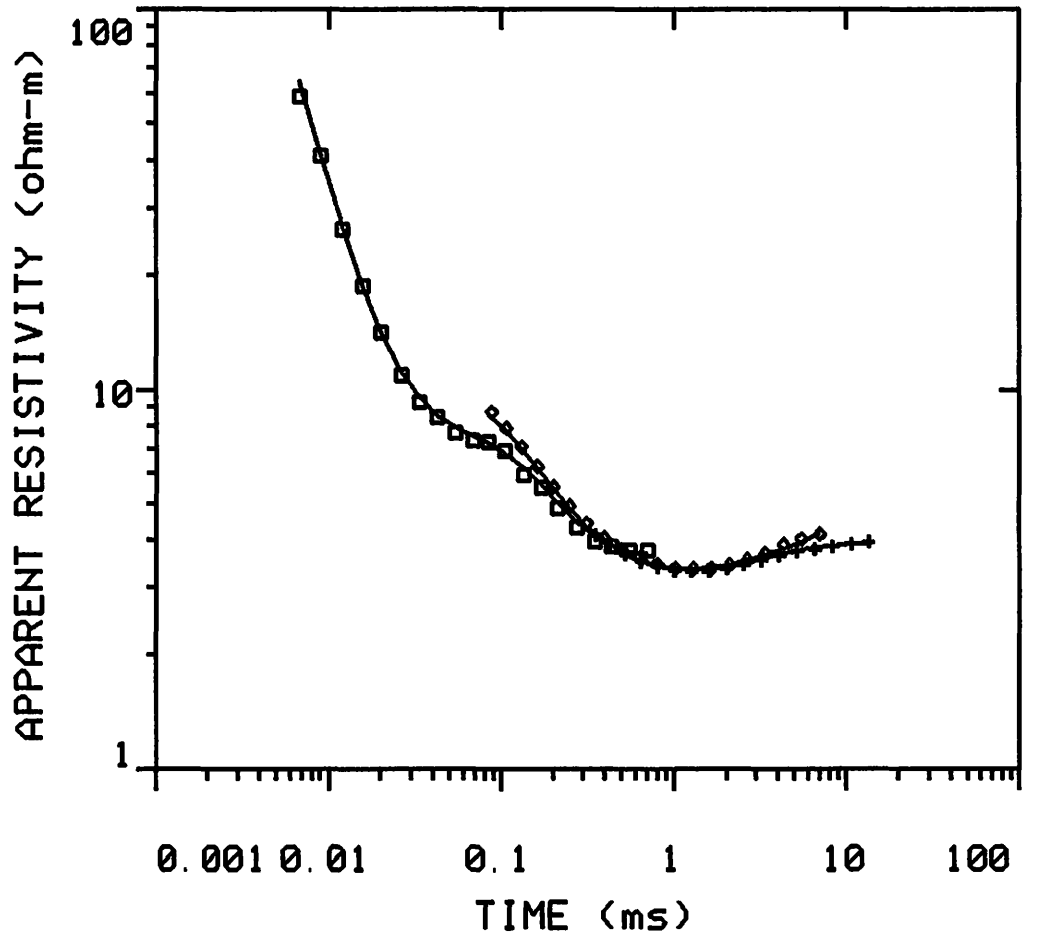
CLIENT: 100x100m DATE: 15-Dec-99
LOCATION: Shvei Zion SOUNDING: z
COUNTY: 158073/264255 (+3.0) ELEVATION: 3.00 m
PROJECT: FITTING ERROR: 1.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 132.0000 SLOPE: NONE

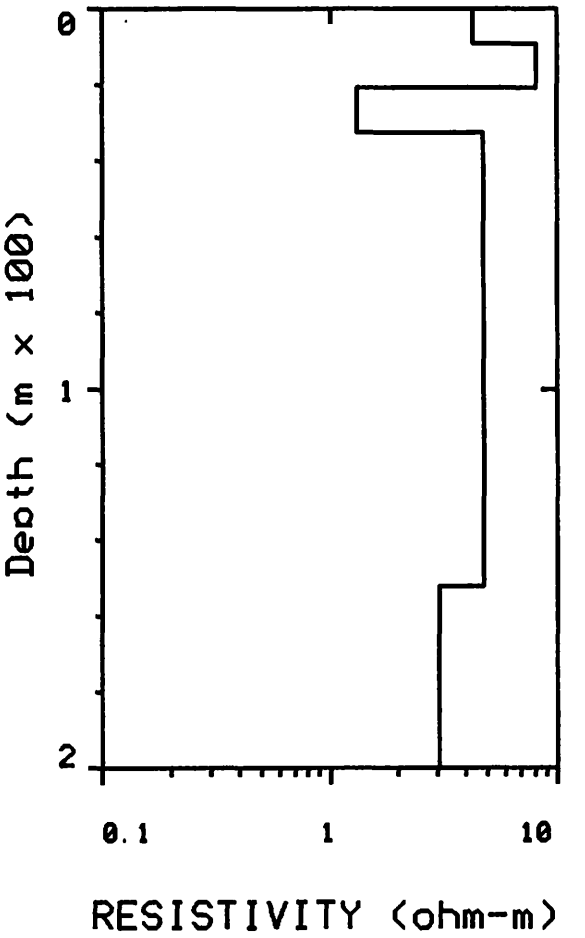
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.327 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			3.00	
1	6.59	19.29	-16.29	2.92
2	13.51	16.80	-33.10	1.24
3	4.03	21.00	-54.11	5.20
4	9.54	39.98	-94.10	4.18
5	4.30			

ALL PARAMETERS ARE FREE





50x50m		FITTING ERROR: 3.2%
I.P.R.G.		Shvei Zion 157813/264233 (+2.6)
Data Set: HS546	Date: 15-Dec-99	
Equipment: Geonics PROTEM	Sounding: z	Azimuth:

DATA SET: HS548

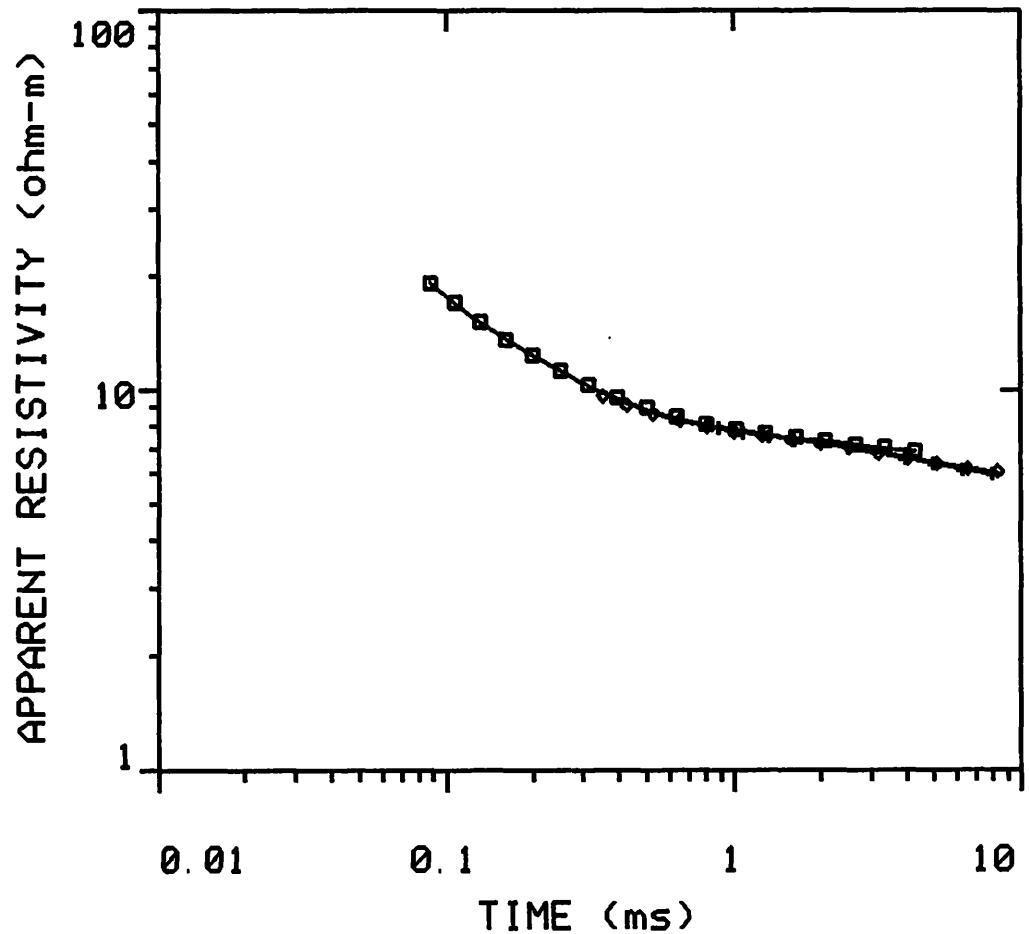
CLIENT: 100x100m DATE: 16-Dec-99
LOCATION: Betzet East SOUNDING: z
COUNTY: 161084/274871 (+17.5) ELEVATION: 17.50 m
PROJECT: FITTING ERROR: 3.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 133.0000 SLOPE: NONE

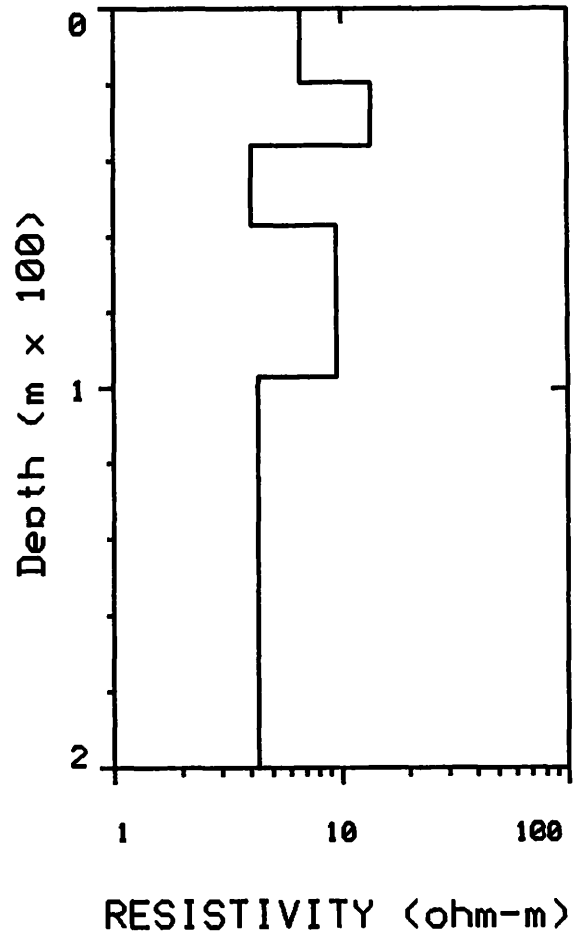
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.311 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			17.50	
1	179.9	9.97	7.52	0.0554
2	27.06	41.93	-34.41	1.54
3	2.55	31.69	-66.11	12.42
4	4.74	66.67	-132.7	14.05
5	2.41			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 1.3%
I.P.R.G.	Shvei Zion 158073/264255 (+3.0)
Data Set: HS547 / Date: 15-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS549

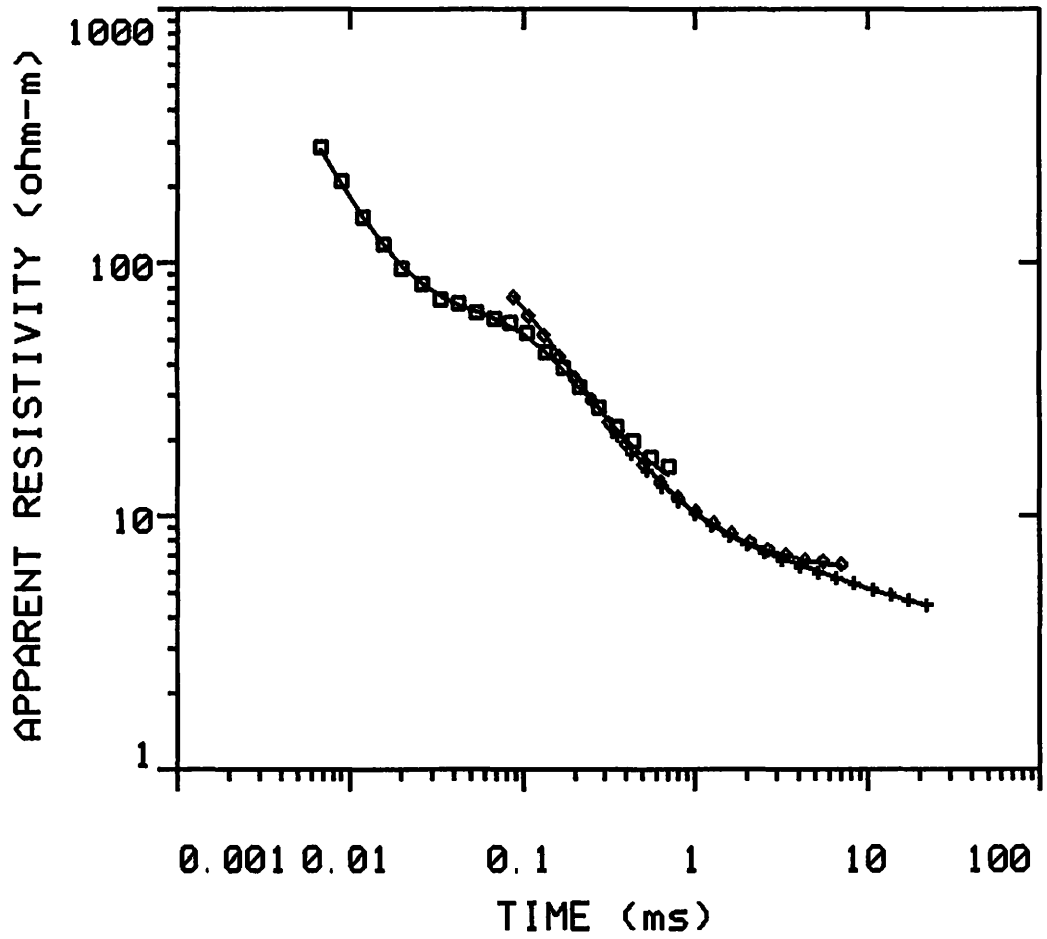
CLIENT: 100x100m DATE: 16-Dec-99
LOCATION: Khatzrot Yosef SOUNDING: z
COUNTY: 157751/262561 (+2.2) ELEVATION: 2.20 m
PROJECT: FITTING ERROR: 1.8% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 134.0000 SLOPE: NONE

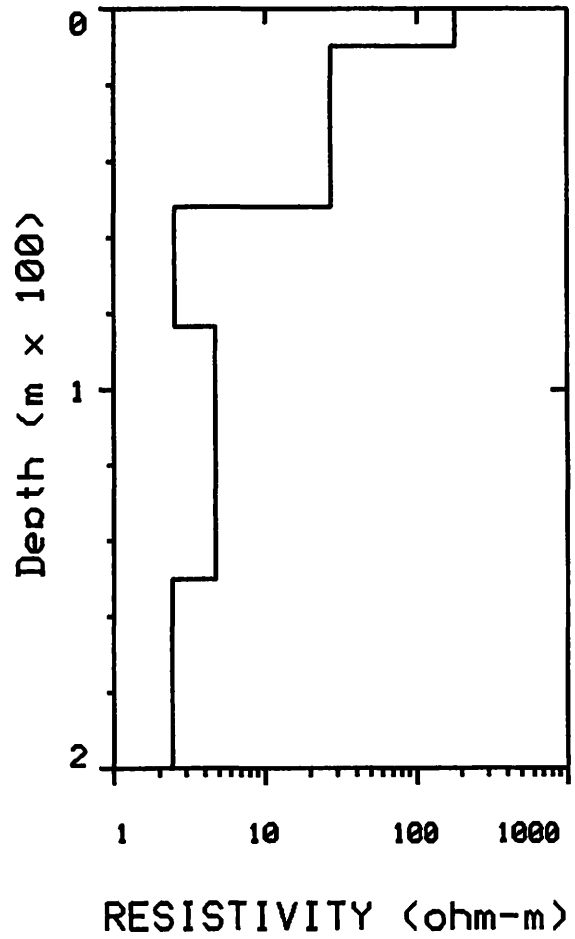
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.821 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			2.20	
1	3.28	21.18	-18.98	6.44
2	5.83	13.34	-32.32	2.28
3	1.02	16.55	-48.88	16.13
4	2.28	33.99	-82.87	14.90
5	0.552	49.85	-132.7	90.22
6	0.173			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.3%
I.P.R.G.	Betzet East 161084/274871 (+17.5)
Data Set: HS548 / Date: 16-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS550

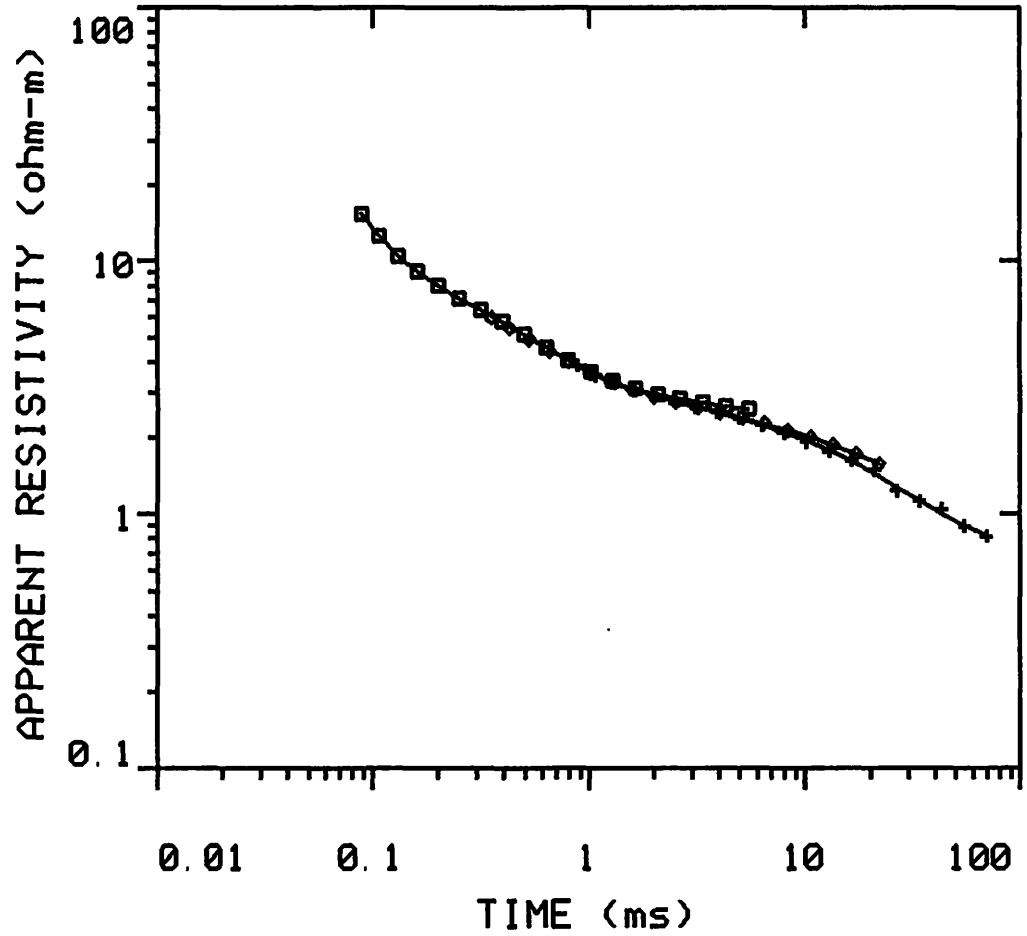
CLIENT: 100x100m DATE: 16-Dec-99
LOCATION: Khatzrot Yosef SOUNDING: z
COUNTY: 158125/262487 (+3.6) ELEVATION: 3.60 m
PROJECT: FITTING ERROR: 3.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 135.0000 SLOPE: NONE

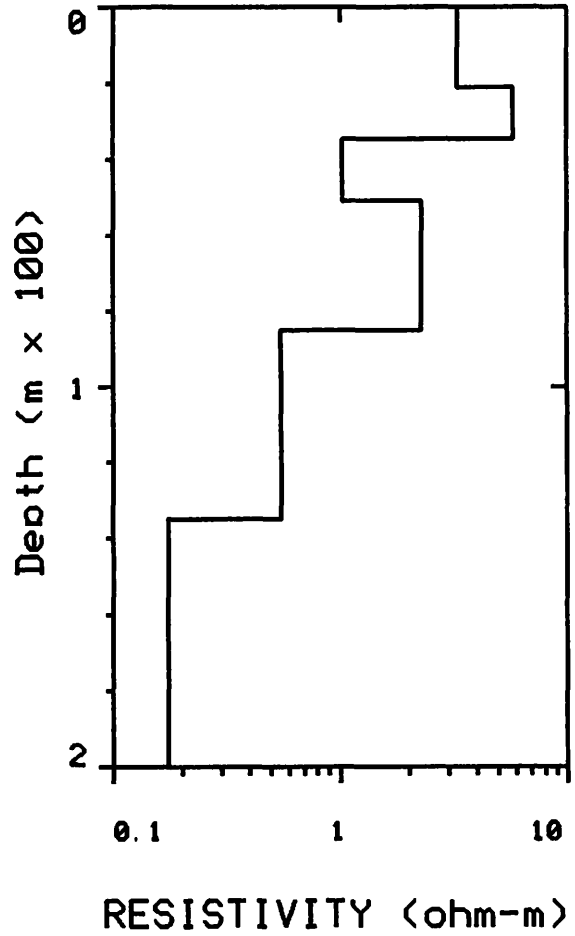
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.304 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			3.60	
1	63.35	5.71	-2.11	0.0902
2	1.28	4.67	-6.79	3.65
3	10.12	7.99	-14.78	0.789
4	19.70	16.95	-31.73	0.860
5	2.08	62.43	-94.16	29.97
6	9.17			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 1.8%
I.P.R.G.	Khatzrot Yosef 157751/262561 (+2.2)
Data Set: HS549 / Date: 16-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS551

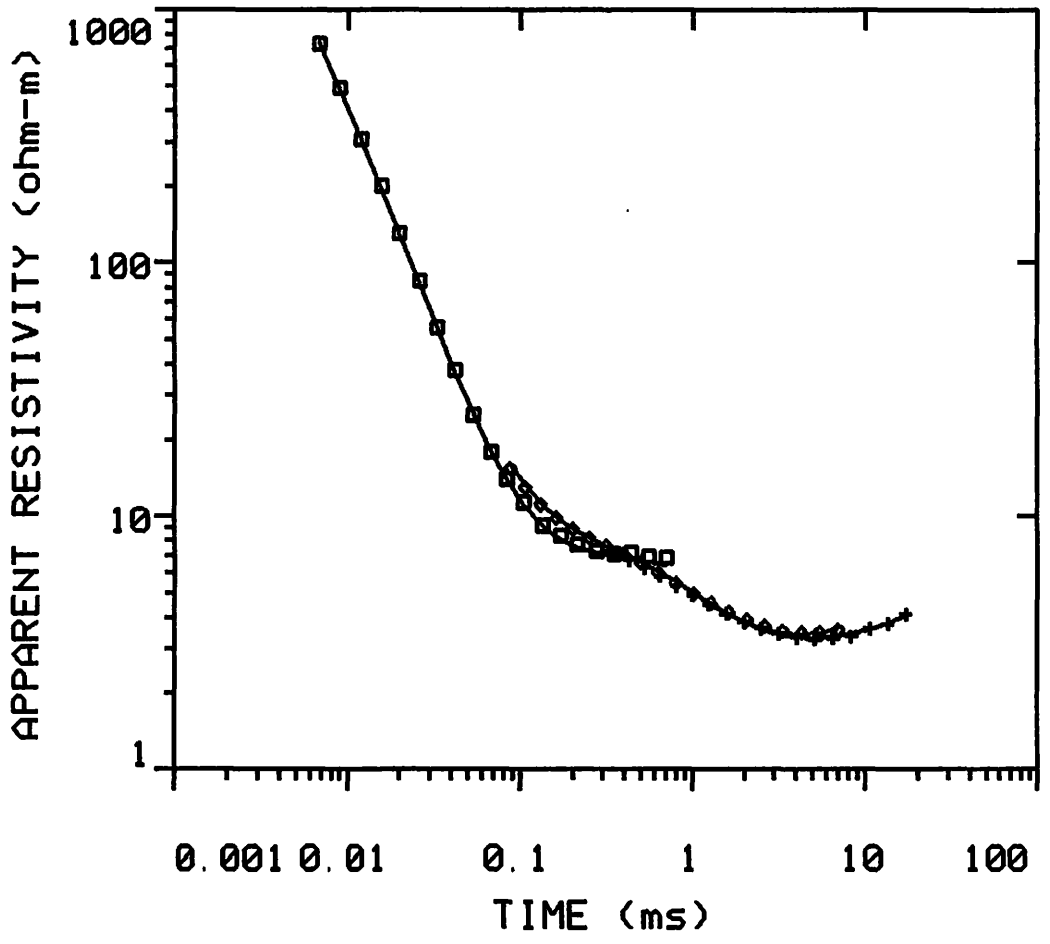
CLIENT: 100x100m DATE: 16-Dec-99
LOCATION: Bustan Hagalil SOUNDING: z
COUNTY: 157835/261625 (+3.4) ELEVATION: 3.40 m
PROJECT: FITTING ERROR: 2.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 136.0000 SLOPE: NONE

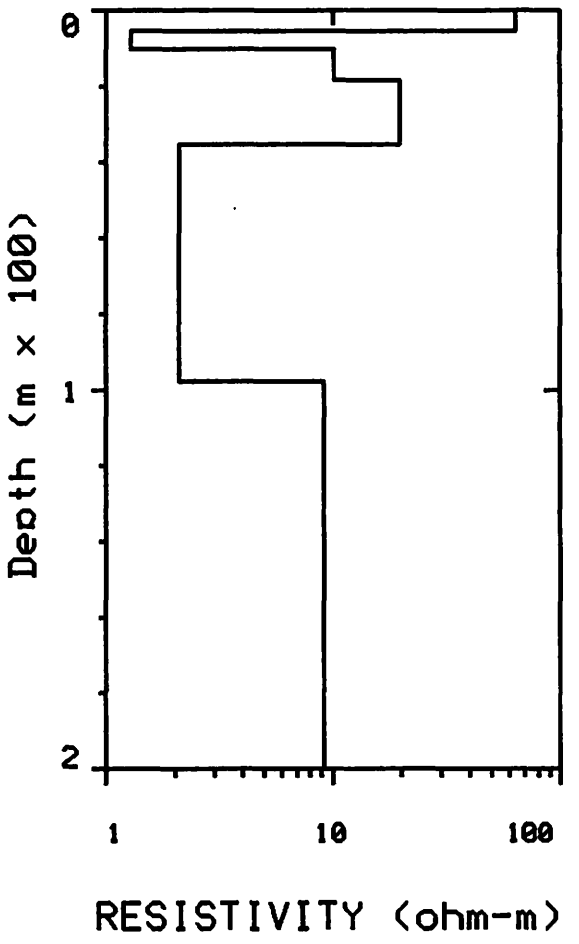
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 2.078 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			3.40	
1	3.34	18.02	-14.62	5.38
2	6.26	26.54	-41.17	4.23
3	2.74	21.50	-62.68	7.83
4	1.01	20.62	-83.30	20.32
5	1.80	38.87	-122.1	21.53
6	0.971			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 3.3%
I.P.R.G.	Khatzrot Yosef 158125/262487 (+3.6)
Data Set: HS550 / Date: 16-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS552

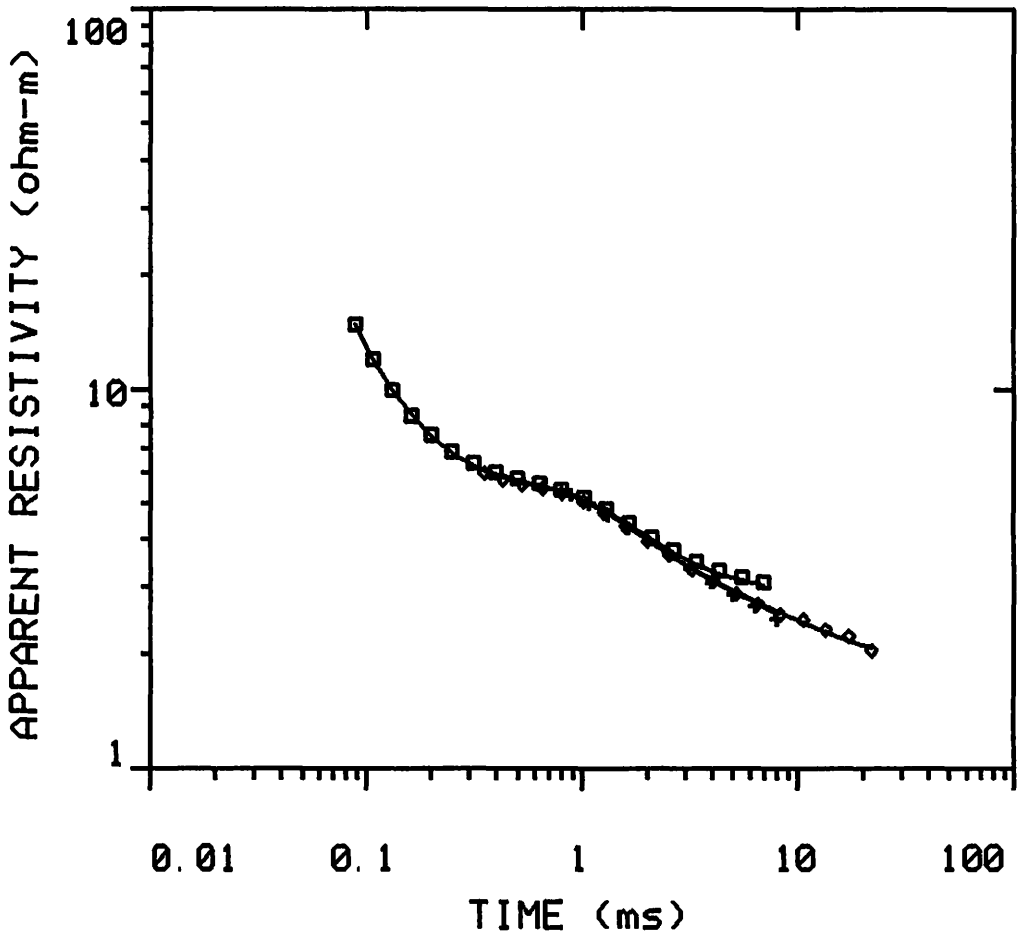
CLIENT: 100x100m DATE: 16-Dec-99
LOCATION: Bustan Hagalil SOUNDING: z
COUNTY: 158350/261007 (+4.1) ELEVATION: 4.10 m
PROJECT: FITTING ERROR: 4.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 137.0000 SLOPE: NONE

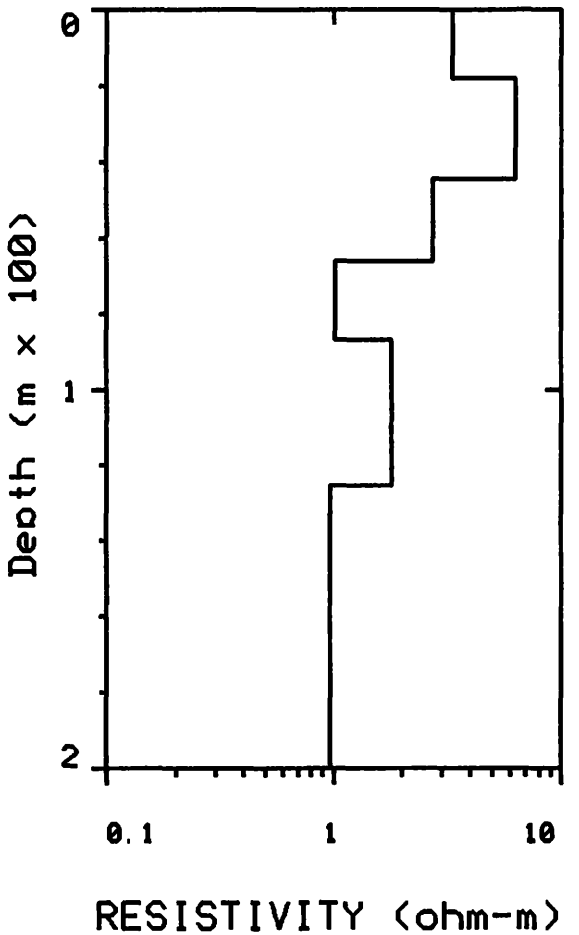
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.348 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			4.10	
1	4.77	5.66	-1.56	1.18
2	13.66	48.56	-50.12	3.55
3	1.28	40.34	-90.47	31.50
4	0.924	101.0	-191.4	109.2
5	35.43			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 2.1%
I.P.R.G.	Bustan Hagalil 157835/261625 (+3.4)
Data Set: HS551 / Date: 16-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS553

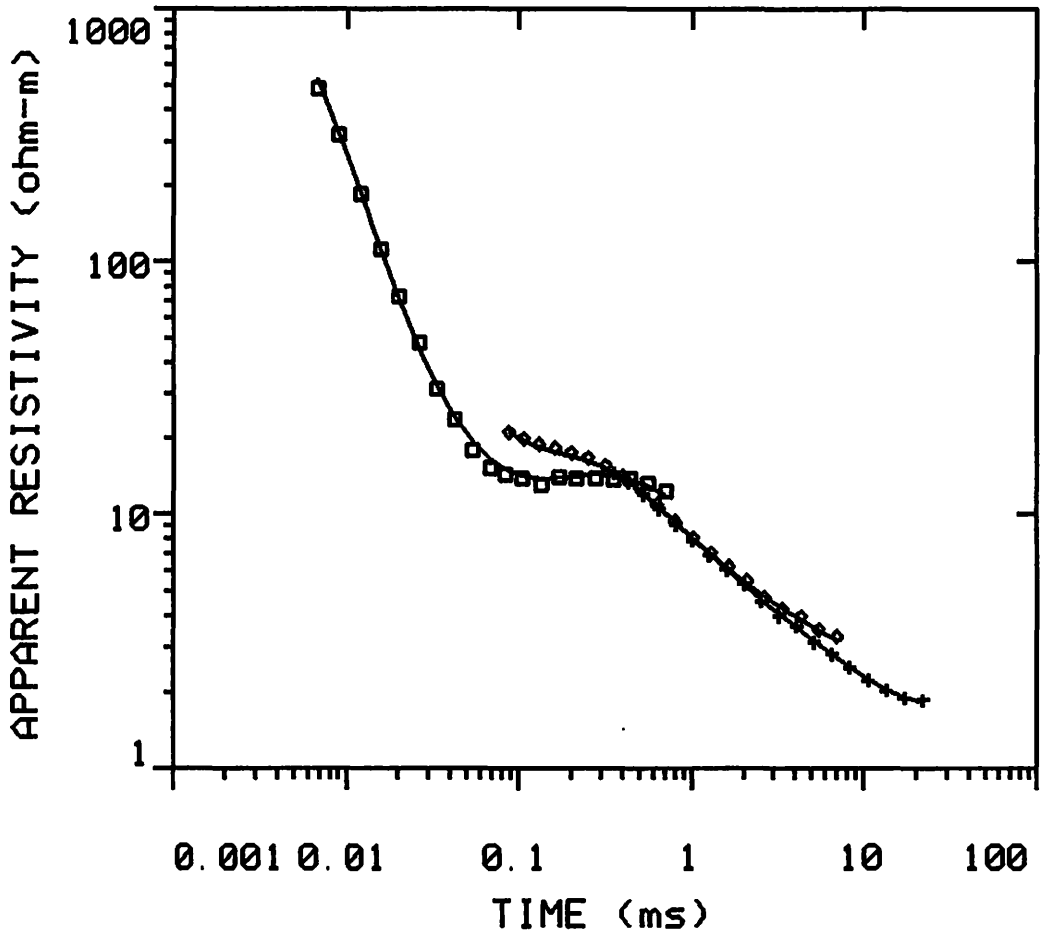
CLIENT: 100x100m DATE: 19-Dec-99
LOCATION: Kfar Galim SOUNDING: z
COUNTY: 146436/240662 (+20.0) ELEVATION: 20.00 m
PROJECT: FITTING ERROR: 4.0% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 138.0000 SLOPE: NONE

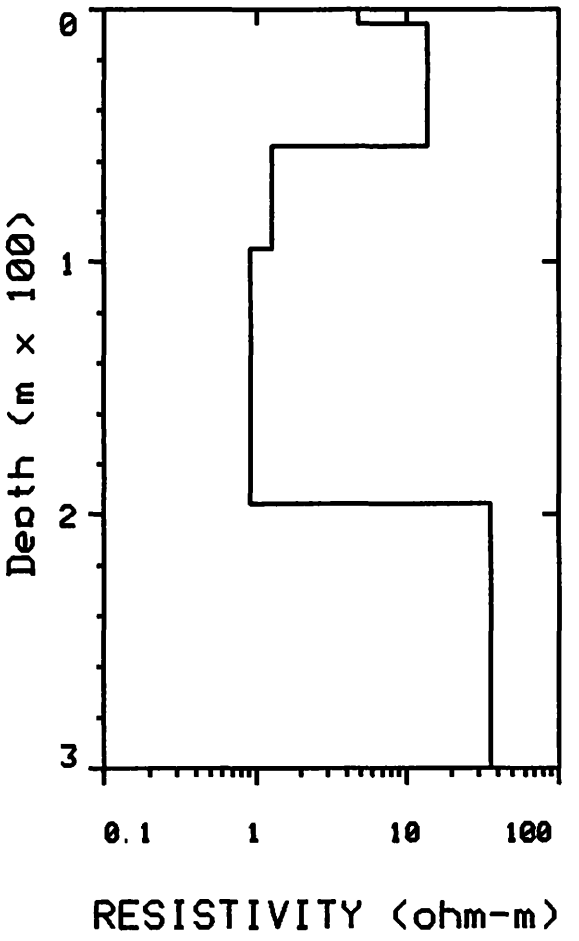
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.974 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			20.00	
1	6.64	15.76	4.23	2.37
2	31.84	25.34	-21.10	0.795
3	7.08	12.44	-33.55	1.75
4	41.56	129.0	-162.6	3.10
5	5.00			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 4.3%
I.P.R.G.	Bustan Hagalil 158350/261007 (+4.1)
Data Set: HS552 / Date: 16-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS554

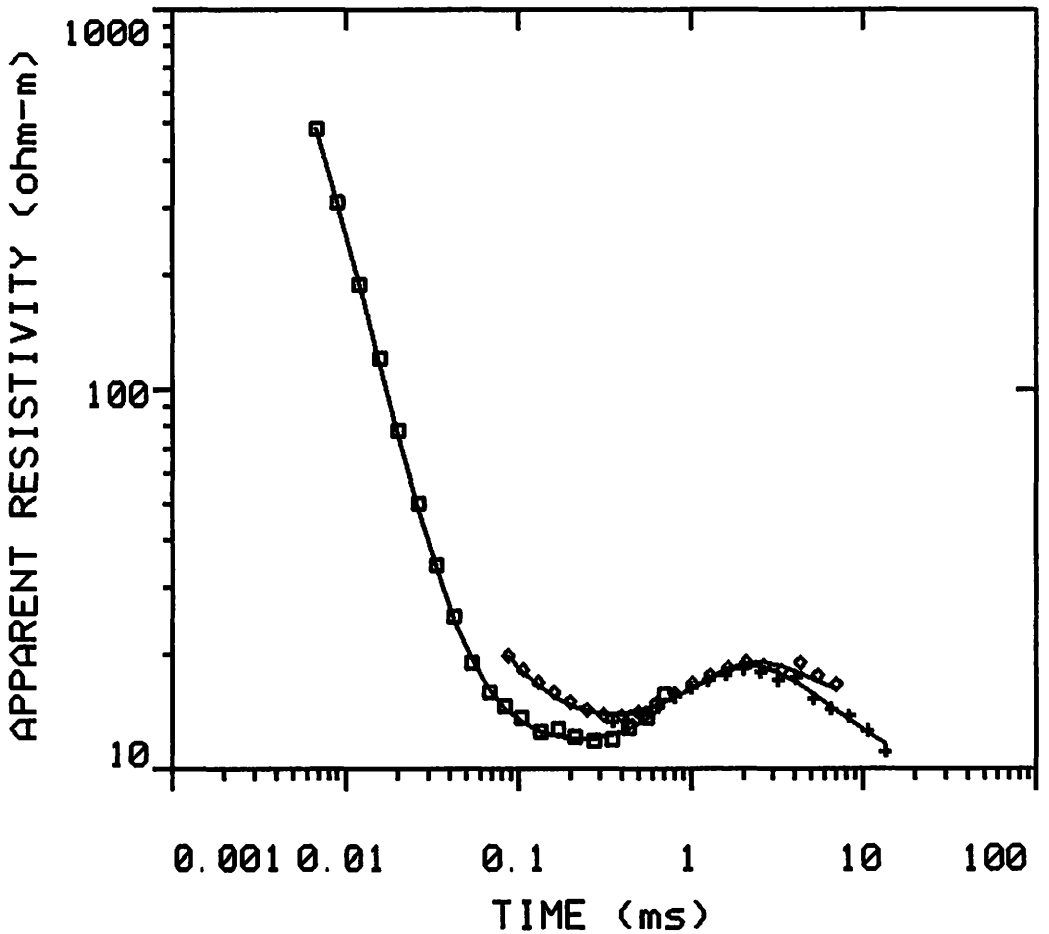
CLIENT: 100x100m DATE: 19-Dec-99
LOCATION: Newe-Yam SOUNDING: z
COUNTY: 144244/231191 (+6.1) ELEVATION: 6.10 m
PROJECT: FITTING ERROR: 4.6% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 139.0000 SLOPE: NONE

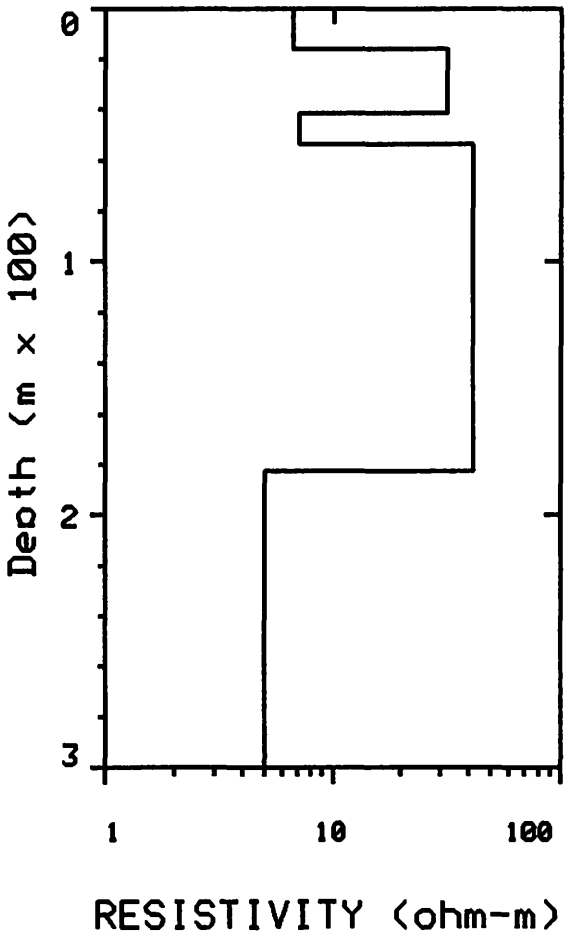
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.607 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			6.10	
1	3.08	11.19	-5.09	3.62
2	97.91	82.28	-87.37	0.840
3	208.8	33.09	-120.4	0.158
4	29.99	31.59	-152.0	1.05
5	4.34	158.9	-310.9	36.61
6	2.05			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 4.0%
I.P.R.G.	Kfar Galim 146436/240662 (+20.0)
Data Set: HS553 / Date: 19-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS555

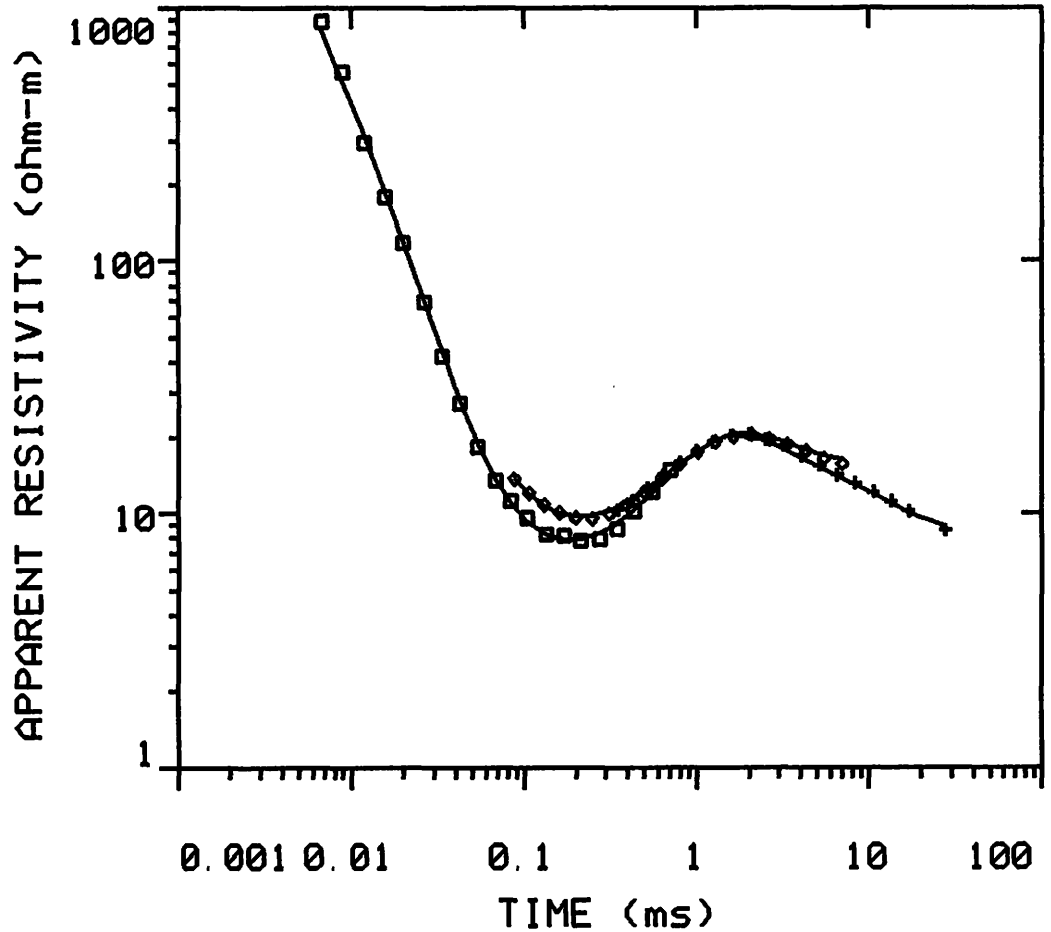
CLIENT: 200x200m DATE: 19-Dec-99
LOCATION: Einot-Shumi SOUNDING: z
COUNTY: 145827/216075 (+18.3) ELEVATION: 18.30 m
PROJECT: FITTING ERROR: 3.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 200.000 m by 200.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 140.0000 SLOPE: NONE

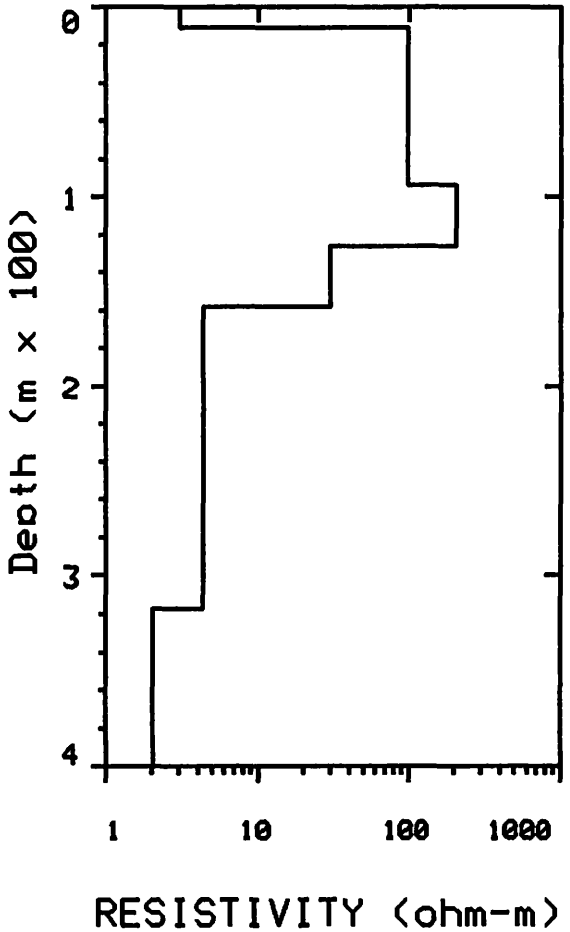
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.223 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			18.30	
1	51.29	92.19	-73.89	1.79
2	29.54	128.2	-202.1	4.33
3	7.37	41.53	-243.6	5.63
4	82.58	86.72	-330.3	1.05
5	150.6	94.70	-425.0	0.628
6	79.24	126.4	-551.5	1.59
7	3.05			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 4.6%
I.P.R.G.	Newe-Yam 144244/231191 (+6.1)
Data Set: HS554 / Date: 19-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS556

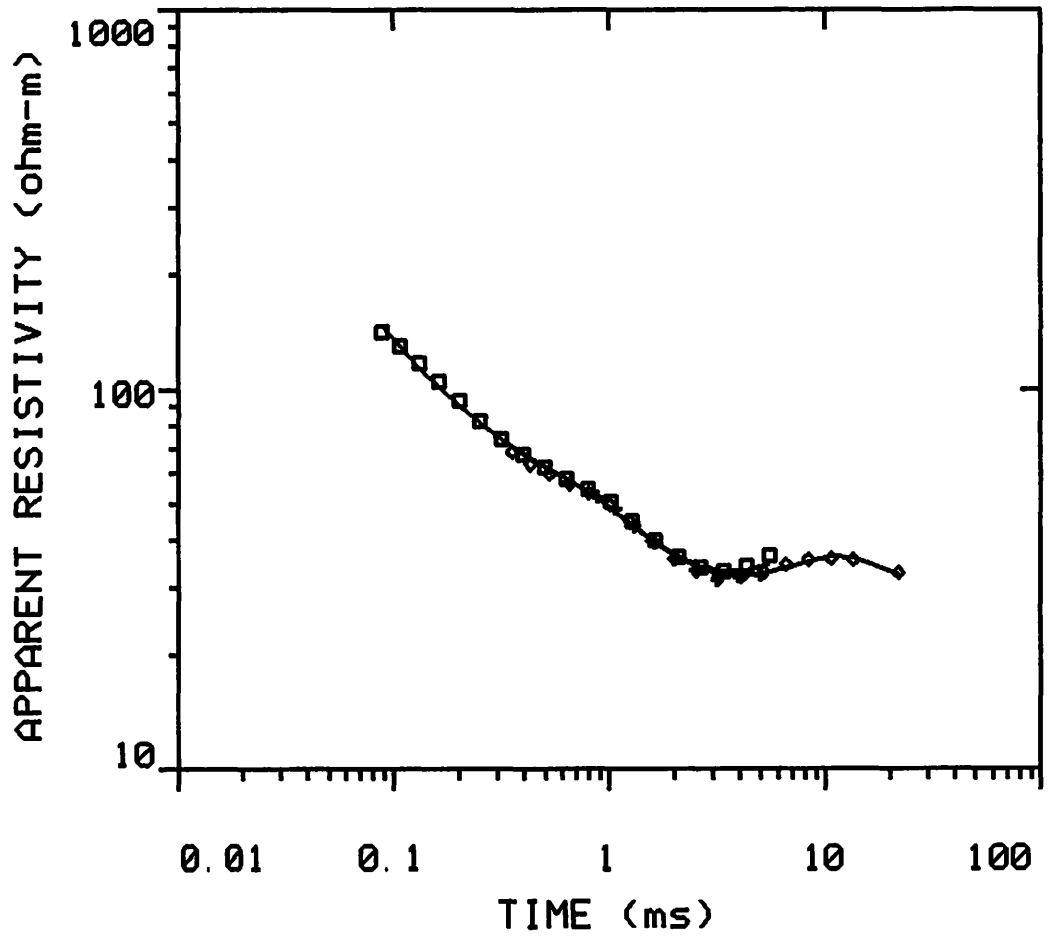
CLIENT: 100x100m DATE: 19-Dec-99
LOCATION: Shefaim SOUNDING: z
COUNTY: 133787/180958 (+33.8) ELEVATION: 33.80 m
PROJECT: FITTING ERROR: 4.2% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 141.0000 SLOPE: NONE

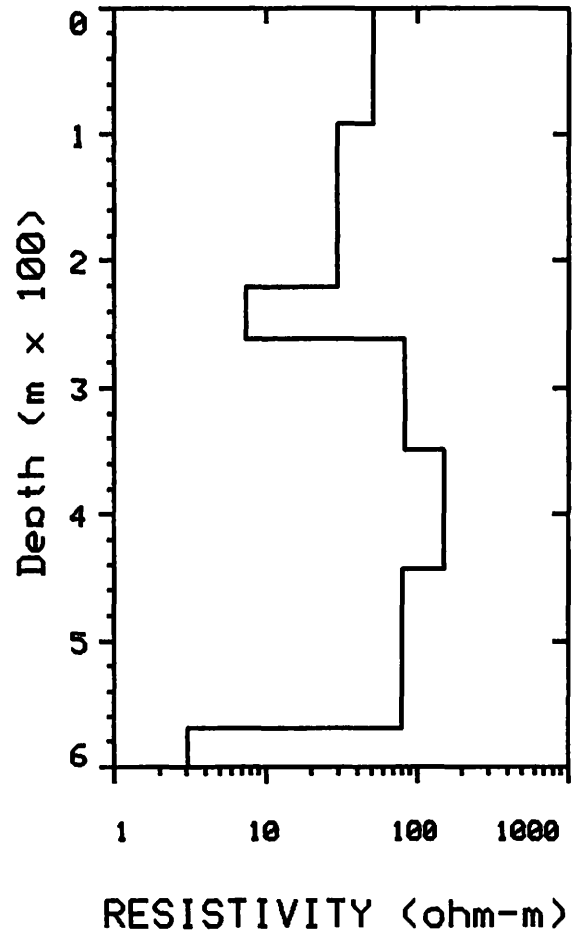
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.225 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			33.80	
1	102.9	32.82	0.971	0.318
2	22.39	29.37	-28.40	1.31
3	94.18	26.62	-55.02	0.282
4	1.27	25.11	-80.14	19.70
5	25.65	70.81	-150.9	2.76
6	1.23			

ALL PARAMETERS ARE FREE





200x200m	FITTING ERROR: 3.2%
I.P.R.G.	Einot-Shumi 145827/216075 (+18.3)
Data Set: HS555 / Date: 19-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS557

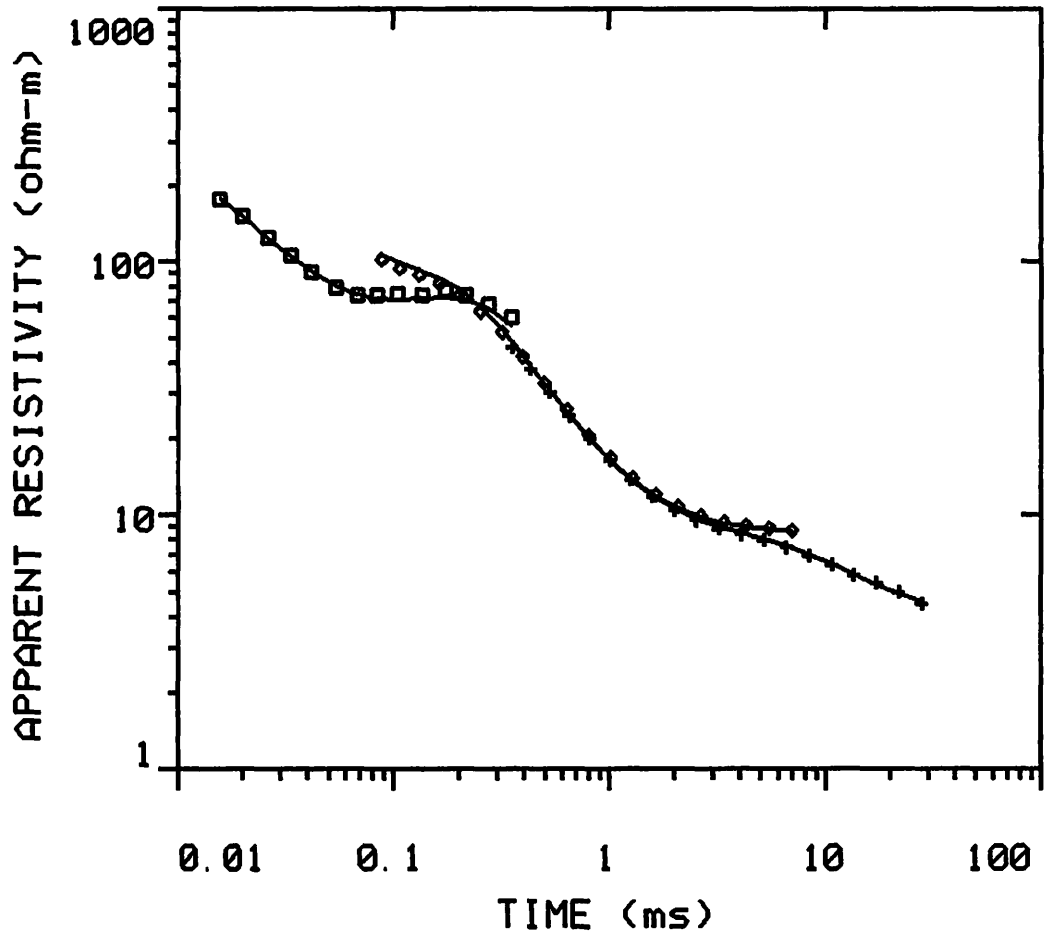
CLIENT: 50x50m DATE: 17-Jan-00
LOCATION: Palmakhim SOUNDING: z
COUNTY: 119737/143936 (+3.2) ELEVATION: 3.20 m
PROJECT: FITTING ERROR: 4.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 151.0000 SLOPE: NONE

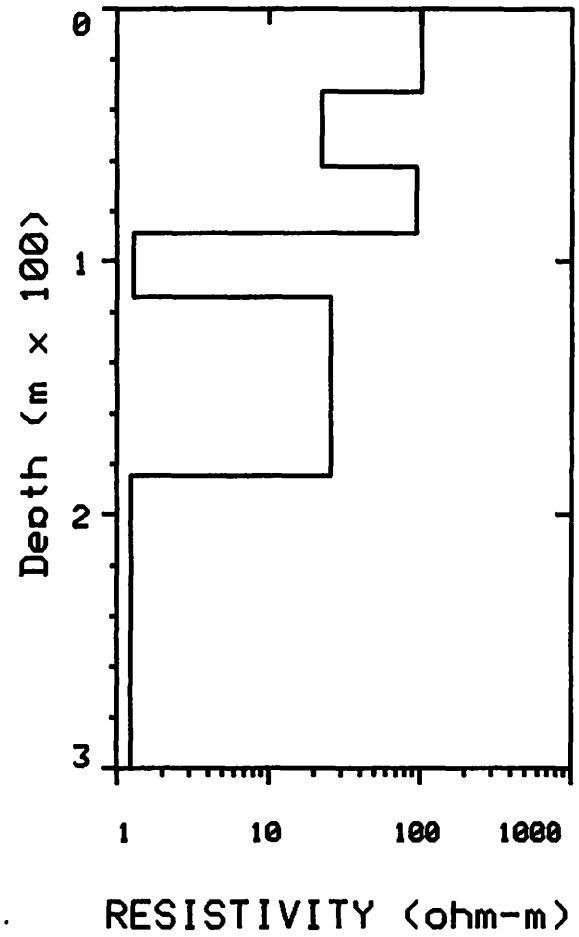
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 4.738 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			3.20	
1	180.9	11.75	-8.55	0.0649
2	12.71	15.41	-23.96	1.21
3	1.39	55.49	-79.45	39.90
4	4.24	19.03	-98.49	4.48
5	45.42	106.4	-204.9	2.34
6	0.957			

ALL PARAMETERS ARE FREE





100x100m	FITTING ERROR: 4.2%
I.P.R.G.	Shefaim 133787/180958 (+33.8)
Data Set: HS556 / Date: 19-Dec-99	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS558

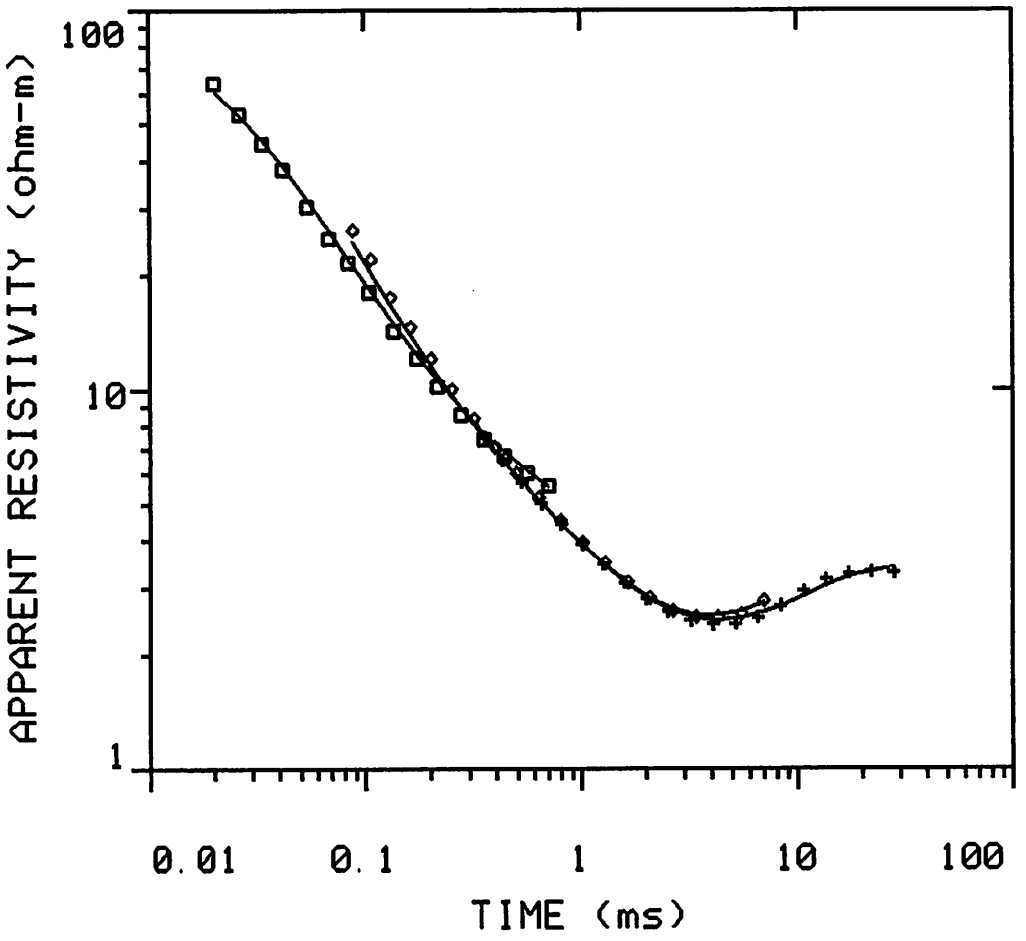
CLIENT: 50x50m DATE: 17-Jan-00
LOCATION: Palmakhim SOUNDING: z
COUNTY: 119874/143763 (+13.9) ELEVATION: 13.90 m
PROJECT: FITTING ERROR: 1.1% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 152.0000 SLOPE: NONE

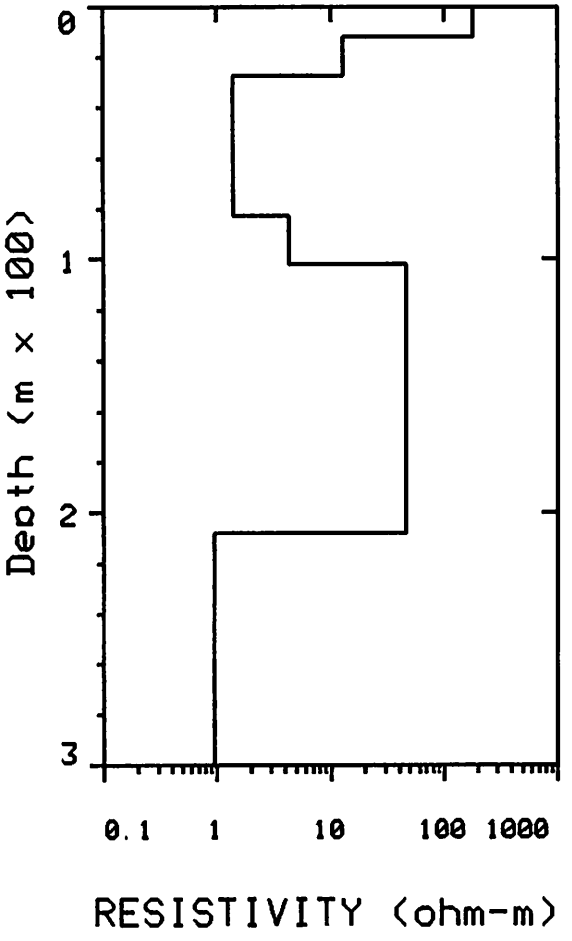
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.092 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			13.90	
1	71.45	28.85	-14.95	0.403
2	18.90	32.95	-47.90	1.74
3	2.15	28.78	-76.69	13.33
4	99.85	79.38	-156.0	0.795
5	3.31			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 4.7%
I.P.R.G.	Palmakhim 119737/143936 (+3.2)
Data Set: HS557 / Date: 17-Jan-00	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS559

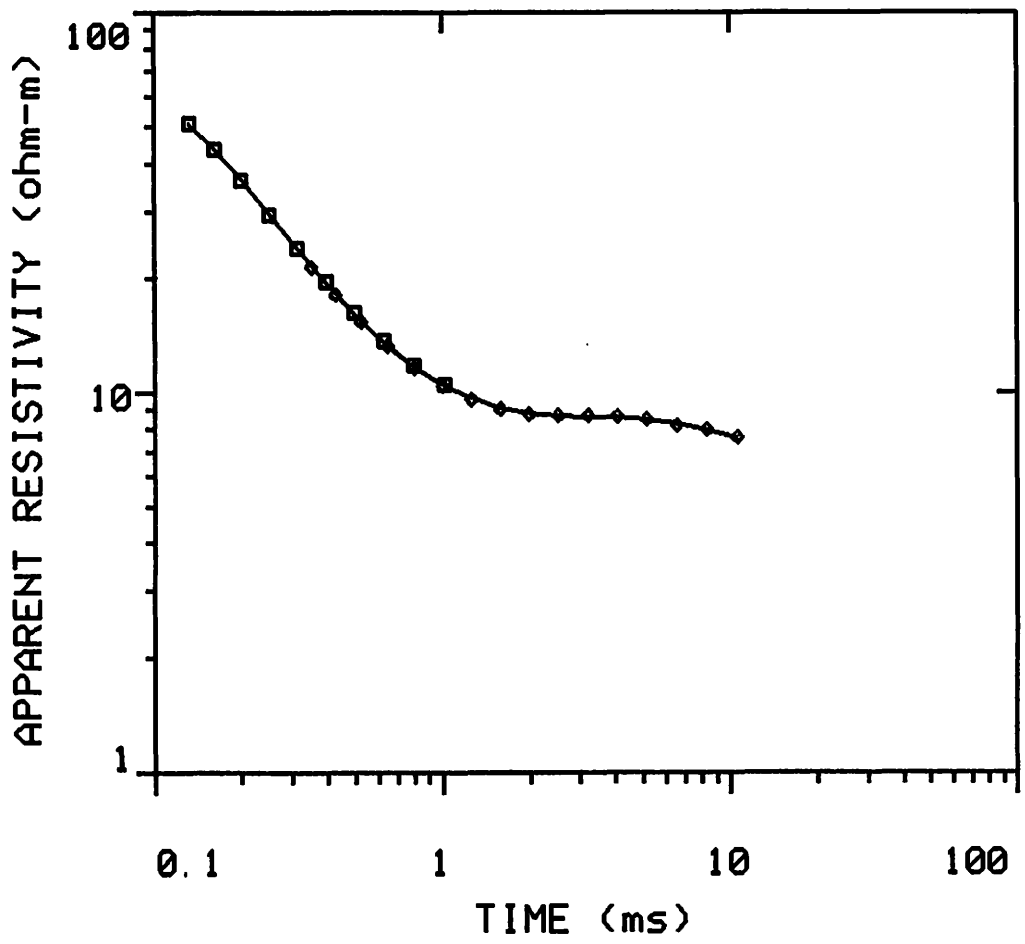
CLIENT: 50x50m DATE: 17-Jan-00
LOCATION: Palmakhim SOUNDING: z
COUNTY: 120988/146965 (+3.5) ELEVATION: 3.50 m
PROJECT: FITTING ERROR: 3.5% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 153.0000 SLOPE: NONE

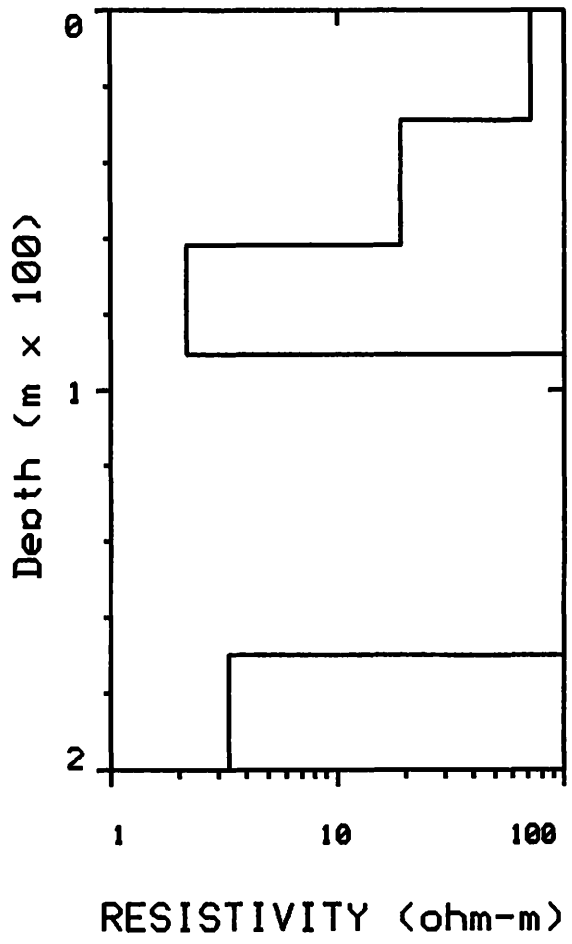
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 3.503 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			3.50	
1	5.66	4.91	-1.41	0.867
2	333.7	21.60	-23.01	0.0647
3	8.62	5.51	-28.53	0.639
4	1.28	24.25	-52.78	18.94
5	2.82	16.35	-69.14	5.78
6	51.52	104.0	-173.2	2.01
7	2.51			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 1.1%
I.P.R.G.	Palmakhim 119874/143763 (+13.9)
Data Set: HS558 / Date: 17-Jan-00	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS560

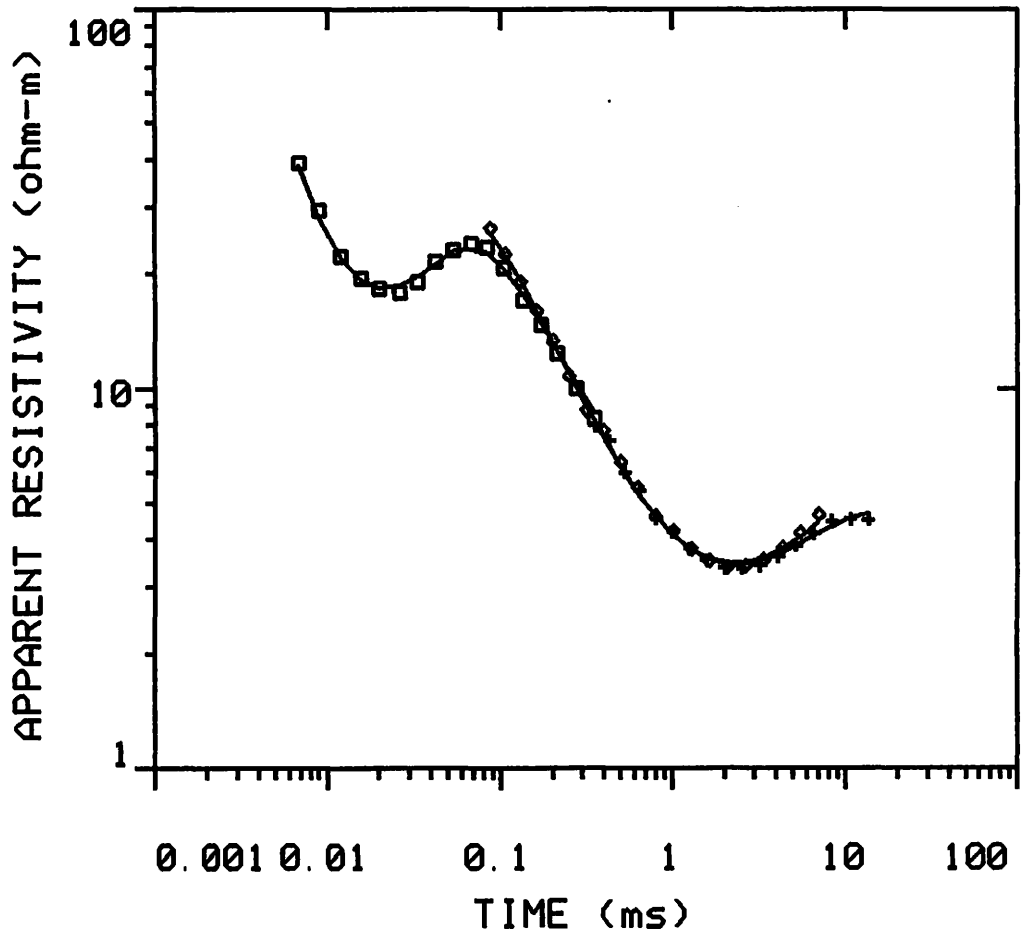
CLIENT: 50x50m DATE: 17-Jan-00
LOCATION: Palmakhim SOUNDING: z
COUNTY: 119416/142330 (+9.6) ELEVATION: 9.60 m
PROJECT: FITTING ERROR: 1.7% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 50.000 m by 50.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 154.0000 SLOPE: NONE

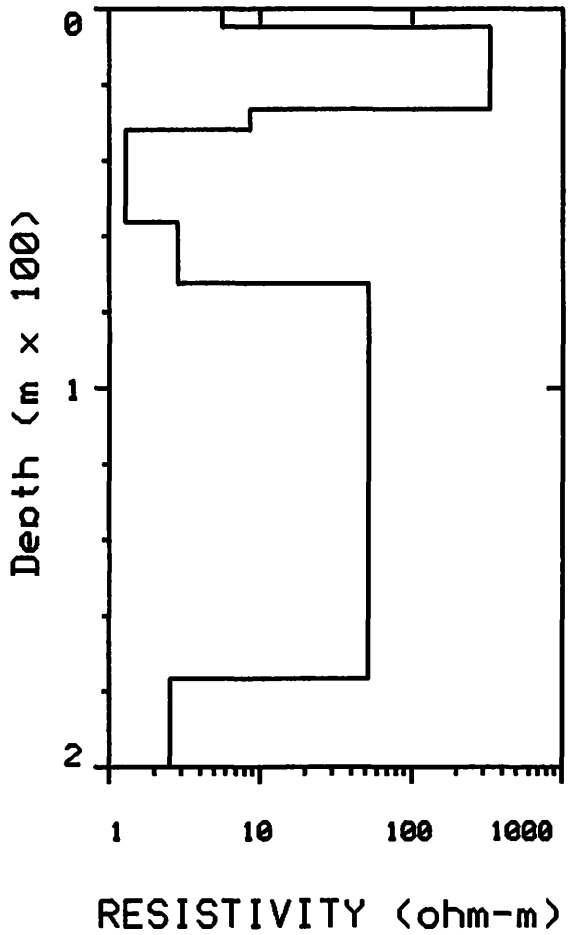
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.744 PERCENT

L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			9.60	
1	15.98	26.67	-17.07	1.66
2	101.7	31.70	-48.78	0.311
3	20.20	23.93	-72.71	1.18
4	2.36	52.06	-124.7	22.03
5	0.533			

ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 3.5%
I.P.R.G.	Palmakhim 120988/146965 (+3.5)
Data Set: HS559 / Date: 17-Jan-00	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:

DATA SET: HS561

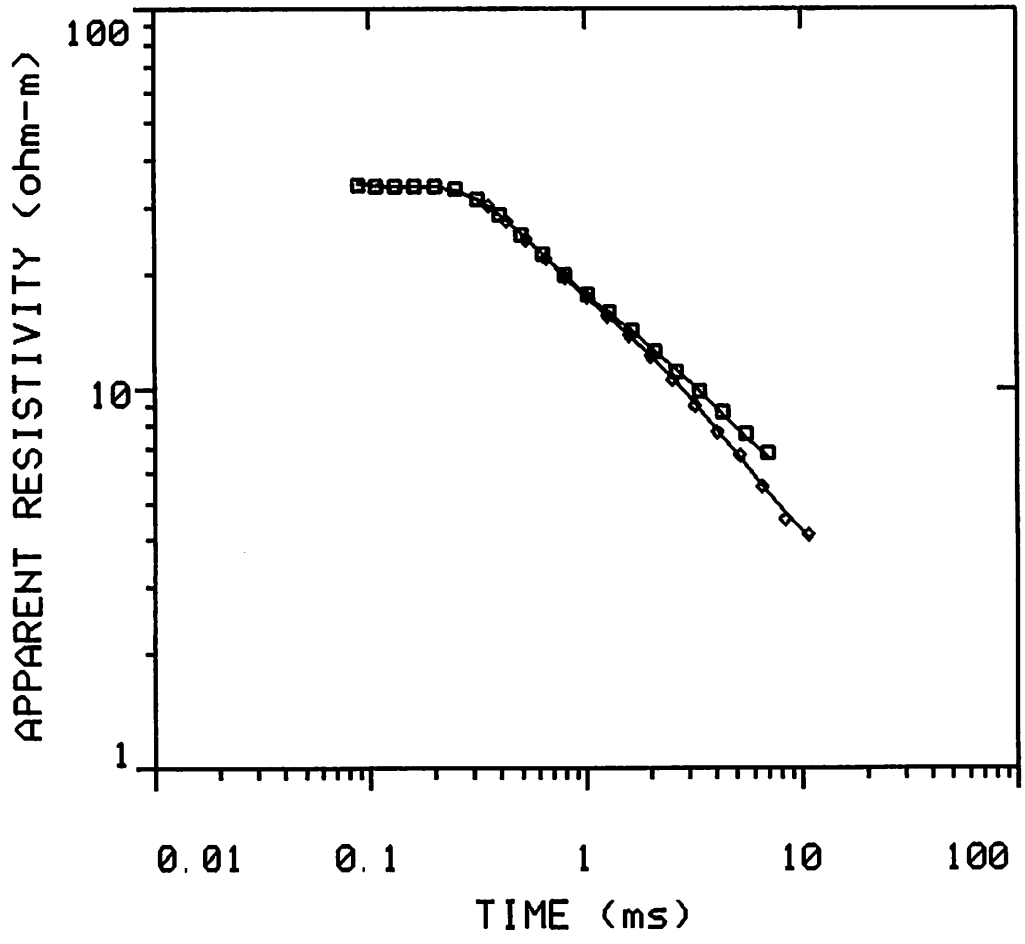
CLIENT: 100x100m DATE: 17-Jan-00
LOCATION: Palmakhim SOUNDING: z
COUNTY: 121194/146902 (+13.4) ELEVATION: 13.40 m
PROJECT: FITTING ERROR: 1.3% EQUIPMENT: Geonics PROTEM
LOOP SIZE: 100.000 m by 100.000 m AZIMUTH:
COIL LOC: 0.000 m (X), 0.000 m (Y) TIME CONSTANT: NONE
SOUNDING COORDINATES: E: 1.0000 N: 155.0000 SLOPE: NONE

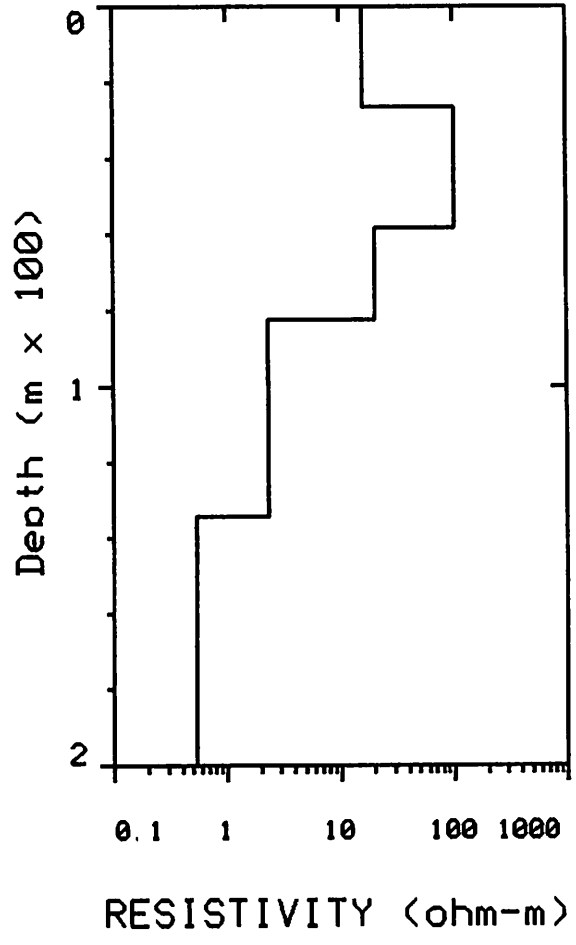
Central Loop Configuration
Geonics PROTEM System

FITTING ERROR: 1.288 PERCENT

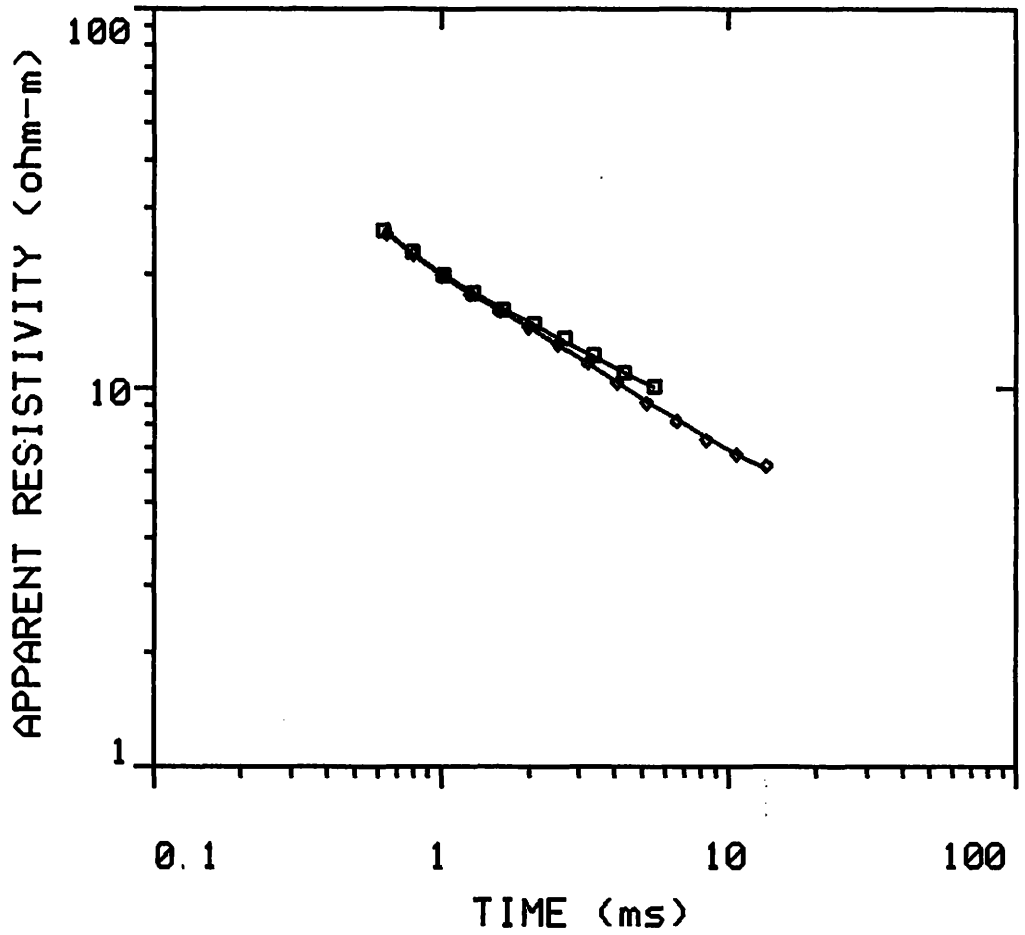
L #	RESISTIVITY (ohm-m)	THICKNESS (meters)	ELEVATION (meters)	CONDUCTANCE (Siemens)
			13.40	
1	51.41	81.22	-67.82	1.57
2	3.13	28.84	-96.67	9.20
3	10.19	40.46	-137.1	3.96
4	2.11			

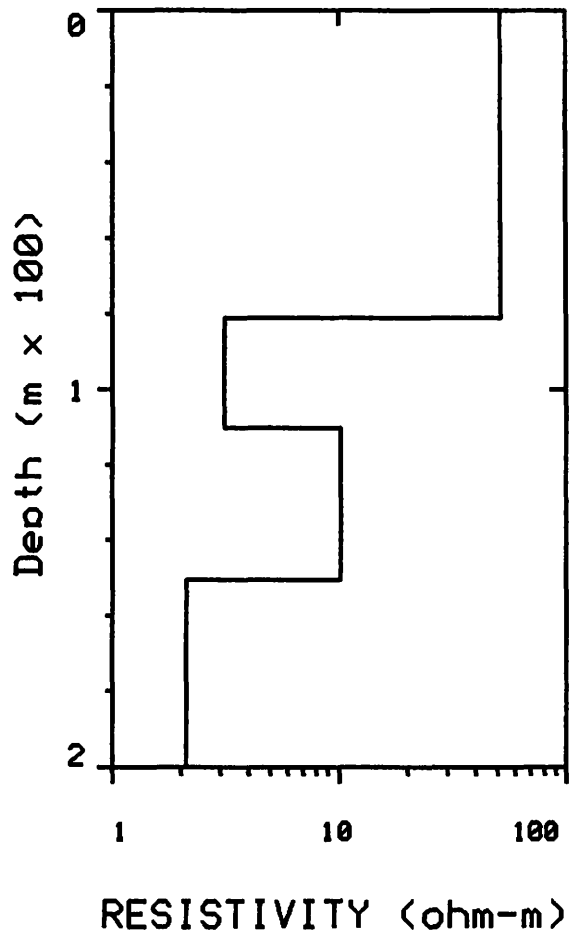
ALL PARAMETERS ARE FREE





50x50m	FITTING ERROR: 1.7%
I.P.R.G.	Palmakhim 119416/142330 (+9.6)
Data Set: HS560 / Date: 17-Jan-00	
Equipment: Geonics PROTEM / Sounding: z	Azimuth:





100x100m	FITTING ERROR: 1.3%
I.P.R.G.	Palmakhim 121194/146902 (+13.4)
Data Set: HS561 / Date: 17-Jan-00	
Equipment: Geonics PROTEM / Sounding: z	Azimuth: